IEEE P802.3.2™/D3.4

Draft Standard for Ethernet YANG Data Model Definition

Prepared by the

LAN/MAN Standards Committee
of the
IEEE Computer Society

This draft is a revision of IEEE Std 802.3.2-2019. Draft D3.4 is prepared for the fourth IEEE SA recirculation ballot. This draft expires 6 months after the date of publication or when the next version is published, whichever comes first.

Abstract: YANG models for IEEE Std 802.3 are defined in this standard. This standard also publishes these models in a machine-readable format.

Keywords: 802.3, 802.3.2, Ethernet, YANG

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When the IEEE-SA Standards Board approved this standard on XX Month 202X, it had the following membership:

Introduction

This introduction is not part of IEEE Std 802.3.2-202x, IEEE Draft Standard for Ethernet YANG Data Model Definitions.

The YANG modules included in this standard provide YANG versions of attributes defined in IEEE Std 802.3™-2022, Clause 30, as well as derivative attributes defined in other management information bases (e.g., SNMP attributes included in IEEE Std 802.3.1, YANG versions of IETF Etherlike MIB attributes, etc.). The YANG modules defined in this standard accommodate IEEE Std 802.3-2022, excluding any currently published or future amendments.

IEEE Std 802.3 will continue to evolve. New Ethernet capabilities are anticipated to be added within the next few years as amendments to this standard.

Contents

IEEE Standard for Ethernet YANG Data Model Definitions

1. Overview

This standard defines YANG modules for various Ethernet functions specified in IEEE Std 802.3.

1.1 Scope

This standard defines YANG data models for IEEE Std 802.3 Ethernet.

1.2 Purpose

The purpose of the standard is to define YANG modules for IEEE Std 802.3 and publish these modules in a machine-readable format.

1.3 Word usage

The word shall indicates mandatory requirements strictly to be followed in order to conform to the standard and from which no deviation is permitted (shall equals is required to).1,2

The word should indicates that among several possibilities one is recommended as particularly suitable, without mentioning or excluding others; or that a certain course of action is preferred but not necessarily required (should equals is recommended that).

The word may is used to indicate a course of action permissible within the limits of the standard (may equals is permitted to).

The word can is used for statements of possibility and capability, whether material, physical, or causal (can equals is able to).

1The use of the word must is deprecated and cannot be used when stating mandatory requirements, must is used only to describe unavoidable situations.

2The use of will is deprecated and cannot be used when stating mandatory requirements, will is only used in statements of fact

1.4 Machine-readable YANG modules

> ***Editor's Note (to be removed prior to publication):***
>
> Yang files contained in https://github.com/YangModels/yang/tree/main/standard/ieee/published/802.3 are IEEE 802.3.2-2019 version and will be updated at the publication time.

The authoritative machine-readable YANG files are attached to this standard. A copy is also available for download at the following URL: https://github.com/YangModels/yang/tree/master/standard/ieee/published/802.3 as text files with a *.yang* extension, e.g., *ieee802-ethernet-interface.yang*. The use of specialized tools to view YANG modules may be useful to create tree, UML image, and HTML outputs from the YANG modules.

Like other languages, YANG (see IETF RFC 7950) has an accepted style for machine-readable files, which was followed during the development of this standard. This formatting may not be preserved when importing the machine-readable YANG modules into the PDF. In case of any discrepancies, the machine-readable files attached to this standard take precedence.

1.5 Summary of YANG-based management framework

The structure of YANG-based management framework closely resembles the structure of the Internet-Standard Management Framework, described in detail in section 7 of IETF RFC 3410.

Managed objects defined using YANG modeling language are hosted on the managed device and accessed through NETCONF (see IETF RFC 7803) or RESTCONF (see IETF RFC 8040). This standard specifies YANG modules that are compliant to YANG 1.1 (see IETF RFC 7950).

1.6 Security considerations

The YANG modules defined in this standard are designed to be accessed via network management protocols, including NETCONF (see IETF RFC 7803) or RESTCONF (see IETF RFC 8040). The lowest NETCONF layer is the secure transport layer, and the mandatory-to-implement secure transport is Secure Shell (SSH) (see IETF RFC 6242) or TLS (see IETF RFC 8446). The lowest RESTCONF layer is HTTPS, and the mandatory-to-implement secure transport is TLS (see IETF RFC 8446).

The NETCONF access control model (see IETF RFC 8341) provides the means to restrict access for particular NETCONF or RESTCONF users to a pre-configured subset of all available NETCONF or RESTCONF protocol operations and content.

There are a number of data nodes defined in these YANG modules that are writable/creatable/deletable, i.e., have the config property set to true, which is the default setting. These data nodes can be considered sensitive or vulnerable in some network environments. Write operations (e.g., edit-config) to these data nodes without proper protection can have a negative effect on network operations.

Some of the readable data nodes in these YANG modules can be considered sensitive or vulnerable in some network environments. It is thus important to control read access (e.g., via get, get-config, or notification) to these data nodes.

Some of the RPC operations in these YANG modules can be considered sensitive or vulnerable in some network environments. Therefore, it is important to control access to these operations.

1.7 YANG module syntax validation

All YANG modules included in this standard are YANG 1.1 (see IETF RFC 7950) compliant and pass automated checks using tools available at the time of publication.

The following open source and/or free versions of YANG validation tools may be used: Pyang (see https://github.com/mbj4668/pyang), ConfD (see http://www.tail-f.com/confd-basic), as well as other YANG model validation tools listed at http://www.yangvalidator.com.

2. Normative references

The following referenced documents are indispensable for the application of this document (i.e., they must be understood and used, so each referenced document is cited in text and its relationship to this document is explained). For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

IEEE Std 802.3™-2022, IEEE Standard for Ethernet.

> ***Editor's Note (to be removed prior to publication):***
>
> IEEE Std 802.3.1 to be updated to correct date once project is complete

IEEE Std 802.3.1™-2013, IEEE Standard for Management Information Base (MIB) Definitions for Ethernet.

IETF RFC 3410, *Introduction and Applicability Statements for Internet Standard Management Framework*, J. Case, R. Mundy, D. Partain, B. Stewart, December 2002.

IETF RFC 6242, *Using the NETCONF Protocol over Secure Shell (SSH)*, Wasserman M, June 2011.

IETF RFC 7803, *Changing the Registration Policy for the NETCONF Capability URNs Registry*, B. Leiba February 2016.

IETF RFC 7950, *The YANG 1.1 Data Modeling Language*, Bjorklund M., August 2016.

IETF RFC 8040, *RESTCONF Protocol*, Bierman A., Bjorklund M., and Watsen K., January 2017.

IETF RFC 8341, *Network Configuration Access Control Model*, A. Bierman and M. Bjorklund, March 2018.

IETF RFC 8407, *Guidelines for Authors and Reviewers of YANG Data Model Documents*, Bierman A., October 2018.

IETF RFC 8446, *The Transport Layer Security (TLS) Protocol Version 1.3*, E. Rescorla, August 2018.

3. Definitions

For the purposes of this document, the following terms and definitions apply. Some terms used in this document are defined in IEEE Std 802.3, and where alternative definitions occur in the IEEE Standards Dictionary, the IEEE Std 802.3 definition should be used. The *IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause.[f]

3.1 data model. model that describes how data are represented and accessed

3.2 YANG module. module that defines a hierarchy of nodes that can be used for NETCONF-based (see IETF RFC 7803) and RESTCONF-based (see IETF RFC 8040) operations

NOTE—With its definitions and the definitions it imports or includes from elsewhere, a module is self-contained and can be compiled.

[f] *IEEE Standards Dictionary Online* is available at: http://dictionary.ieee.org/.

4. Abbreviations

This standard contains the following abbreviations:

CO	Central Office
CPE	Customer Premise Equipment
CSMA/CD	carrier sense multiple access with collision detection
DTE	data terminal equipment
EPON	Ethernet passive optical networks
IEEE	Institute of Electrical and Electronics Engineers
IETF	Internet Engineering Task Force
LLID	Link Local Identifier
MPCP	Multi-Point Control Protocol
NETCONF	Network Configuration Protocol
OAM	Operations, Administration, and Maintenance
ONU	Optical Network Unit
OLT	Optical Line terminal
PoE	Power over Ethernet
PoDL	Power over Data Line
RESTCONF	RESTful Configuration Protocol
TDM	Time Division Multiplexing
TDMA	Time Division Multiple Access
WDM	Wavelength Division Multiplexing
YANG	Yet Another Next Generation

5. Ethernet YANG Module

5.1 YANG module structure

Four modules defined in this clause are focused on the configuration and monitoring of IEEE Std 802.3 Ethernet interfaces.

ieee802-ethernet-interface YANG module contains definitions of current attributes used widely in the industry in current products,

ieee802-ethernet-interface-half-duplex YANG module contains definitions of half-duplex attributes.

ieee802-ethernet-lldp YANG module contains definitions for configuring LLDP for IEEE Std 802.3 compliant interfaces.

ieee802-ethernet-mac-merge modules contain definition for configuration of MAC Merge for IEEE Std 802.1Qcw frame preemption.

ieee802-ethernet-phy YANG module contain definition for Ethernet Phy, derived from IANA-MAU-MIB REVISION 202309070000Z, the YANG and definitions correct as per release of 2024-11-09.

ieee802-ethernet-phy-type YANG module contain definition for Ethernet Phy Types, derived IANA-MAU-MIB REVISION 202309070000Z using smidump, the YANG and definitions are correct as per 2024-11-09.

NOTE—ieee802-ethernet-phy-type this module may in future be maintained by an external organization (IANA).

This standard does not have a normative requirement for data nodes of the base ietf-interfaces YANG module, but the following data nodes are supported: name, description, type, enabled, admin-status, oper-status, if-index, and phys-address.

5.2 Mapping of IEEE Std 802.3, Clause 30 managed objects

This subclause contains the mapping between YANG data nodes included in *ieee802-ethernet-interface* (see Table 5–1), *ieee802-ethernet-interface-half-duplex* (see Table 5–4), ieee802-ethernet-mac-merge (see Table 5–5), and *ieee802-ethernet-lldp* (see Table 5–6) YANG modules, managed objects, and attributes defined in IEEE Std 802.3, Clause 30.

Table 5–1—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-interface* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-interface* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
oAutoNegoti-aion	acAutoNegAdminControl	30.6.1.2.2	interfaces/interface/ethernet/	auto-negotiation/enable	R/W
	aAutoNegAutoConfig	30.6.1.1.4		negotiation-status	R
N/A	N/A			flow-control/pause/direction	R/W
oMACControl-FunctionEntity	aPAUSEMACCtrlFramesReceived	30.3.4.3		flow-control/pause/statistics/in-frames-pause	R
	aPAUSEMACCtrlFramesTransmitted	30.3.4.2		flow-control/pause/statistics/out-frames-pause	R
N/A	dot3HCOutPFCFrames				
N/A	N/A			flow-control/force-flow-control	R/W
N/A	N/A			speed	R/W
oMACEntity	aDuplexStatus	30.3.1.1.32		duplex	R/W
	aMaxFrameLength	30.3.1.1.37		max-frame-length	R
	aSlowProtocolFrameLimit	30.3.1.1.38		frame-limit-slow-protocol	R
oEXTENSION	aEXTENSIONMACCtrlStatus	30.3.8.3		mac-control-extension-control	R
N/A	N/A			capabilities/auto-negotiation	R

Table 5–1—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-interface* YANG data nodes *(continued)*

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-interface* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
oMACEntity	aFramesReceivedOK	30.3.1.1.5	interfaces/interface/ethernet/statistics/frame	in-frames	R
	aMulticastFramesReceivedOK	30.3.1.1.21		in-multicast-frames	R
	aBroadcastFramesReceivedOK	30.3.1.1.22		in-broadcast-frames	R
	aFrameCheckSequenceErrors + aAlignmentErrors	30.4.3.1.6, 30.4.3.1.7		in-error-fcs-frames	R
oMACEntity	aFrameTooLongErrors	30.3.1.1.25		in-error-oversize-frames	R
	aFramesLostDueToIntMACRcvError	30.3.1.1.15		in-error-mac-internal-frames	R
	aFramesTransmittedOK	30.3.1.1.2		out-frames	R
	aMulticastFramesXmittedOK	30.3.1.1.18		out-multicast-frames	R
	aBroadcastFramesXmittedOK	30.3.1.1.19		out-broadcast-frames	R
	aFramesLostDueToIntMACXmitError	30.3.1.1.12		out-error-mac-internal-frames	R
oPHYEntity	aSymbolErrorDuringCarrier	30.3.2.1.5	interfaces/interface/ethernet/statistics/phy	in-error-symbol	R
	aReceiveLPITransitions	30.3.2.1.11	interfaces/interface/ethernet/statistics/phy/lpi	in-lpi-transitions	R
	aReceiveLPIMicroseconds	30.3.2.1.9		in-lpi-time	R
	aTransmitLPITransitions	30.3.2.1.10		out-lpi-transitions	R
	aTransmitLPIMicroseconds	30.3.2.1.8		out-lpi-time	R

Table 5–1—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-interface* YANG data nodes *(continued)*

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-interface* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
oMACControlEntity	aUnsupportedOpcodesReceived	30.3.3.5	interfaces/interface/ethernet/statistics/mac-control	in-frames-mac-control-unknown	R
oEXTENSION	aEXTENSIONMACCtrlFramesReceived	30.3.8.2		in-frames-mac-control-extension	R
	aEXTENSIONMACCtrlFramesTransmitted	30.3.8.1		out-frames-mac-control-extension	R

Table 5–2—Mapping between IETF RFC 2819 managed objects and *ieee802-ethernet-interface* YANG data nodes

IETF RFC 2819 Attribute(s)	Corresponding *ieee802-ethernet-interface* YANG data nodes		
	Container(s)	**Data node(s)**	**R/W**
no direct object[a]	interfaces/interface/ethernet/statistics/frame	in-total-frames	R
etherStatsOctets		in-total-octets	R
etherStatsUndersizePkts + etherStatsFragments		in-error-undersize-frames	R

[a] Can be calculated as: aFramesReceivedOK + aFrameCheckSequenceErrors + aAlignmentErrors + aFrameTooLongErrors + aFramesLostDueToIntMACRcvError.

Table 5–3—Mapping between IETF RFC 3635 managed objects and *ieee802-ethernet-interface* YANG data nodes

ETHERLIKE MIB Attribute(s)	Corresponding *ieee802-ethernet-interface* YANG data nodes		
	Container(s)	**Data node(s)**	**R/W**
dot3HCInPFCFrames	interfaces/interface/ethernet/	flow-control/pfc{ethernet-pfc} / statistics/in-frames-pfc	R
dot3HCOutPFCFrames		flow-control/pfc{ethernet-pfc} / statistics/out-frames-pfc	R

Table 5–4—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-interface-half-duplex* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-interface-half-duplex* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
oMACEntity	aRateControlAbility	30.3.1.1.33	interfaces/interface/ethernet	dynamic-rate-control	R/W
			interfaces/interface/ethernet/capability	dynamic-rate-control-supported	R
oPHYEntity	aSQETestErrors	30.3.2.1.4	interfaces/interface/ethernet/statistics/frame/ csmacd{csma-cd}	in-errors-sqe-test	R
oMACEntity	aSingleCollisionFrames	30.3.1.1.3		out-frames-collision-single	R
	aMultipleCollisionFrames	30.3.1.1.4		out-frames-collision-multiple	R
	aFramesWithDeferredXmissions	30.3.1.1.9		out-frames-deferred	R
	aFramesAbortedDueToXSColls	30.3.1.1.11		out-frames-collisions-excessive	R
	aLateCollisions	30.3.1.1.10		out-collisions-late	R
	aCarrierSenseErrors	30.3.1.1.13		out-errors-carrier-sense	R
	aCollisionFrames	30.3.1.1.30		collision-histogram/collision-count	R
				collision-histogram/collision-count-frames	R

Table 5–5—Mapping between IEEE Std 802.3, 30.14 managed objects and *ieee802-ethernet-mac-merge* YANG data nodes

IEEE Std 802.3, 30.14		Reference	Corresponding *ieee802-ethernet-mac-merge* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
oMacMer-geEntity	aMACMergeSupport	30.14.1.1	Interfaces/interface/ethernet/mac-merge/admin-status	merge-support	R
	aMACMergeStatusVerify	30.14.1.2		verify-status	R
	aMACMergeStatusTx	30.14.1.5		status-tx	R
	aMACMergeEnableTx	30.14.1.3	Interfaces/interface/ethernet/mac-merge/admin-control	merge-enable-tx	R/W
	aMACMergeVerifyDisableTx	30.14.1.4		verify-disable-tx	R/W
	aMACMergeVerifyTime	30.14.1.6		verify-time	R/W
	aMACMergeAddFragSize	30.14.1.7		frag-size	R/W
	aMACMergeFrameAssErrorCount	30.14.1.8	Interfaces/interface/ethernet/mac-merge/statistics	assembly-error-count	R
	aMACMergeFrameSmdErrorCount	30.14.1.9		smd-error-count	R
	aMACMergeFrameAssOkCount	30.14.1.10		assembly-ok-count	R
	aMACMergeFragCountRx	30.14.1.11		fragment-count-rx	R
	aMACMergeFragCountTx	30.14.1.12		fragment-count-tx	R
	aMACMergeHoldCount	30.14.1.13		hold-count	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-lldp* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
oLldpXdot3Config	aLldpXdot3PortConfigTLVsTxEnable	30.12.1.1.1	lldp/port	tlvs-port-config-enable	R/W
oLldpXdot3LocSystemsGroup	aLldpXdot3LocPortAutoNegSupported	30.12.2.1.1		auto-negotiation-supported	R
	aLldpXdot3LocPortAutoNegEnabled	30.12.2.1.2		auto-negotiation-enabled	R
	aLldpXdot3LocPortAutoNegAdvertisedCap	30.12.2.1.3		auto-negotiation-cap	R
	aLldpXdot3LocPortOperMauType	30.12.2.1.4		operational-mau-type	R
	aLldpXdot3LocPowerPortClass	30.12.2.1.5		power-port-class	R
	aLldpXdot3LocPowerMDISupported	30.12.2.1.6		mdi-power-supported	R
	aLldpXdot3LocPowerMDIEnabled	30.12.2.1.7		mdi-power-enabled	R
	aLldpXdot3LocPowerPairControllable	30.12.2.1.8		power-pair-controlable	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-lldp* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
	aLldpXdot3LocPowerPairs	30.12.2.1.9		power-pairs	R
	aLldpXdot3LocPowerClass	30.12.2.1.10		local-power-class	R
	aLldpXdot3LocLinkAggStatus	30.12.2.1.11		link-aggregation-status	R
	aLldpXdot3LocLinkAggPortId	30.12.2.1.12		aggregation-port-id	R
	aLldpXdot3LocMaxFrameSize	30.12.2.1.13		local-max-frame-size	R
	aLldpXdot3LocPowerType	30.12.2.1.14		power-type	R
	aLldpXdot3LocPowerSource	30.12.2.1.15		power-source	R
	aLldpXdot3LocPowerPriority	30.12.2.1.16		local-power-priority	R/W
	aLldpXdot3LocPDRequestedPowerValue	30.12.2.1.17		pd-requested-power-value	R
	aLldpXdot3LocPDRequestedPowerValueA	30.12.2.1.18		pd-requested-power-value-a	R
	aLldpXdot3LocPDRequestedPowerValueB	30.12.2.1.19		pd-requested-power-value-b	R
	aLldpXdot3LocPSEAllocatedPowerValue	30.12.2.1.20		pse-allocated-power-value	R
	aLldpXdot3LocPSEAllocatedPowerValueA	30.12.2.1.21		pse-allocated-power-value-a	R
	aLldpXdot3LocPSEAllocatedPowerValueB	30.12.2.1.22		pse-allocated-power-value-b	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-lldp* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
	aLldpXdot3LocPSEPoweringStatus	30.12.2.1.23		pse-powering-status	R
	aLldpXdot3LocPDPoweredStatus	30.12.2.1.24		pd-powered-status	R
	aLldpXdot3LocPowerPairsExt	30.12.2.1.25		power-pairs-ext	R
	aLldpXdot3LocPowerClassExtA	30.12.2.1.26		power-class-ext-A	R
	aLldpXdot3LocPowerClassExtB	30.12.2.1.27		power-class-ext-B	R
	aLldpXdot3LocPowerClassExt	30.12.2.1.28		power-class-ext	R
	aLldpXdot3LocPowerTypeExt	30.12.2.1.29		power-type-ext	R
	aLldpXdot3LocPDLoad	30.12.2.1.30		pd-load	R
	aLldpXdot3LocPD4PID	30.12.2.1.31		pd-4pid	R
	aLldpXdot3LocPSEMaxAvailPower	30.12.2.1.32		pse-max-avail-power	R
	aLldpXdot3LocPSEAutoclassSupport	30.12.2.1.33		pse-autoclass-support	R
	aLldpXdot3LocAutoclassCompleted	30.12.2.1.34		autoclass-completed	R
	aLldpXdot3LocAutoclassRequest	30.12.2.1.35		autoclass-request	R
	aLldpXdot3LocPowerDownRequest	30.12.2.1.36		power-down-request	R
	aLldpXdot3LocPowerDownTime	30.12.2.1.37		power-down-time	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-lldp* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
	aLldpXdot3LocMeasVoltageSupport	30.12.2.1.38		meas-voltage-support	R
	aLldpXdot3LocMeasCurrentSupport	30.12.2.1.39		meas-current-support	R
	aLldpXdot3LocMeasPowerSupport	30.12.2.1.40		meas-power-support	R
	aLldpXdot3LocMeasEnergySupport	30.12.2.1.41		meas-energy-support	R
	aLldpXdot3LocMeasurementSource	30.12.2.1.42		measurement-source	R
	aLldpXdot3LocMeasVoltageRequest	30.12.2.1.43		meas-voltage-request	R
	aLldpXdot3LocMeasCurrentRequest	30.12.2.1.44		meas-current-request	R
	aLldpXdot3LocMeasCurrentRequest	30.12.2.1.45		meas-power-request	R
	aLldpXdot3LocMeasEnergyRequest	30.12.2.1.46		meas-energy-request	R
	aLldpXdot3LocMeasVoltageValid	30.12.2.1.47		meas-voltage-valid	R
	aLldpXdot3LocMeasCurrentValid	30.12.2.1.48		meas-current-valid	R
	aLldpXdot3LocMeasPowerValid	30.12.2.1.49		meas-power-valid	R
	aLldpXdot3LocMeasEnergyValid	30.12.2.1.50		meas-energy-valid	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-lldp* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
	aLldpXdot3LocMeasVoltageUncertainty	30.12.2.1.51		meas-voltage-uncertainty	R
	aLldpXdot3LocMeasCurrentUncertainty	30.12.2.1.52		meas-current-uncertainty	R
	aLldpXdot3LocMeasPowerUncertainty	30.12.2.1.53		meas-power-uncertainty	R
	aLldpXdot3LocMeasEnergyUncertainty	30.12.2.1.54		meas-energy-uncertainty	R
	aLldpXdot3LocVoltageMeasurement	30.12.2.1.55		voltage-measurement	R
	aLldpXdot3LocCurrentMeasurement	30.12.2.1.56		current-measurement	R
	aLldpXdot3LocPowerMeasurement	30.12.2.1.57		power-measurement	R
	aLldpXdot3LocEnergyMeasurement	30.12.2.1.58		energy-measurement	R
	aLldpXdot3LocPSEPowerPriceIndex	30.12.2.1.59		pse-power-price-index	R
	aLldpXdot3LocResponseTime	30.12.2.1.60		local-response	R
	aLldpXdot3LocReady	30.12.2.1.61		local-system-ready	R
	aLldpXdot3LocTxTwSys	30.12.2.1.62		tx-system-value	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-lldp* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
	aLldpXdot3LocTxTwSysEcho	30.12.2.1.63		tx-system-value-echo	R
	aLldpXdot3LocRxTwSys	30.12.2.1.64		rx-system-value	R
	aLldpXdot3LocRxTwSysEcho	30.12.2.1.65		rx-system-value-echo	R
	aLldpXdot3LocFbTwSys	30.12.2.1.66		fallback-system-value	R
	aLldpXdot3TxDllReady	30.12.2.1.67		tx-dll-ready	R
	aLldpXdot3RxDllReady	30.12.2.1.68		rx-dll-ready	R
	aLldpXdot3LocDllEnabled	30.12.2.1.69		dll-ready	R
	aLldpXdot3LocTxFw	30.12.2.1.70		tx-system-fw	R
	aLldpXdot3LocTxFwEcho	30.12.2.1.71		tx-system-fw-echo	R
	aLldpXdot3LocRxFw	30.12.2.1.72		rx-system-fw	R
	aLldpXdot3LocRxFwEcho	30.12.2.1.73		rx-system-fw-echo	R
	aLldpXdot3LocPreemptSupported	30.12.2.1.74		preemption-supported	R
	aLldpXdot3LocPreemptEnabled	30.12.2.1.75		preemption-enabled	R
	aLldpXdot3LocPreemptActive	30.12.2.1.76		preemption-active	R
	aLldpXdot3LocAddFragSize	30.12.2.1.77		additional-fragment-size	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-lldp* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
oLldpXdot3RemSystemsGroup	aLldpXdot3RemPortAutoNegSupported	30.12.3.1.1	lldp/port/remote-systems-data	auto-negotiation-supported	R
	aLldpXdot3RemPortAutoNegEnabled	30.12.3.1.2		auto-negotiation-enabled	R
	aLldpXdot3RemPortAutoNegAdvertisedCap	30.12.3.1.3		auto-negotiation-cap	R
	aLldpXdot3RemPortOperMauType	30.12.3.1.4		operational-mau-type	R
	aLldpXdot3RemPowerPortClass	30.12.3.1.5		power-port-class	R
	aLldpXdot3RemPowerMDISupported	30.12.3.1.6		mdi-power-supported	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-lldp* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
	aLldpXdot3RemPowerMDIEnabled	30.12.3.1.7		mdi-power-enabled	R
	aLldpXdot3RemPowerPairControllable	30.12.3.1.8		power-pair-controlable	R
	aLldpXdot3RemPowerPairs	30.12.3.1.9		power-pairs	R
	aLldpXdot3RemPowerClass	30.12.3.1.10		power-class	R
	aLldpXdot3RemLinkAggStatus	30.12.3.1.11		link-aggregation-status	R
	aLldpXdot3RemLinkAggPortId	30.12.3.1.12		aggregation-port-id	R
	aLldpXdot3RemMaxFrameSize	30.12.3.1.13		local-max-frame-size	R
	aLldpXdot3RemPowerType	30.12.3.1.14		power-type	R
	aLldpXdot3RemPowerSource	30.12.3.1.15		power-source	R
	aLldpXdot3RemPowerPriority	30.12.3.1.16		power-priority	RW
	aLldpXdot3RemPDRequestedPowerValue	30.12.3.1.17		pd-requested-power-value	R
	aLldpXdot3RemPDRequestedPowerValueA	30.12.3.1.18		pd-requested-power-value-a	R
	aLldpXdot3RemPDRequestedPowerValueB	30.12.3.1.19		pd-requested-power-value-b	R
	aLldpXdot3RemPSEAllocatedPowerValue	30.12.3.1.20		pse-allocated-power-value	R
	aLldpXdot3RemPSEAllocatedPowerValueA	30.12.3.1.21		pse-allocated-power-value-a	R
	aLldpXdot3RemPSEAllocatedPowerValueB	30.12.3.1.22		pse-allocated-power-value-b	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-lldp* YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aLldpXdot3RemPSEPoweringStatus	30.12.3.1.23		pse-powering-status	R
	aLldpXdot3RemPDPoweredStatus	30.12.3.1.24		pd-powered-status	R
	aLldpXdot3RemPowerPairsExt	30.12.3.1.25		power-pairs-ext	R
	aLldpXdot3RemPowerClassExtA	30.12.3.1.26		power-class-ext-A	R
	aLldpXdot3RemPowerClassExtB	30.12.3.1.27		power-class-ext-B	R
	aLldpXdot3RemPowerClassExt	30.12.3.1.28		power-class-ext	R
	aLldpXdot3RemPowerTypeExt	30.12.3.1.29		power-type-ext	R
	aLldpXdot3RemPDLoad	30.12.3.1.30		pd-load	R
	aLldpXdot3RemPD4PID	30.12.3.1.31		pd-4pid	R
	aLldpXdot3RemPSEMaxAvailPower	30.12.3.1.32		pse-max-avail-power	R
	aLldpXdot3RemPSEAutoclassSupport	30.12.3.1.33		pse-autoclass-support	R
	aLldpXdot3RemAutoclassCompleted	30.12.3.1.34		autoclass-completed	R
	aLldpXdot3RemAutoclassRequest	30.12.3.1.35		autoclass-request	R
	aLldpXdot3RemPowerDownRequest	30.12.3.1.36		power-down-request	R
	aLldpXdot3RemPowerDownTime	30.12.3.1.37		power-down-time	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-lldp* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
	aLldpXdot3RemMeasVoltageSupport	30.12.3.1.38		meas-voltage-support	R
	aLldpXdot3RemMeasCurrentSupport	30.12.3.1.39		meas-current-support	R
	aLldpXdot3RemMeasPowerSupport	30.12.3.1.40		meas-power-support	R
	aLldpXdot3RemMeasEnergySupport	30.12.3.1.41		meas-energy-support	R
	aLldpXdot3RemMeasurementSource	30.12.3.1.42		measurement-source	R
	aLldpXdot3RemMeasVoltageRequest	30.12.3.1.43		meas-voltage-request	R
	aLldpXdot3RemMeasCurrentRequest	30.12.3.1.44		meas-current-request	R
	aLldpXdot3RemMeasCurrentRequest	30.12.3.1.45		meas-power-request	R
	aLldpXdot3RemMeasEnergyRequest	30.12.3.1.46		meas-energy-request	R
	aLldpXdot3RemMeasVoltageValid	30.12.3.1.47		meas-voltage-valid	R
	aLldpXdot3RemMeasCurrentValid	30.12.3.1.48		meas-current-valid	R
	aLldpXdot3RemMeasPowerValid	30.12.3.1.49		meas-power-valid	R
	aLldpXdot3RemMeasEnergyValid	30.12.3.1.50		meas-energy-valid	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-lldp* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
	aLldpXdot3RemMeasVoltageUncertainty	30.12.3.1.51		meas-voltage-uncertainty	R
	aLldpXdot3RemMeasCurrentUncertainty	30.12.3.1.52		meas-current-uncertainty	R
	aLldpXdot3RemMeasPowerUncertainty	30.12.3.1.53		meas-power-uncertainty	R
	aLldpXdot3RemMeasEnergyUncertainty	30.12.3.1.54		meas-energy-uncertainty	R
	aLldpXdot3RemVoltageMeasurement	30.12.3.1.55		voltage-measurement	R
	aLldpXdot3RemCurrentMeasurement	30.12.3.1.56		current-measurement	R
	aLldpXdot3RemPowerMeasurement	30.12.3.1.57		power-measurement	R
	aLldpXdot3RemEnergyMeasurement	30.12.3.1.58		energy-measurement	R
	aLldpXdot3RemPSEPowerPriceIndex	30.12.3.1.59		pse-power-price-index	R
	aLldpXdot3RemTxTwSys	30.12.3.1.60		tx-system-value	R
	aLldpXdot3RemTxTwSysEcho	30.12.3.1.61		tx-system-value-echo	R
	aLldpXdot3RemRxTwSys	30.12.3.1.62		rx-system-value	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-lldp* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
	aLldpXdot3RemRxTwSysEcho	30.12.3.1.63		rx-system-value-echo	R
	aLldpXdot3RemFbTwSys	30.12.3.1.64		fallback-system-value	R
	aLldpXdot3RemTxFw	30.12.3.1.65		tx-system-fw	R
	aLldpXdot3RemTxFwEcho	30.12.3.1.66		tx-system-fw-echo	R
	aLldpXdot3RemRxFw	30.12.3.1.67		rx-system-fw	R
	aLldpXdot3RemRxFwEcho	30.12.3.1.68		rx-system-fw-echo	R
	aLldpXdot3RemPreemptSupported	30.12.3.1.69		preemption-supported	R
	aLldpXdot3RemPreemptEnabled	30.12.3.1.70		preemption-enabled	R
	aLldpXdot3RemPreemptActive	30.12.3.1.71		preemption-active	R
	aLldpXdot3RemAddFragSize	30.12.3.1.72		additonal-fragment-size	R

5.3 YANG module definition[g]

The YANG module tree hierarchy uses terms defined in IETF RFC 8407.

5.3.1 Tree hierarchy

5.3.1.1 ieee802-ethernet-interface

```
module: ieee802-ethernet-interface

  augment /if:interfaces/if:interface:
    +--rw ethernet
       +--rw auto-negotiation!
       |  +--rw enable?               boolean
       |  +--ro negotiation-status?   enumeration
       +--ro phy-type?                        identityref
       +--ro pmd-type?                        identityref
       +--rw duplex?                          duplex-type
       o--rw speed?                           eth-if-speed-type
       x--rw flow-control
       |  +--rw pause {ethernet-pause}?
       |  |  +--rw direction?    pause-fc-direction-type
       |  |  +--ro statistics
       |  |     +--ro in-frames-pause?    yang:counter64
       |  |     +--ro out-frames-pause?   yang:counter64
       |  +--rw pfc {ethernet-pfc}?
       |  |  +--rw enable?       boolean
       |  |  x--ro statistics
       |  |     +--ro in-frames-pfc?    yang:counter64
       |  |     +--ro out-frames-pfc?   yang:counter64
       |  +--rw force-flow-control?   boolean
       +--ro max-frame-length?                uint16
       +--ro mac-control-extension-control?   boolean
       +--ro frame-limit-slow-protocol?       uint64
       +--ro capabilities
       |  +--ro auto-negotiation?   boolean
```

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```
+--rw ethernet-pause {ethernet-pause}?
|  +--rw control-and-status
|  |  +--rw pause-admin-control?    pause-fc-direction-type
|  |  +--ro pause-oper-status?      pause-fc-direction-type
|  |  +--rw link-delay-allowance?   uint32
|  |  +--ro pfc-enable-status?      boolean
|  +--ro statistics
|     +--ro in-frames-pause?    yang:counter64
|     +--ro out-frames-pause?   yang:counter64
+--ro statistics
   +--ro frame
   |  +--ro in-total-frames?                 yang:counter64
   |  +--ro in-total-octets?                 yang:counter64
   |  +--ro in-frames?                       yang:counter64
   |  +--ro in-multicast-frames?             yang:counter64
   |  +--ro in-broadcast-frames?             yang:counter64
   |  +--ro in-error-fcs-frames?             yang:counter64
   |  x--ro in-error-undersize-frames?       yang:counter64
   |  +--ro in-error-oversize-frames?        yang:counter64
   |  +--ro in-error-mac-internal-frames?    yang:counter64
   |  +--ro out-frames?                      yang:counter64
   |  +--ro out-multicast-frames?            yang:counter64
   |  +--ro out-broadcast-frames?            yang:counter64
   |  +--ro out-error-mac-internal-frames?   yang:counter64
   +--ro phy
   |  +--ro in-error-symbol?   yang:counter64
   |  +--ro lpi
   |     +--ro in-lpi-transitions?    yang:counter64
   |     +--ro in-lpi-time?           decimal64
   |     +--ro out-lpi-transitions?   yang:counter64
   |     +--ro out-lpi-time?          decimal64
   +--ro mac-control
      +--ro in-frames-mac-control-unknown?      yang:counter64
      +--ro in-frames-mac-control-extension?    yang:counter64
      +--ro out-frames-mac-control-extension?   yang:counter64
```

5.3.1.2 ieee802-ethernet-interface-half-duplex

```
module: ieee802-ethernet-interface-half-duplex

  augment /if:interfaces/if:interface/ieee802-eth-if:ethernet:
    o--rw dynamic-rate-control?   dynamic-rate-control-type {dynamic-rate-control}?
  augment /if:interfaces/if:interface/ieee802-eth-if:ethernet/ieee802-eth-if:capabilities:
    o--ro dynamic-rate-control-supported?   boolean {dynamic-rate-control}?
  augment /if:interfaces/if:interface/ieee802-eth-if:ethernet/ieee802-eth-if:statistics/ieee802-eth-if:frame:
    +--ro csma-cd {csma-cd}?
       +--ro in-errors-sqe-test?               yang:counter64
       +--ro out-frames-collision-single?      yang:counter64
       +--ro out-frames-collision-multiple?    yang:counter64
       +--ro out-frames-deferred?              yang:counter64
       +--ro out-frames-collisions-excessive?  yang:counter64
       +--ro out-collisions-late?              yang:counter64
       +--ro out-errors-carrier-sense?         yang:counter64
       +--ro collision-histogram* [collision-count]
          +--ro collision-count           yang:counter64
          +--ro collision-count-frames?   yang:counter64
```

5.3.1.3 ieee802-ethernet-mac-merge

```
module: ieee802-ethernet-mac-merge

  augment /if:interfaces/if:interface/ieee802-eth-if:ethernet:
    +--rw mac-merge {mac-merge}?
       +--rw admin-control
       |  +--rw merge-enable-tx?     enumeration
       |  +--rw verify-disable-tx?   enumeration
       |  +--rw verify-time?         uint16
       |  +--rw frag-size?           uint8
       +--ro admin-status
       |  +--ro merge-support?   enumeration
       |  +--ro verify-status?   enumeration
       |  +--ro status-tx?       enumeration
       +--ro statistics
          +--ro assembly-error-count?   yang:counter64
```

```
        +--ro smd-error-count?        yang:counter64
        +--ro assembly-ok-count?      yang:counter64
        +--ro fragment-count-rx?      yang:counter64
        +--ro fragment-count-tx?      yang:counter64
        +--ro hold-count?             yang:counter64
```

5.3.1.4 ieee802-ethernet-lldp

```
module: ieee802-ethernet-lldp

  augment /lldp:lldp/lldp:port:
    +--ro auto-negotiation-supported?    boolean
    +--ro auto-negotiation-enabled?      boolean
    x--ro auto-negotiation-cap?          binary
    x--ro operational-mau-type?          int32
    +--rw auto-negotiation-cap-bits?     ieee802-phy:auto-neg-ability-bits
    +--rw operational-pmd-type?          identityref
    +--rw power-port-class?              port-class-type
    +--rw mdi-power-supported?           boolean
    +--rw mdi-power-enabled?             boolean
    +--rw power-pair-controlable?        boolean
    +--rw power-pairs?                   pse-pinout-type
    +--rw local-power-class?             pse-power-class-type
    o--ro link-aggregation-status?       bits
    o--ro aggregation-port-id?           int32
    o--ro local-max-frame-size?          int32
    +--ro power-type?                    bits
    +--rw power-support?                 bits
    +--ro power-source?                  power-source-type
    +--ro pd-requested-power-value?      int32
    +--ro pd-requested-power-value-a?    int32
    +--ro pd-requested-power-value-b?    int32
    +--ro pse-allocated-power-value?     int32
    +--ro pse-allocated-power-value-a?   int32
    +--ro pse-allocated-power-value-b?   int32
    +--ro pse-powering-status?           powering-status-type
    +--ro pd-powered-status?             powered-status-type
    +--ro power-pairs-ext?               power-pairs-type
```

```
+--ro power-class-ext-A?            power-class-ext-AB-type
+--ro power-class-ext-B?            power-class-ext-AB-type
+--ro power-class-ext?              power-class-ext-type
+--ro power-type-ext?               power-type
+--ro pd-load?                      boolean
+--ro pd-4pid?                      boolean
+--ro pse-max-avail-power?          int32
+--ro pse-autoclass-support?        boolean
+--ro autoclass-completed?          boolean
+--ro autoclass-request?            boolean
+--rw power-down-request?           int32
+--rw power-down-time?              int32
+--rw meas-voltage-support?         boolean
+--rw meas-current-support?         boolean
+--rw meas-power-support?           boolean
+--rw meas-energy-support?          boolean
o--rw measurement-source?           bits
+--ro meas-voltage-request?         boolean
+--ro meas-current-request?         boolean
+--ro meas-power-request?           boolean
+--ro meas-energy-request?          boolean
+--ro meas-voltage-valid?           boolean
+--ro meas-current-valid?           boolean
+--ro meas-power-valid?             boolean
+--ro meas-energy-valid?            boolean
+--ro meas-voltage-uncertainty?     int32
+--ro meas-current-uncertainty?     int32
+--ro meas-power-uncertainty?       int32
+--ro meas-energy-uncertainty?      int32
+--ro voltage-measurement?          int32
+--ro current-measurement?          int32
+--ro power-measurement?            int32
+--ro energy-measurement?           int32
+--ro pse-power-price-index?        int32
+--ro local-response?               int32
+--ro local-system-ready?           boolean
+--ro tx-system-value?              int32
+--ro tx-system-value-echo?         int32
+--ro rx-system-value?              int32
```

```
  +--ro rx-system-value-echo?          int32
  +--ro fallback-system-value?         int32
  +--ro tx-dll-ready?                  boolean
  +--ro rx-dll-ready?                  boolean
  +--ro dll-enabled?                   boolean
  +--ro tx-system-fw?                  boolean
  +--ro tx-system-fw-echo?             boolean
  +--ro rx-system-fw?                  boolean
  +--ro rx-system-fw-echo?             boolean
  +--ro preemption-supported?          boolean
  +--ro preemption-enabled?            boolean
  +--ro preemption-active?             boolean
  +--ro additional-fragment-size?      int32
augment /lldp:lldp/lldp:port:
  +--rw tlvs-port-config-enable?    bits
  +--rw local-power-priority?       power-priority-type
  +--rw local-measurement-source?   measurement-source-type
augment /lldp:lldp/lldp:port/lldp:remote-systems-data:
  +--ro auto-negotiation-supported?    boolean
  +--ro auto-negotiation-enabled?      boolean
  o--ro auto-negotiation-cap?          binary
  o--ro operational-mau-type?          int32
  +--ro auto-negotiation-cap-bits?     ieee802-phy:auto-neg-ability-bits
  +--ro operational-pmd-type?          identityref
  +--ro power-port-class?              port-class-type
  +--ro mdi-power-supported?           boolean
  +--ro mdi-power-enabled?             boolean
  +--ro power-pair-controlable?        boolean
  +--ro power-pairs?                   pse-pinout-type
  +--ro power-class?                   pse-power-class-type
  o--ro link-aggregation-status?       bits
  o--ro aggregation-port-id?           int32
  o--ro local-max-frame-size?          int32
  o--ro power-type?                    bits
  +--ro power-source?                  power-source-type
  +--ro power-priority?                power-priority-type
  +--ro pd-requested-power-value?      int32
  +--ro pd-requested-power-value-a?    int32
  +--ro pd-requested-power-value-b?    int32
```

```
+--ro pse-allocated-power-value?     int32
+--ro pse-allocated-power-value-a?   int32
+--ro pse-allocated-power-value-b?   int32
+--ro pse-powering-status?           powering-status-type
+--ro pd-powered-status?             powered-status-type
+--ro power-pairs-ext?               power-pairs-type
+--ro power-class-ext-A?             power-class-ext-AB-type
+--ro power-class-ext-B?             power-class-ext-AB-type
+--ro power-class-ext?               power-class-ext-type
+--ro power-type-ext?                power-type
+--ro pd-load?                       boolean
+--ro pd-4pid?                       boolean
+--ro pse-max-avail-power?           int32
+--ro pse-autoclass-support?         boolean
+--ro autoclass-completed?           boolean
+--ro autoclass-request?             boolean
+--ro power-down-request?            int32
+--ro power-down-time?               int32
+--ro meas-voltage-support?          boolean
+--ro meas-current-support?          boolean
+--ro meas-power-support?            boolean
+--ro meas-energy-support?           boolean
o--ro measurement-source?            bits
+--ro rem-measurement-source?        measurement-source-type
+--ro meas-voltage-request?          boolean
+--ro meas-current-request?          boolean
+--ro meas-power-request?            boolean
+--ro meas-energy-request?           boolean
+--ro meas-voltage-valid?            boolean
+--ro meas-current-valid?            boolean
+--ro meas-power-valid?              boolean
+--ro meas-energy-valid?             boolean
+--ro meas-voltage-uncertainty?      int32
+--ro meas-current-uncertainty?      int32
+--ro meas-power-uncertainty?        int32
+--ro meas-energy-uncertainty?       int32
+--ro voltage-measurement?           int32
+--ro current-measurement?           int32
+--ro power-measurement?             int32
```

```
    +--ro energy-measurement?            int32
    +--ro pse-power-price-index?         int32
    +--ro tx-system-value?               int32
    +--ro tx-system-value-echo?          int32
    +--ro rx-system-value?               int32
    +--ro rx-system-value-echo?          int32
    +--ro fallback-system-value?         int32
    +--ro tx-system-fw?                  boolean
    +--ro tx-system-fw-echo?             boolean
    +--ro rx-system-fw?                  boolean
    +--ro rx-system-fw-echo?             boolean
    +--ro preemption-supported?          boolean
    +--ro preemption-enabled?            boolean
    +--ro preemption-active?             boolean
    +--ro additional-fragment-size?      int32
```

59.3.1.5 ieee802-ethernet-phy

```
module: ieee802-ethernet-phy

  augment /if:interfaces/if:interface:
    +--rw phy-table
    |  +--rw control
    |  |  +--rw phy-reset?           boolean
    |  |  +--rw phy-admin-control?   enumeration
    |  |  +--rw fec-control! {fec-support}?
    |  |     +--rw fec-mode-control?        enumeration
    |  |     +--rw rs-fec-bypass-control?   enabled-state
    |  +--ro status
    |  |  +--ro phy-type?                identityref
    |  |  +--ro pmd-type?                identityref
    |  |  +--ro pmd-type-list?           ieee802-phy:pmd-type-list
    |  |  +--ro phy-status?              enumeration
    |  |  +--ro phy-media-available?     ieee802-phy:media-available-type
    |  |  +--ro jabber-state?            enumeration
    |  |  +--ro jabbering-state-enters?  yang:counter64
    |  |  +--ro fec-status! {fec-support}?
```

```
|  |  |  +--ro fec-ability?                    bits
|  |  |  +--ro fec-mode-status?                enumeration
|  |  |  +--ro rs-fec-bypass-enabled-status?   enabled-state
|  |  +--ro baset-snr! {baset-snr-support}?
|  |  |  +--ro snr-op-margin-chnl-a?   snr-value
|  |  |  +--ro snr-op-margin-chnl-b?   snr-value
|  |  |  +--ro snr-op-margin-chnl-c?   snr-value
|  |  |  +--ro snr-op-margin-chnl-d?   snr-value
|  |  +--ro time-sync! {time-sync-support}?
|  |     +--ro time-sync-ability?        bits
|  |     +--ro time-sync-delay-tx-max?   uint32
|  |     +--ro time-sync-delay-tx-min?   uint32
|  |     +--ro time-sync-delay-rx-max?   int32
|  |     +--ro time-sync-delay-rx-min?   int32
|  +--ro statistics
|  |  +--ro phy-media-losts?            yang:counter64
|  |  +--ro jabbering-state-enters?     yang:counter64
|  |  +--ro false-carriers?             yang:counter64
|  |  +--ro pcs-coding-violations?      yang:counter64
|  |  +--ro fast-retrain-count-local?   counter-1000-second
|  |  +--ro fast-retrain-count-peer?    counter-1000-second
|  +--ro pcs-lane-statistics {pcs-lane-statistics-support}?
|     +--ro pcs-lane* [pcs-lane-index]
|        +--ro pcs-lane-index            uint32
|        +--ro fec-corrected-blocks?     counter-pcs-lane
|        +--ro fec-uncorrected-blocks?   counter-pcs-lane
+--rw auto-neg-table {auto-negotiation-support}?
   +--rw control
   |  +--rw auto-neg-control?   boolean
   |  +--rw auto-neg-restart?   boolean
   +--ro status
      +--ro auto-neg-enabled?               boolean
      +--ro auto-neg-remote-signaling?      boolean
      +--ro auto-neg-status?                enumeration
      +--ro auto-neg-capability-bits?       ieee802-phy:auto-neg-ability-bits
      +--ro auto-neg-cap-advertised-bits?   ieee802-phy:auto-neg-ability-bits
      +--ro auto-neg-cap-received-bits?     ieee802-phy:auto-neg-ability-bits
```

5.3.2 YANG modules

In the YANG module definition, should any discrepancy between the text of the description for individual YANG nodes in the YANG modules of this clause and the corresponding definition in 5.2 through 5.3 of this clause occur, the definitions in the YANG modules attached to this standard shall take precedence.

An ASCII text version of the Ethernet YANG module can be found at the following URL:[h]

> ***Editor's Note (to be removed prior to publication):***
>
> *Yang files contained in https://github.com/YangModels/yang/tree/main/standard/ieee/published/802.3 are IEEE 802.3.2-2019 version and will be updated at the publication time.*

https://github.com/YangModels/yang/tree/master/standard/ieee/published/802.3

5.3.2.1 Ethernet interface module

> ***Editor's Note (to be removed prior to publication):***
>
> *Pretty printing of ieee802-ethernet-interface.yang file may change the appearance by adding whitespace and reformatting lines*

> ***Editor's Note (to be removed prior to publication):***
>
> IEEE Std 802.3.1 and IEEE Std 802.3.2 to be updated at the publication time

```
module ieee802-ethernet-interface {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-interface";
  prefix ieee802-eth-if;

  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
  }
  import iana-if-type {
    prefix ianaift;
    reference
      "http://www.iana.org/assignments/yang-parameters/
       iana-if-type@2023-01-26.yang";
  }
  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import ieee802-ethernet-phy-type {
    prefix ieee802-phy;
    reference
      "IEEE Std 802.3-2022";
  }
```

[h]Copyright release for YANG modules: Users of this standard may freely reproduce the YANG module contained in this subclause so that it can be used for its intended purpose.

```
organization
  "IEEE Std 802.3 Ethernet Working Group
   Web URL: http://www.ieee802.org/3/";
contact
  "Web URL: http://www.ieee802.org/3/";
description
  "This module contains YANG definitions for configuring IEEE Std
   802.3 Ethernet Interfaces.
   In this YANG module, 'Ethernet interface' can be interpreted
   as referring to 'IEEE Std 802.3 compliant Ethernet
   interfaces'.";

revision 2025-04-17 {
  description
    "Updates under IEEE Std 802.3.2-202x, Draft D3.3";
  reference
    "IEEE Std 802.3-2022 and IEEE Std 802.3.1-202X, unless dated
     explicitly";
}

feature ethernet-pfc {
  description
    "This device supports Ethernet priority flow-control.";
}

feature ethernet-pause {
  description
    "This device supports Ethernet PAUSE.";
}

typedef eth-if-speed-type {
  type decimal64 {
    fraction-digits 3;
  }
  units "Gb/s";
  status obsolete;
  description
    "Used to represent the configured, negotiated, or actual
     speed of an Ethernet interface in Gigabits per second
     (Gb/s), accurate to 3 decimal places (i.e., accurate to 1
     Mb/s).";
}

typedef duplex-type {
  type enumeration {
    enum full {
      description
        "Full duplex.";
    }
    enum half {
      description
        "Half duplex.";
    }
    enum unknown {
      description
        "Link is currently disconnected or initializing.";
    }
  }
  default "full";
```

```
    description
      "Used to represent the configured, negotiated, or actual
       duplex mode of an Ethernet interface.";
    reference
      "IEEE Std 802.3, 30.3.1.1.32, aDuplexStatus";
  }

  typedef pause-fc-direction-type {
    type enumeration {
      enum disabled {
        description
          "Flow-control disabled in both ingress and egress
           directions.";
      }
      enum ingress-only {
        description
          "PAUSE frame based flow control is enabled in the ingress
           direction only.";
      }
      enum egress-only {
        description
          "PAUSE frame based flow control is enabled in the egress
           direction only.";
      }
      enum bi-directional {
        description
          "PAUSE frame based flow control is enabled in both
           ingress and egress directions.";
      }
      enum undefined {
        description
          "Link is currently disconnected or initializing.";
      }
    }
    description
      "Used to represent the configured, negotiated, or actual
       PAUSE frame-based flow control setting.";
    reference
      "IEEE Std 802.3.1, dot3PauseAdminMode and dot3PauseOperMode";
  }

  augment "/if:interfaces/if:interface" {
    when "derived-from-or-self(if:type, 'ianaift:ethernetCsmacd')" {
      description
        "Applies to all Ethernet interfaces.";
    }
    description
      "Augment interface model with Ethernet interface
       specific configuration nodes.";
    container ethernet {
      description
        "Contains all Ethernet interface related configuration.";
      container auto-negotiation {
        presence "The presence of this container indicates that
                  auto-negotiation is supported on this Ethernet
                  interface.";
        description
          "Contains auto-negotiation transmission parameters
```

```
            This container contains a data node that allows the
            advertised duplex value in the negotiation to be
            restricted.

            If not specified then the default behavior for the
            duplex data node is to negotiate all available values for
            the particular type of Ethernet PHY associated with the
            interface.";
        reference
          "IEEE Std 802.3, Clause 28 and Annexes 28A-D";
        leaf enable {
          type boolean;
          default "true";
          description
            "Controls whether auto-negotiation is enabled or
             disabled.
             For interface types that support auto-negotiation then
             it defaults to being enabled.

             For interface types that do not support
             auto-negotiation, the related configuration data is
             ignored.";
        }
        leaf negotiation-status {
          when "../enable = 'true'";
          type enumeration {
            enum in-progress {
              description
                "The auto-negotiation protocol is running and
                 negotiation is currently in-progress.";
            }
            enum complete {
              description
                "The auto-negotiation protocol has completed
                 successfully.";
            }
            enum failed {
              description
                "The auto-negotiation protocol has failed.";
            }
            enum unknown {
              description
                "The auto-negotiation status is not currently known,
                 this could be because it is still negotiating or the
                 protocol cannot run
                 (e.g., if no medium is present).";
            }
            enum no-negotiation {
              description
                "No auto-negotiation is executed.
                 The auto-negotation function is either not
                 supported on this interface or has not been
                 enabled.";
            }
          }
          config false;
          description
            "The status of the auto-negotiation protocol.";
          reference
```

```
            "IEEE 802.3, 30.6.1.1.4, aAutoNegAutoConfig";
        }
      }
      leaf phy-type {
        type identityref {
          base ieee802-phy:phy-type;
        }
        config false;
        description
          "A value that uniquely identifies the IEEE 802.3 PHY type
           of the interface.";
        reference
          "IEEE Std 802.3, 30.3.2.1.2 aPhyType";
      }
      leaf pmd-type {
        type identityref {
          base ieee802-phy:pmd-type;
        }
        config false;
        description
          "A value that uniquely identifies the IEEE 802.3 PMD type
           of the interface.";
        reference
          "IEEE Std 802.3, 30.5.1.1.2 aMAUType";
      }
      leaf duplex {
        type duplex-type;
        description
          "Operational duplex mode of the Ethernet interface.";
        reference
          "IEEE Std 802.3, 30.3.1.1.32 aDuplexStatus";
      }
      leaf speed {
        type eth-if-speed-type;
        units "Gb/s";
        status obsolete;
        description
          "Operational speed (data rate) of the Ethernet interface.
           The default value is implementation-dependent.
           Supersceeded by speed in /if:interfaces/if:interface";
      }
      /* deprecated flow-control container */
      container flow-control {
        status deprecated;
        description
          "Holds the different types of Ethernet PAUSE frame based
           flow control that can be enabled.
           Supersceeded by new container - pause.";
        container pause {
          if-feature "ethernet-pause";
          description
            "IEEE Std 802.3 PAUSE frame based PAUSE frame based
             flow control.";
          reference
            "IEEE Std 802.3, Annex 31B";
          leaf direction {
            type pause-fc-direction-type;
            description
              "Indicates which direction PAUSE frame based flow
```

```
              control is enabled in, or whether it is disabled.
              The default flow-control settings are vendor
              specific. If auto-negotiation is enabled, then PAUSE
              based flow-control is negotiated by default.
              The default value is implementation-dependent.";
        }
        container statistics {
          config false;
          description
            "Contains the number of PAUSE frames received or
             transmitted.

             Discontinuities in the values of counters in
             this container can occur at re-initialization of
             the management system, and at other times as
             indicated by the value of the 'discontinuity-time'
             leaf defined in the ietf-interfaces YANG module
             (IETF RFC 8343).";
          leaf in-frames-pause {
            type yang:counter64;
            units "frames";
            description
              "A count of PAUSE MAC Control frames transmitted on
               this Ethernet interface.";
            reference
              "IEEE Std 802.3, 30.3.4.3
               aPAUSEMACCtrlFramesReceived";
          }
          leaf out-frames-pause {
            type yang:counter64;
            units "frames";
            description
              "A count of PAUSE MAC Control frames transmitted on
               this Ethernet interface.";
            reference
              "IEEE Std 802.3, 30.3.4.2
               aPAUSEMACCtrlFramesTransmitted";
          }
        }
      }
      container pfc {
        if-feature "ethernet-pfc";
        description
          "IEEE Std 802.3 Priority-based flow control.";
        reference
          "IEEE Std 802.3, Annex 31D";
        leaf enable {
          type boolean;
          description
            "True indicates that IEEE Std 802.3 priority-based
             flow control is enabled, false indicates that
             IEEE Std 802.3 priority-based flow control is
             disabled. For interfaces that have auto-negotiation,
             the priority-based flow control is enabled by
             default.";
        }
        container statistics {
          config false;
          status deprecated;
```

```
            description
              "This container collects all statistics for
               Ethernet interfaces.

               Discontinuities in the values of counters in
               this container can occur at re-initialization of
               the management system, and at other times as
               indicated by the value of the 'discontinuity-time'
               leaf defined in the ietf-interfaces YANG module
               (IETF RFC 8343).";
            leaf in-frames-pfc {
              type yang:counter64;
              units "frames";
              description
                "Deprecated in-frames-pfc as not defined in base
                 standard. A count of PFC MAC Control frames
                 received on this Ethernet interface.";
              reference
                "IEEE Std 802.3.1, dot3HCInPFCFrames";
            }
            leaf out-frames-pfc {
              type yang:counter64;
              units "frames";
              description
                "Deprecated out-frames-pfc as not defined in base
                 standard.A count of PFC MAC Control frames
                 transmitted on this interface.";
              reference
                "IEEE Std 802.3.1, dot3HCInPFCFrames";
            }
          }
        }
        leaf force-flow-control {
          type boolean;
          default "false";
          description
            "Explicitly forces the local PAUSE frame based flow
             control settings regardless of what has been
             negotiated.

             Since the auto-negotiation of flow-control settings
             does not allow all sane combinations to be negotiated
             (e.g., consider a device that is only capable of
             sending PAUSE frames connected to a peer device that
             is only capable of receiving and acting on PAUSE
             frames) and failing to agree on the flow-control
             settings does not cause the auto-negotiation to fail
             completely, then it is sometimes useful to be able to
             explicitly enable particular PAUSE frame based flow
             control settings on the local device regardless of
             what is being advertised or negotiated.";
          reference
            "IEEE Std 802.3, Table 28B-3";
        }
      }
      leaf max-frame-length {
        type uint16;
        units "octets";
        config false;
```

```
        description
          "This indicates the MAC frame length (including FCS bytes)
           at which frames are dropped for being too long.";
        reference
          "IEEE Std 802.3, 30.3.1.1.37 aMaxFrameLength";
      }
      leaf mac-control-extension-control {
        type boolean;
        config false;
        description
          "A value that identifies the current EXTENSION
           MAC Control function, as specified in
           IEEE Std 802.3, Annex 31C.";
        reference
          "IEEE Std 802.3, 30.3.8.3 aEXTENSIONMACCtrlStatus
           IEEE Std 802.3.1, dot3ExtensionMacCtrlStatus ";
      }
      leaf frame-limit-slow-protocol {
        type uint64;
        units "f/s";
        default "10";
        config false;
        description
          "The maximum number of Slow Protocol frames of a given
           subtype that can be transmitted in a one second
           interval.
           The default value is 10.";
        reference
          "IEEE Std 802.3, 30.3.1.1.38 aSlowProtocolFrameLimit";
      }
      container capabilities {
        config false;
        description
          "Container all Ethernet interface specific capabilities.";
        leaf auto-negotiation {
          type boolean;
          description
            "Indicates whether auto-negotiation may be configured on
             this interface.";
        }
      }
      container ethernet-pause {
        if-feature "ethernet-pause";
        description
          "IEEE Std 802.3 PAUSE flow control.

           Discontinuities in the values of counters in
           this container can occur at re-initialization of
           the management system, and at other times as
           indicated by the value of the 'discontinuity-time'
           leaf defined in the ietf-interfaces YANG module
           (IETF RFC 8343).";
        reference
          "IEEE Std 802.3,30.3.4 PAUSE entity managed object class and
           Clause 31,Annex 31B)";
        container control-and-status {
          description
            "PAUSE function control and status objects.";
          leaf pause-admin-control {
```

```
            type pause-fc-direction-type;
            default "disabled";
            description
              "What PAUSE functionality will run on this interface when
               Auto-Negotiation is not implemented or disabled.";
            reference
              "IEEE Std 802.3, ????";
          }
          leaf pause-oper-status {
            type pause-fc-direction-type;
            config false;
            description
              "What PAUSE functionality is running on this
               interface, as a result of  Auto-Negotiation or
               setting pause-admin-control.";
            reference
              "IEEE Std 802.3, ????";
          }
          leaf link-delay-allowance {
            type uint32;
            description
              "The value in bit-times of the allowance made by the
               MAC Control PAUSE entity for round-trip propagation
               delay.
               The default value is implementation-dependent.";
          }
          leaf pfc-enable-status {
            type boolean;
            config false;
            description
              "Is IEEE 802.1 Priority-based Flow Control(PFC)
               enabled on the interface. If PFC is enabled, then
               802.3 PAUSE flow control is disabled.";
            reference
              "IEEE Std 802.3, 30.3.3.6 aPFCEnableStatus";
          }
        }
        container statistics {
          config false;
          description
            "PAUSE frame counters.";
          leaf in-frames-pause {
            type yang:counter64;
            units "frames";
            description
              "A count of MAC Control PAUSE frames transmitted on
               this Ethernet interface.";
            reference
              "IEEE Std 802.3, 30.3.4.3
               aPAUSEMACCtrlFramesReceived";
          }
          leaf out-frames-pause {
            type yang:counter64;
            units "frames";
            description
              "A count of MAC Control PAUSE frames transmitted on
               this Ethernet interface.";
            reference
              "IEEE Std 802.3, 30.3.4.2
```

```
              aPAUSEMACCtrlFramesTransmitted";
        }
      }
    }
    container statistics {
      config false;
      description
        "Contains statistics specific to Ethernet interfaces.

         Discontinuities in the values of counters in the
         container can occur at re-initialization of the
         management system, and at other times as indicated by
         the value of the 'discontinuity-time' leaf defined in
         the ietf-interfaces YANG module (IETF RFC 8343).";
      container frame {
        description
          "Contains frame statistics specific to Ethernet
           interfaces.

           All octet frame lengths include the 4 byte FCS.

           Error counters are only reported once. The count
           represented by an instance of this object is
           incremented when the frameCheckError status is
           returned by the MAC service to the MAC Client.
           Received frames for which multiple error conditions
           pertain are, according to the conventions of
           IEEE Std 802.3 Layer Management, counted exclusively
           according to the error status presented to the MAC
           Client.

           A frame that is counted by an instance of this object
           is also counted by the corresponding instance of
           'in-errors' leaf defined in the ietf-interfaces YANG
           module (IETF RFC 8343).";
        leaf in-total-frames {
          type yang:counter64;
          units "frames";
          description
            "The total number of frames (including bad frames)
             received on the Ethernet interface.

             This counter is calculated by summing the following
             IEEE Std 802.3, Clause 30 counters:
             aFramesReceivedOK +
             aFrameCheckSequenceErrors +
             aAlignmentErrors +
             aFrameTooLongErrors +
             aFramesLostDueToIntMACRcvError";
          reference
            "IEEE Std 802.3, Clause 30 counters, as specified
             in the description above.";
        }
        leaf in-total-octets {
          type yang:counter64;
          units "octets";
          description
            "The total number of octets of data (including those
             in bad frames) received on the Ethernet interface.
```

```
                Includes the 4-octet FCS.";
            reference
              "IETF RFC 2819, etherStatsOctets";
          }
          leaf in-frames {
            type yang:counter64;
            units "frames";
            description
              "A count of frames (including unicast, multicast and
               broadcast) that have been successfully received on
               the Ethernet interface.

               This count does not include frames received with
               frame-too-long, FCS, length or alignment errors, or
               frames lost due to internal MAC sublayer error.";
            reference
              "IEEE Std 802.3, 30.3.1.1.5 aFramesReceivedOK";
          }
          leaf in-multicast-frames {
            type yang:counter64;
            units "frames";
            description
              "A count of multicast frames that have been
               successfully received on the Ethernet interface.

                This counter represents a subset of the frames
                counted by in-frames.

                This count does not include frames received with
                frame-too-long, FCS, length or alignment errors, or
                frames lost due to internal MAC sublayer error.";
            reference
              "IEEE Std 802.3, 30.3.1.1.21
               aMulticastFramesReceivedOK";
          }
          leaf in-broadcast-frames {
            type yang:counter64;
            units "frames";
            description
              "A count of broadcast frames that have been
               successfully received on the Ethernet interface.

               This counter represents a subset of the frames
               counted by in-frames.

               This count does not include frames received with
               frame-too-long, FCS, length or alignment errors, or
               frames lost due to internal MAC sublayer error.";
            reference
              "IEEE Std 802.3, 30.3.1.1.22
               aBroadcastFramesReceivedOK";
          }
          leaf in-error-fcs-frames {
            type yang:counter64;
            units "frames";
            description
              "A count of receive frames that are of valid length,
                        but do not pass the FCS check, regardless of whether
```

```
                    or not the frames are an integral number of octets
                    in length.

                    This counter is calculated by summing the following
                    counters:
                      aFrameCheckSequenceErrors +
                      aAlignmentErrors";
        reference
          "IEEE Std 802.3, 30.3.1.1.6 aFrameCheckSequenceErrors;
           IEEE Std 802.3, 30.3.1.1.7 aAlignmentErrors";
      }
      leaf in-error-undersize-frames {
        type yang:counter64;
        units "frames";
        status deprecated;
        description
          "Deprecated in-error-undersize-frames as not defined
           in base standard. A count of frames received on a
           particular Ethernet interface that are less than
           64 bytes in length, and are discarded.

           This counter is incremented regardless of whether
           the frame passes the FCS check.";
        reference
          "IETF RFC 2819, etherStatsUndersizePkts and
           etherStatsFragments";
      }
      leaf in-error-oversize-frames {
        type yang:counter64;
        units "frames";
        description
          "A count of frames received on a particular Ethernet
           interface that exceed the maximum permitted frame
           size, that is specified in max-frame-length, and are
           discarded.

           This counter is incremented regardless of whether
           the frame passes the FCS check.";
        reference
          "IEEE Std 802.3, 30.3.1.1.25
           aFrameTooLongErrors";
      }
      leaf in-error-mac-internal-frames {
        type yang:counter64;
        units "frames";
        description
          "A count of frames for which reception on a
           particular Ethernet interface fails due to an
           internal MAC sublayer receive error.

           A frame is only counted by an instance of this
           object if it is not counted by the corresponding
           instance of either the in-error-fcs-frames,
           in-error-undersize-frames, or
           in-error-oversize-frames.  The precise meaning of
           the count represented by an instance of this object
           is implementation-specific.

           In particular, an instance of this object may
```

```
            represent a count of receive errors on a particular
            Ethernet interface that are not otherwise counted.";
        reference
          "IEEE Std 802.3, 30.3.1.1.15
           aFramesLostDueToIntMACRcvError";
      }
      leaf out-frames {
        type yang:counter64;
        units "frames";
        description
          "A count of frames (including unicast, multicast and
           broadcast) that have been successfully transmitted
           on the Ethernet interface.";
        reference
          "IEEE Std 802.3, 30.3.1.1.2 aFramesTransmittedOK";
      }
      leaf out-multicast-frames {
        type yang:counter64;
        units "frames";
        description
          "A count of multicast frames that have been
           successfully transmitted on the Ethernet interface.

           This counter represents a subset of the frames
           counted by out-frames.";
        reference
          "IEEE Std 802.3, 30.3.1.1.18
            aMulticastFramesXmittedOK";
      }
      leaf out-broadcast-frames {
        type yang:counter64;
        units "frames";
        description
          "A count of broadcast frames that have been
           successfully transmitted on the Ethernet interface.

           This counter represents a subset of the frames
           counted by out-frames.";
        reference
          "IEEE Std 802.3, 30.3.1.1.19
           aBroadcastFramesXmittedOK";
      }
      leaf out-error-mac-internal-frames {
        type yang:counter64;
        units "frames";
        description
          "A count of frames for which transmission on a
           particular Ethernet interface fails due to an
           internal MAC sublayer transmit error.

           The precise meaning of the count represented by an
           instance of this object is implementation-specific.
           In particular, an instance of this object may
           represent a count of transmission errors on a
           particular Ethernet interface that are not otherwise
           counted.";
        reference
          "IEEE Std 802.3, 30.3.1.1.12
           aFramesLostDueToIntMACXmitError";
```

```
          }
        }
        container phy {
          description
            "Ethernet statistics related to the PHY layer.";
          leaf in-error-symbol {
            type yang:counter64;
            units "errors";
            description
              "A count of the number of symbol errors that have
               occurred.

               For the precise definition of when the symbol error
               counter is incremented, please see the 'description'
               text associated with aSymbolErrorDuringCarrier,
               specified in IEEE Std 802.3, 30.3.2.1.5.";
            reference
              "IEEE Std 802.3, 30.3.2.1.5
               aSymbolErrorDuringCarrier";
          }
          container lpi {
            description
              "Physical Ethernet statistics for the energy
               efficiency related low power idle indications.";
            leaf in-lpi-transitions {
              type yang:counter64;
              units "transitions";
              description
                "The number of times the link partner transitioned to
                 Low Power Idle.
                 This counter has a maximum per second increment rate
                 of:
                    50 thousand for 100 Mb/s;
                    90 thousand for 1000 Mb/s
                    230 thousand for 10 Gb/s.";
              reference
                "IEEE Std 802.3, 30.3.2.1.11
                 aReceiveLPITransitions";
            }
            leaf in-lpi-time {
              type decimal64 {
                fraction-digits 6;
              }
              units "seconds";
              description
                "The amount of time the link partner has
                 been in Low Power Idle.";
              reference
                "IEEE Std 802.3, 30.3.2.1.9
                 aReceiveLPIMicroseconds";
            }
            leaf out-lpi-transitions {
              type yang:counter64;
              units "transitions";
              description
                "The number of times this port transitioned to
                 Low Power Idle.
                 This counter has a maximum per second increment rate
                 of:
```

```
                50 thousand for 100 Mb/s;
                90 thousand for 1000 Mb/s
                230 thousand for 10 Gb/s.";
          reference
            "IEEE Std 802.3, 30.3.2.1.10
             aTransmitLPITransitions";
        }
        leaf out-lpi-time {
          type decimal64 {
            fraction-digits 6;
          }
          units "seconds";
          description
            "The  amount of time in this port has
             been in Low Power Idle.";
          reference
            "IEEE Std 802.3, 30.3.2.1.8
             aTransmitLPIMicroseconds";
        }
      }
    }
    container mac-control {
      description
        "A group of statistics specific to MAC Control
         operation of selected Ethernet interfaces.";
      reference
        "IEEE Std 802.3.1, dot3ExtensionTable";
      leaf in-frames-mac-control-unknown {
        type yang:counter64;
        units "frames";
        description
          "A count of MAC Control frames with an unsupported
           opcode received on this Ethernet interface.

           Frames counted against this counter are also counted
           against in-discards defined in the ietf-interfaces
           YANG module (IETF RFC 8343).";
        reference
          "IEEE Std 802.3, 30.3.3.5
           aUnsupportedOpcodesReceived";
      }
      leaf in-frames-mac-control-extension {
        type yang:counter64;
        units "frames";
        description
          "The count of Extension MAC Control frames received
            on this Ethernet interface.";
        reference
          "IEEE Std 802.3, 30.3.8.2
           aEXTENSIONMACCtrlFramesReceived";
      }
      leaf out-frames-mac-control-extension {
        type yang:counter64;
        units "frames";
        description
          "The count of Extension MAC Control frames
           transmitted on this Ethernet interface.";
        reference
          "IEEE Std 802.3, 30.3.8.1
```

```
                aEXTENSIONMACCtrlFramesTransmitted";
          }
        }
      }
    }
  }
}
```

5.3.2.2 Ethernet interface module (half-duplex)

> ***Editor's Note (to be removed prior to publication):***
>
> *Pretty printing of ieee802-ethernet-interface-half-duplex.yang file may change the appearance by adding whitespace and reformatting lines*

> ***Editor's Note (to be removed prior to publication):***
>
> IEEE Std 802.3.1 and IEEE Std 802.3.2 to be updated at the publication time

```
module ieee802-ethernet-interface-half-duplex {
  yang-version 1.1;
  namespace
    "urn:ieee:std:802.3:yang:ieee802-ethernet-interface-half-duplex";
  prefix ieee802-eth-half-duplex;

  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
  }
  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import iana-if-type {
    prefix ianaift;
    reference
      "http://www.iana.org/assignments/yang-parameters/
       iana-if-type@2023-01-26.yang";
  }
  import ieee802-ethernet-interface {
    prefix ieee802-eth-if;
  }

  organization
    "IEEE Std 802.3 Ethernet Working Group
     Web URL: http://www.ieee802.org/3/";
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
    "This module contains YANG definitions for half duplex
     Ethernet interfaces, e.g., 10BASE-T1S.";

  revision 2025-04-17 {
    description
      "Updates under IEEE Std 802.3.2-202x, Draft D3.3";
```

```
    reference
      "IEEE Std 802.3-2022, unless dated explicitly";
  }

  feature dynamic-rate-control {
    status obsolete;
    description
      "This feature indicates that the device supports Ethernet
       interfaces lowering the average data rate of the MAC
       sublayer, with frame granularity, by using Rate Control to
       dynamically increase the inter-packet gap for some types of
       Ethernet interface.
       Only valid for Ethernet interfaces operating at speeds
       (data rates)above 1000 Mb/s.";
    reference
      "IEEE Std 802.3, 30.3.1.1.33 aRateControlAbility";
  }

  feature csma-cd {
    description
      "This feature indicates that the device supports Ethernet
       interfaces running at half-duplex using CSMA/CD.";
  }

  typedef dynamic-rate-control-type {
    type enumeration {
      enum disabled {
        description
          "Dynamic rate control is disabled";
      }
      enum sonet-oc192 {
        value 2;
        description
          "Dynamic rate control is enabled for a 10 Gb/s Ethernet
           interface to SONET/SDH OC192/STM64.";
      }
    }
    default "disabled";
    status obsolete;
    description
      "Allowed values for dynamic-rate-control.";
    reference
      "IEEE Std 802.3, 4.4.2 ipgStretchRatio and 30.3.1.1.34
       aRateControlStatus";
  }

  augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet" {
    when "derived-from-or-self(../if:type, 'ianaift:ethernetCsmacd')
          and ieee802-eth-if:duplex = 'half'" {
      description
        "Applies to half-duplex Ethernet interfaces.";
    }
    status obsolete;
    description
      "Augment with Ethernet interface configuration parameters
       for half-duplex operation.";
    leaf dynamic-rate-control {
      if-feature "dynamic-rate-control";
      type dynamic-rate-control-type;
```

```
      status obsolete;
      description
        "Enables dynamic rate control and specifies what speed
         (data rate) the dynamic rate control is operating at.
         The value of this attribute is constrained by the MAC
         data rate and hardware support.
         The default value is implementation-dependent.";
      reference
        "IEEE Std 802.3, 30.3.1.1.34 aRateControlStatus";
    }
  }

  augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet/"
        + "ieee802-eth-if:capabilities" {
    when "derived-from-or-self(../../if:type,
          'ianaift:ethernetCsmacd')
           and ../ieee802-eth-if:duplex = 'half'" {
      description
        "Applies to half-duplex Ethernet interfaces";
    }
    status obsolete;
    description
      "Augment with configuration capabilities for half-duplex
       Ethernet interface.";
    leaf dynamic-rate-control-supported {
      if-feature "dynamic-rate-control";
      type boolean;
      default "false";
      status obsolete;
      description
        "Indicates whether the Ethernet interface supports lowering
         the average data rate of the MAC sublayer, with frame
         granularity, by using Rate Control to dynamically increase
         the inter-packet gap.
         Only valid for Ethernet interfaces operating at speeds
         (data rates) above 1000 Mb/s.";
      reference
        "IEEE Std 802.3, 30.3.1.1.33 aRateControlAbility";
    }
  }

  augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet/"
        + "ieee802-eth-if:statistics/ieee802-eth-if:frame" {
    when "derived-from-or-self(../../../if:type,
          'ianaift:ethernetCsmacd') and
              ../../ieee802-eth-if:duplex = 'half'" {
      description
        "Applies to half-duplex Ethernet interfaces.";
    }
    description
      "Augment with statistics for half-duplex Ethernet interface.";
    container csma-cd {
      if-feature "csma-cd";
      description
        "Holds counters that are specific to CDMA/CD half-duplex
         operation of Ethernet interfaces.
         This counter does not increment on Ethernet interfaces
         operating at speeds (data rates) greater than 10 Mb/s, or
         on Ethernet interfaces operating in full-duplex mode.
```

```
           Discontinuities in the value of this counter can occur at
           re-initialization of the management system, and at other
           times as indicated by the value of the
           'discontinuity-time' leaf defined in the ietf-interfaces
           YANG module (IETF RFC 8343).";
      leaf in-errors-sqe-test {
        type yang:counter64;
        units "errors";
        description
          "A count of times that the SQE TEST ERROR is received on
           a particular interface. The SQE TEST ERROR is set in
           accordance with the rules for verification of the SQE
           detection mechanism in the PLS Carrier Sense Function as
           described in IEEE Std 802.3, 7.2.4.6.";
        reference
          "IEEE Std 802.3, 7.2.4.6, and 30.3.2.1.4 aSQETestErrors";
      }
      leaf out-frames-collision-single {
        type yang:counter64;
        units "frames";
        description
          "A count of frames that are involved in a single
           collision, and are subsequently transmitted
           successfully. A frame that is counted by an instance of
           this object is also counted by the corresponding
           instance of either 'out-unicast-frames',
           'out-broadcast-frames', or 'out-multicast-frames',
           and is not counted by the corresponding instance of the
           'out-frames-collision-multiple'.";
        reference
          "IEEE Std 802.3, 30.3.1.1.3 aSingleCollisionFrames";
      }
      leaf out-frames-collision-multiple {
        type yang:counter64;
        units "frames";
        description
          "A count of frames that are involved in multiple
           collisions, and are subsequently transmitted
           successfully. A frame that is counted by an instance of
           this object is also counted by the corresponding
           instance of either 'out-unicast-frames',
           'out-broadcast-frames', or 'out-multicast-frames', and
           is not counted by the corresponding instance of the
           'out-frames-collision-single'.";
        reference
          "IEEE Std 802.3, 30.3.1.1.4 aMultipleCollisionFrames";
      }
      leaf out-frames-deferred {
        type yang:counter64;
        units "frames";
        description
          "A count of frames for which the first transmission
           attempt on a particular Ethernet interface is delayed
           because the medium is busy.
           A deferred frame that is not subject to any number of
           collisions is not counted by an instance of
           'out-frames-collision-single' or
           'out-frames-collision-multiple' objects.";
        reference
```

```
        "IEEE Std 802.3, 30.3.1.1.9
         aFramesWithDeferredXmissions";
    }
    leaf out-frames-collisions-excessive {
      type yang:counter64;
      units "frames";
      description
        "A count of frames for which transmission on a particular
         Ethernet interface fails due to excessive collisions.";
      reference
        "IEEE Std 802.3, 30.3.1.1.11 aFramesAbortedDueToXSColls";
    }
    leaf out-collisions-late {
      type yang:counter64;
      units "collisions";
      description
        "The number of times that a collision is detected on a
         particular Ethernet interface later than one slotTime
         into the transmission of a packet.
         A (late) collision included in a count represented by an
         instance of this object is also considered as a
         (generic) collision for purposes of other
         collision-related statistics.";
      reference
        "IEEE Std 802.3, 30.3.1.1.10 aLateCollisions";
    }
    leaf out-errors-carrier-sense {
      type yang:counter64;
      units "errors";
      description
        "The number of times that the carrier sense condition was
         lost or never asserted when attempting to transmit a
         frame on a particular Ethernet interface.
         The count represented by an instance of this object is
         incremented at most once per transmission attempt, even
         if the carrier sense condition fluctuates during a
         transmission attempt.";
      reference
        "IEEE Std 802.3, 30.3.1.1.13 aCarrierSenseErrors";
    }
    list collision-histogram {
      key "collision-count";
      description
        "A collection of collision histograms for a particular
         interface.";
      reference
        "IEEE Std 802.3, 30.3.1.1.30 aCollisionFrames";
      leaf collision-count {
        type yang:counter64;
        units "collisions";
        description
          "The number of per-frame media collisions for which a
           particular collision histogram cell represents the
           frequency on a particular interface.";
      }
      leaf collision-count-frames {
        type yang:counter64;
        units "frames";
        description
```

```
            "A count of individual MAC frames for which the
             transmission (successful or otherwise) on a particular
             interface occurs after the frame has experienced
             exactly the number of collisions in the associated
             dot3CollCount object.
             For example, a frame which is transmitted on an
             interface after experiencing exactly 4 collisions
             would be indicated by incrementing only
             collision-count-frames object associated with the
             collision-count value of 4. No other instance of
             collision-count-frames would be incremented in this
             example.";
        }
      }
    }
  }
}
```

5.3.2.3 Ethernet MAC merge module

> ***Editor's Note (to be removed prior to publication):***
>
> *Pretty printing of ieee802-mac-merge.yang file may change the appearance by adding whitespace and reformatting lines*

> ***Editor's Note (to be removed prior to publication):***
>
> IEEE Std 802.3.1 and IEEE Std 802.3.2 to be updated at the publication time

```
module ieee802-ethernet-mac-merge {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-mac-merge";
  prefix mac-merge;

  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
  }
  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import ieee802-ethernet-interface {
    prefix ieee802-eth-if;
    reference
      "IEEE Std 802.3.2-2019";
  }

  organization
    "IEEE Std 802.3 Ethernet Working Group
     Web URL: http://www.ieee802.org/3/";
  contact
    "Web URL: http://www.ieee802.org/3/";
```

```
  description
    "The Yang model for managing devices that support the MAC Merge
     (aka preemption) sublayer as defined in Clause 99. Unless
     otherwise indicated, the references in this model module are to
         IEEE Std 802.3-2022.";

  revision 2025-04-17 {
    description
      "Updates under IEEE Std 802.3.2-202x, Draft D3.3";
    reference
      "IEEE Std 802.3-2022, unless dated explicitly";
  }

  feature mac-merge {
    description
      "MAC Merge is supported.";
    reference
      "IEEE Std 802.3-2022";
  }

  augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet" {
    if-feature "mac-merge";
    description
      "Interfaces supporting MAC Merge.";
    container mac-merge {
      description
        "Container for MAC Merge objects.";
      container admin-control {
        description
          "Container for MAC Merge control objects.";
        leaf merge-enable-tx {
          type enumeration {
            enum Disabled {
              description
                "Transmit preemption is disabled";
            }
            enum Enabled {
              description
                "Transmit preemption is enabled";
            }
          }
          default "Disabled";
          description
            "Is MAC Merge enabled when transmitting?
             It may be modified using an edit operation.";
          reference
            "IEEE Std 802.3, 30.14.1.3";
        }
        leaf verify-disable-tx {
          type enumeration {
            enum Disabled {
              description
                "Verify is disabled";
            }
            enum Enabled {
              description
                "Verify is enabled";
            }
          }
```

```
          default "Disabled";
          description
            "Does MAC Merge check to make sure the link partner
             supports MAC Merge?
             The check is peformed by sending a verify request and
             getting a verify response.
             It may be modified using an edit operation.;";
          reference
            "IEEE Std 802.3, 30.14.1.4";
        }
        leaf verify-time {
          type uint16 {
            range "1..128";
          }
          units "milliseconds";
          default "10";
          description
            "The retry timer for verification attempts.
             The valid range is 1 to 128 inclusive.
             The default value is 10.";
          reference
            "IEEE Std 802.3, 30.14.1.6";
        }
        leaf frag-size {
          type uint8 {
            range "0..3";
          }
          default "0";
          description
            "This controls minimum frame size for packet fragments.
             It's specified in units of 64 bytes above the minimum
             of 64 bytes. The equation is:
             minimum-size = 64 + (frag-size*64)-4";
          reference
            "IEEE Std 802.3, 30.14.1.7";
        }
      }
      container admin-status {
        config false;
        description
          "Container for MAC Merge status objects.";
        leaf merge-support {
          type enumeration {
            enum Supported {
              description
                "MAC Merge is supported on the device";
            }
            enum NotSupported {
              description
                "MAC Merge is not supported on the device";
            }
          }
          description
            "Is MAC Merge is supported?";
          reference
            "IEEE Std 802.3, 30.14.1.1";
        }
        leaf verify-status {
          type enumeration {
```

```
            enum unknown {
              description
                "unknown";
            }
            enum initial {
              description
                "initial";
            }
            enum verifying {
              description
                "verifying";
            }
            enum succeeded {
              description
                "succeeded";
            }
            enum failed {
              description
                "failed";
            }
            enum disabled {
              description
                "disabled";
            }
          }
          description
            "The status of MAC Merge Verification.";
          reference
            "IEEE Std 802.3, 30.14.1.2";
        }
        leaf status-tx {
          type enumeration {
            enum unknown {
              description
                "unknown";
            }
            enum inactive {
              description
                "inactive";
            }
            enum active {
              description
                "active";
            }
          }
          description
            "The status of MAC Merge in the transmit direction.";
          reference
            "IEEE Std 802.3, 30.14.1.5";
        }
      }
      container statistics {
        config false;
        description
          "Container for MAC Merge counters.
           Discontinuities in the values of counters in
           this container can occur at re-initialization of
           the management system, and at other times as
           indicated by the value of the 'discontinuity-time'
```

```
              leaf defined in the ietf-interfaces YANG module
              (IETF RFC 8343).";
          leaf assembly-error-count {
            type yang:counter64;
            description
              "A count of MAC frames with reassembly errors.";
            reference
              "IEEE Std 802.3, 30.14.1.8";
          }
          leaf smd-error-count {
            type yang:counter64;
            description
              "A count of received MAC frames / MAC frame fragments
               rejected with an unexpected preemption header.";
            reference
              "IEEE Std 802.3, 30.14.1.9";
          }
          leaf assembly-ok-count {
            type yang:counter64;
            description
              "A count of preemptable MAC frames that were successfully
               reassembled and delivered to MAC.";
            reference
              "IEEE Std 802.3, 30.14.1.10";
          }
          leaf fragment-count-rx {
            type yang:counter64;
            description
              "The number of preemptable frame fragments recieved.";
            reference
              "IEEE Std 802.3, 30.14.1.11";
          }
          leaf fragment-count-tx {
            type yang:counter64;
            description
              "The number of preemptable frame fragments transmitted.";
            reference
              "IEEE Std 802.3, 30.14.1.12";
          }
          leaf hold-count {
            type yang:counter64;
            description
              "The number of MAC Merge hold requests (i.e., to stop
               transmissions of preemptable traffic) received from
               the MAC";
            reference
              "IEEE Std 802.3, 30.14.1.13";
          }
        }
      }
    }
  }
```

5.3.2.4 Ethernet LLDP module

> ***Editor's Note (to be removed prior to publication):***
>
> Pretty printing of ieee802-ethernet-lldp.yang file may change the appearance by adding whitespace and reformatting lines

> ***Editor's Note (to be removed prior to publication):***
>
> IEEE Std 802.3.1 and IEEE Std 802.3.2 to be updated at the publication time

```
module ieee802-ethernet-lldp {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-lldp";
  prefix ieee802-eth-lldp;

  import ieee802-dot1ab-lldp {
    prefix lldp;
    reference
      "IEEE Std 802.1ABcu-2021";
  }
  import ieee802-ethernet-phy-type {
    prefix ieee802-phy;
    reference
      "IEEE Std 802.3-2022";
  }

  organization
    "IEEE Std 802.3 Ethernet Working Group
     Web URL: http://www.ieee802.org/3/";
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
    "This module contains YANG definitions for configuring LLDP for
     802.3 Ethernet Interfaces.
     In this YANG module, 'Ethernet interface' can be interpreted
     as referring to 'IEEE Std 802.3 compliant Ethernet
     interfaces'.";

  revision 2025-04-17 {
    description
      "Updates under IEEE Std 802.3.2-202x, Draft D3.3";
    reference
      "IEEE Std 802.3-2022, unless dated explicitly";
  }

  typedef port-class-type {
    type enumeration {
      enum p-class-pse {
        value 0;
        description
          "Power Sourcing Equipment";
      }
      enum p-class-pd {
        value 1;
        description
          "Powered Device";
      }
```

```
    }
    description
      "Enumeration for the power port class";
    reference
      "IEEE Std 802.3, 30.12.2.1.5";
  }

  typedef pse-pinout-type {
    type enumeration {
      enum signal {
        value 0;
        description
          "PSE Pinout Alternative A";
      }
      enum spare {
        value 1;
        description
          "PSE Pinout Alternative B";
      }
    }
    description
      "Enumeration for the pinout alternatives used for PD
       detection and power ";
    reference
      "IEEE Std 802.3, 30.12.2.1.9";
  }

  typedef pse-power-class-type {
    type enumeration {
      enum class0 {
        value 0;
        description
          "Class 0 PD";
      }
      enum class1 {
        value 1;
        description
          "Class 1 PD";
      }
      enum class2 {
        value 2;
        description
          "Class 2 PD";
      }
      enum class3 {
        value 3;
        description
          "Class 3 PD";
      }
      enum class4 {
        value 4;
        description
          "Class 4 PD";
      }
    }
    description
      "Enumeration for the PD class";
    reference
      "IEEE Std 802.3, 30.12.2.1.10";
```

```
  }

  typedef power-class-ext-AB-type {
    type enumeration {
      enum singlesig {
        value 0;
        description
          "Single-signature PD or 2-pair only PSE";
      }
      enum class1 {
        value 1;
        description
          "Class 1";
      }
      enum class2 {
        value 2;
        description
          "Class 2";
      }
      enum class3 {
        value 3;
        description
          "Class 3";
      }
      enum class4 {
        value 4;
        description
          "Class 4";
      }
      enum class5 {
        value 5;
        description
          "Class 5";
      }
    }
    description
      "Enumeration for the assigned power class ";
    reference
      "IEEE Std 802.3, 30.12.3.1.26";
  }

  typedef power-class-ext-type {
    type enumeration {
      enum dualsig {
        value 0;
        description
          "Dual-signature PD";
      }
      enum class1 {
        value 1;
        description
          "Class 1";
      }
      enum class2 {
        value 2;
        description
          "Class 2";
      }
      enum class3 {
```

```
          value 3;
          description
            "Class 3";
        }
        enum class4 {
          value 4;
          description
            "Class 4";
        }
        enum class5 {
          value 5;
          description
            "Class 5";
        }
        enum class6 {
          value 6;
          description
            "Class 6";
        }
        enum class7 {
          value 7;
          description
            "Class 7";
        }
        enum class8 {
          value 8;
          description
            "Class 8";
        }
      }
      description
        "Enumeration for the assigned power class ";
      reference
        "IEEE Std 802.3, 30.12.3.1.28";
    }

    typedef power-type {
      type enumeration {
        enum type4dualsigPD {
          value 0;
          description
            "Type 4 dual-signature PD";
        }
        enum type4singlesigPD {
          value 1;
          description
            "Type 4 single-signature PD";
        }
        enum type3dualsigPD {
          value 2;
          description
            "Type 3 dual-signature PD";
        }
        enum type3singlesigPD {
          value 3;
          description
            "Type 3 single-signature PD";
        }
        enum type4PSE {
```

```
        value 4;
        description
          "Type 4 PSE";
      }
      enum type3PSE {
        value 5;
        description
          "Type 3 PSE";
      }
    }
    description
      "Enumeration for the PD class";
    reference
      "IEEE Std 802.3, 30.12.2.1.29";
  }

  typedef power-priority-type {
    type enumeration {
      enum low {
        value 0;
        description
          "low priority PD";
      }
      enum high {
        value 1;
        description
          "high priority PD";
      }
      enum critical {
        value 2;
        description
          "critical priority PD";
      }
      enum unknown {
        value 3;
        description
          "priority unknown";
      }
    }
    description
      "Enumeration for possible priorities of a PD system";
    reference
      "IEEE Std 802.3, 30.12.2.1.16";
  }

  typedef power-source-type {
    type enumeration {
      enum pse-primary {
        value 0;
        description
          "PSE powered by a primary power source";
      }
      enum pse-backup {
        value 1;
        description
          "PSE powered by a backup power source";
      }
      enum pse-unknwon {
        value 2;
```

```
        description
          "PSE powered by an unkown power source";
      }
      enum pd-pse-and-local {
        value 3;
        description
          "PD powered by a PSE and locally";
      }
      enum pd-local-only {
        value 4;
        description
          "PD powered only locally";
      }
      enum pd-pse-only {
        value 5;
        description
          "PD powered by PD only";
      }
      enum pd-unknown {
        value 6;
        description
          "PD powered by an uknown source";
      }
    }
    description
      "Enumeration for the power sources of the
       remote system. When the remote system is a PSE, it
       indicates whether it is being powered by a primary
       power source; a backup power source; or unknown.
       When the remote system is a PD, it indicates whether
       it is being powered by a PSE and locally;
       locally only; by a PSE only; or unknown.";
    reference
      "IEEE Std 802.3, 30.12.2.1.15";
  }

  typedef powering-status-type {
    type enumeration {
      enum 4PdualsigPD {
        value 0;
        description
          "4-pair powering a dual-signature PD";
      }
      enum 4PsinglesigPD {
        value 1;
        description
          "4-pair powering a single-signature PD";
      }
      enum 2P {
        value 2;
        description
          "2-pair powering";
      }
    }
    description
      "Enumeration for the power status of the PSE";
    reference
      "IEEE Std 802.3, 30.12.2.1.23";
  }
```

```
  typedef powered-status-type {
    type enumeration {
      enum 4PdualsigPD {
        value 0;
        description
          "4-pair powered dual-signature PD";
      }
      enum 2PdualsigPD {
        value 1;
        description
          "2-pair powered dual-signature PD";
      }
      enum singlesigPD {
        value 2;
        description
          "powered single-signature PD";
      }
    }
    description
      "Enumeration for the power status of the PSE";
    reference
      "IEEE Std 802.3, 30.12.2.1.24";
  }

  typedef power-pairs-type {
    type enumeration {
      enum altA {
        value 0;
        description
          "Alternative A";
      }
      enum altB {
        value 1;
        description
          "Alternative B";
      }
      enum both {
        value 2;
        description
          "both";
      }
    }
    description
      "Enumeration for the PSE Pinout Alternative";
    reference
      "IEEE Std 802.3, 30.12.2.1.25";
  }

  typedef measurement-source-type {
    type enumeration {
      enum noRequest {
        description
          "No Request";
      }
      enum altA {
        description
          "Pairset Alternative A / Mode A";
      }
```

```
        enum altB {
          description
            "Pairset Alternative B / Mode B";
        }
        enum total {
          description
            "Port total";
        }
      }
      description
        "PoE measurement source";
      reference
        "IEEE Std 802.3, 30.12.2.1.42 and 30.12.3.1.42";
    }

    augment "/lldp:lldp/lldp:port" {
      description
        "Augments port with 802.3 port read-only objects";
      leaf auto-negotiation-supported {
        type boolean;
        config false;
        description
          "True if the port supports Auto-negotiation";
        reference
          "IEEE Std 802.3, 30.12.2.1.1";
      }
      leaf auto-negotiation-enabled {
        type boolean;
        config false;
        description
          "True if Auto-negotiation is enabled";
        reference
          "IEEE Std 802.3, 30.12.2.1.2";
      }
      /* Deprecated MAU objects */
      leaf auto-negotiation-cap {
        type binary {
          length "2";
        }
        config false;
        status deprecated;
        description
          "A read-only 2-octet value that contains the value (bitmap)
           of the ifMauAutoNegCapAdvertisedBits object
           (defined in IETF RFC 4836) which is associated with the
           given port on the local system.
           Superseeded by auto-negotiation-cap-bits.";
        reference
          "IEEE Std802.3, 30.12.2.1.3";
      }
      leaf operational-mau-type {
        type int32;
        config false;
        status deprecated;
        description
          "32-bit integer value that indicates the operational MAU
           type of the given port.
           Superseeded by operational-pmd-type.";
        reference
```

```
          "IEEE Std 802.3, 30.12.2.1.4";
      }
      leaf auto-negotiation-cap-bits {
        type ieee802-phy:auto-neg-ability-bits;
        description
          "The capabilities advertised by the local Auto-Negotiation
           entity.";
        reference
          "IEEE Std 802.3, 30.6.1.1.6,
           aAutoNegadvertisedTechnologyAbility.";
      }
      leaf operational-pmd-type {
        type identityref {
          base ieee802-phy:pmd-type;
        }
        description
          "Operational PMD type of the given port";
        reference
          "IEEE Std 802.3, Clause 30.5.1.1.2 aMAUType";
      }
      /* Clause 30 & 145 PoE related LLDP definitions */
      leaf power-port-class {
        type port-class-type;
        description
          "A read-only value that identifies the port Class of the
           given port";
        reference
          "IEEE Std 802.3, 30.12.2.1.5";
      }
      leaf mdi-power-supported {
        type boolean;
        description
          "True if MDI power is supported";
        reference
          "IEEE Std 802.3, 30.12.2.1.6";
      }
      leaf mdi-power-enabled {
        type boolean;
        description
          "True if MDI power is enabled";
        reference
          "IEEE Std 802.3, 30.12.2.1.7";
      }
      leaf power-pair-controlable {
        type boolean;
        description
          "True if the pair selection can be controlled";
        reference
          "IEEE Std 802.3, 30.12.2.1.8";
      }
      leaf power-pairs {
        type pse-pinout-type;
        description
          "Indicates which pinout alernative is used for PD detection
           and power";
        reference
          "IEEE Std 802.3, 30.12.2.1.9";
      }
      leaf local-power-class {
```

```
        type pse-power-class-type;
        description
          "PD Power Class";
        reference
          "IEEE Std 802.3, 30.12.2.1.10";
      }
      /* Obsolete Link Aggregation objects
          Link aggregation has not been part of 802.3 since IEEE 802.1AX-2008
          */
      leaf link-aggregation-status {
        type bits {
          bit aggregation-capability {
            position 0;
            description
              "IEEE Std 802.3, 79.3.3.1";
          }
          bit aggregation-status {
            position 1;
            description
              "IEEE Std 802.3, 79.3.3.1";
          }
        }
        config false;
        status obsolete;
        description
          "The bitmap value which contains the link aggregation
           capabilities and the current aggregation
           status of the link";
        reference
          "IEEE Std 802.3, 30.12.2.1.11";
      }
      leaf aggregation-port-id {
        type int32;
        config false;
        status obsolete;
        description
          "The unique identifier allocated to this Aggregation Port
           by the local System.";
        reference
          "IEEE Std 802.3, 30.12.2.1.12";
      }
      leaf local-max-frame-size {
        type int32;
        config false;
        status obsolete;
        description
          "An integer value indicating the maximum supported frame
           size in octets on the given port of the local system.";
        reference
          "IEEE Std 802.3, 30.12.2.1.13";
      }
      leaf power-type {
        type bits {
          bit type1-or-greater {
            position 0;
            description
              "0-type1, 1-greater than type1";
          }
          bit pse-or-pd {
```

```
          position 1;
          description
            "0-pse, 1-pd";
        }
      }
      config false;
      description
        "A read-only attribute that returns a bit string indicating
         whether the local system is a PSE or a PD and whether it
         is Type 1 or greater than Type 1. The first bit indicates
         Type 1 or greater than Type 1. The second bit indicates
         PSE or PD. A PSE sets this bit to indicate a PSE. A PD sets
         this bit to indicate a PD. See also aLldpXdot3LocPowerTypeExt.
         Superseeded by power-support";
      reference
        "IEEE Std 802.3, 30.12.2.1.14";
    }
    leaf power-support {
      type bits {
        bit pse-pd {
          description
            "0-PD, 1-PSE";
        }
        bit pse-power-support {
          description
            "0-not supported, 1-supported";
        }
        bit pse-power-state {
          description
            "0-disabled 1-enabled";
        }
        bit pse-pairs-selection {
          description
            "0-pair selection not controllable
             1-pair selection  controllable";
        }
      }
      description
        "A read-only attribute that returns a bit string indicating
         whether local system PoE support.";
      reference
        "IEEE Std 802.3, 79.3.2.1 MDI power support";
    }
    leaf power-source {
      type power-source-type;
      config false;
      description
        "Indicates the power sources of the local system. A PSE
         indicates whether it is being powered by a primary power
         source; a backup power source; or unknown. A PD indicates
         whether it is being powered by a PSE and locally; by a PSE
         only; or unknown.;";
      reference
        "IEEE Std 802.3, 30.12.2.1.15";
    }
    leaf pd-requested-power-value {
      type int32;
      config false;
      description
```

```
        "PD requested power value. For a PD, it is the power value
         that the PD has currently requested from the remote
         system. For a PSE, it is the power value that the PSE
         mirrors back to the remote system";
      reference
        "IEEE Std 802.3, 30.12.2.1.17";
    }
    leaf pd-requested-power-value-a {
      type int32;
      config false;
      description
        "A read-only attribute that returns the PD requested power
         value for the Mode A pairset in units of 0.1 W. For a PD,
         it's the power value that the PD has currently requested
         from the remote system for the Mode A pairset. For a PSE,
         it is the power value for the Alternative A pairset that
         the PSE echoes back to the remote system";
      reference
        "IEEE Std 802.3, 30.12.2.1.18";
    }
    leaf pd-requested-power-value-b {
      type int32;
      config false;
      description
        "A read-only attribute that returns the PD requested power
         value for the Mode B pairset in units of 0.1 W.
         For a PD, it is the power value that the PD has currently
         requested from the remote system for the Mode B pairset.
         For a PSE, it is the power value for the Alternative B
         pairset that the PSE echoes back to the remote system";
      reference
        "IEEE Std 802.3, 30.12.2.1.19";
    }
    leaf pse-allocated-power-value {
      type int32;
      config false;
      description
        "PSE allocated power value. For a PSE, it is the power
         value that the PSE has currently allocated to the remote
         system. For a PD, it is the power value that the PD mirrors
         back to the remote syste";
      reference
        "IEEE Std 802.3, 30.12.2.1.20";
    }
    leaf pse-allocated-power-value-a {
      type int32;
      config false;
      description
        " PSE allocated power value for the Alternative A pairset
          in units of 0.1 W.For a PSE, it is the power value for the
          Alternative A pairset that the PSE has currently allocated
          to the remote system. For a PD, it is the power value for
          the Mode A pairset that the PD echoes back to the remote
          system.";
      reference
        "IEEE Std 802.3, 30.12.2.1.21";
    }
    leaf pse-allocated-power-value-b {
      type int32;
```

```
          config false;
          description
            "PSE allocated power value for the Alternative B pairset
             in units of 0.1 W. For a PSE, it is the power value for the
             Alternative B pairset that the PSE has currently
             allocated to the remote system. For a PD, it is the power
             value for the Mode B pairset that the PD echoes back to the
             remote system.";
          reference
            "IEEE Std 802.3, 30.12.2.1.22";
        }
        leaf pse-powering-status {
          type powering-status-type;
          config false;
          description
            "A read only value that indicates the powering status of
             the PSE. For a PD, the contents of this
             attribute are undefined.";
          reference
            "IEEE Std 802.3, 30.12.2.1.23";
        }
        leaf pd-powered-status {
          type powered-status-type;
          config false;
          description
            "A read only value that indicates the powering status of
             the PD. For a PSE, the contents of this attribute are
             undefined";
          reference
            "IEEE Std 802.3, 30.12.2.1.24";
        }
        leaf power-pairs-ext {
          type power-pairs-type;
          config false;
          description
            "A read-only value that identifies the supported PSE
             Pinout Alternative specified in 145.2.4. For a PSE, this
             attribute contains the value of the aPSEPowerPairs
              attribute (see 30.9.1.1.4). For a PD, the contents of this
              attribute are undefined";
          reference
            "IEEE Std 802.3, 30.12.2.1.25";
        }
        leaf power-class-ext-A {
          type power-class-ext-AB-type;
          config false;
          description
            "For a dual-signature PD, a read-only value that indicates
             the requested Class for Mode A during Physical Layer
             Classification (see 145.3.6). For a single-signature PD, a
             read-only value set to â€šsinglesigâ€™.For a PSE connected to
             a dual-signature PD, a read-only value that indicates the
             currently assigned Class for Mode A (see 145.2.8). For a
             PSE connected to a single-signature PD or a PSE that
             operates only in 2-pair mode, a read-only value set to
             â€šsinglesigâ€™";
          reference
            "IEEE Std 802.3, 30.12.2.1.26";
        }
```

```
    leaf power-class-ext-B {
      type power-class-ext-AB-type;
      config false;
      description
        "For a dual-signature PD, a read-only value that indicates
         the requested Class for Mode B during Physical Layer
         Classification (see 145.3.6). For a single-signature PD,
         a read-only value set to â€šinglesigâ€™.For a PSE connected to
         a dual-signature PD, a read-only value that indicates the
         currently assigned Class for Mode B (see 145.2.8). For a
         PSE connected to a single-signature PD or a PSE that
         operates only in 2-pair mode, a read-only value set to
         â€šinglesigâ€™";
      reference
        "IEEE Std 802.3, 30.12.2.1.27";
    }
    leaf power-class-ext {
      type power-class-ext-type;
      config false;
      description
        "For a single-signature PD, a read-only value that
         indicates the requested Class during Physical
         Layer Classification (see 145.3.6). For a dual-signature
         PD, a read-only value set to â€đualsigâ€™.
         For a PSE connected to a single-signature PD or a PSE that
         operates only in 2-pair mode, a read-only value that
         indicates the currently assigned Class (see 145.2.8).
         For a PSE connected to a dual-signature PD,
         a read-only value set to â€đualsigâ€™.â€™";
      reference
        "IEEE Std 802.3, 30.12.2.1.28";
    }
    leaf power-type-ext {
      type power-type;
      config false;
      description
        "A read-only attribute that returns a value to indicate if
         the local system is a Type 3 or Type 4 PSE or PD and, in
         the case of a Type 3 or Type 4 PD, if it is a
         single-signature PD or a dual-signature
         PD";
      reference
        "IEEE Std 802.3, 30.12.2.1.29";
    }
    leaf pd-load {
      type boolean;
      config false;
      description
        "For a dual-signature PD, a read-only attribute that
         returns whether the load of a dual-signature PD is
         electrically isolated, as defined in 79.3.2.10.2. For a
         single-signature PD or a PSE, the value of this attribute
         is FALSE";
      reference
        "IEEE Std 802.3, 30.12.2.1.30";
    }
    leaf pd-4pid {
      type boolean;
      config false;
```

```
          description
            "A read-only Boolean attribute indicating whether the local
             PD system supports powering of both PD Modes.";
          reference
            "IEEE Std 802.3, 30.12.2.1.31";
        }
        leaf pse-max-avail-power {
          type int32;
          config false;
          description
            "A read-only attribute that returns the local PSE maximum
             available power value in units of 0.1 W";
          reference
            "IEEE Std 802.3, 30.12.2.1.32";
        }
        leaf pse-autoclass-support {
          type boolean;
          config false;
          description
            "Indicates whether the local PSE system supports Autoclass.";
          reference
            "IEEE Std 802.3, 30.12.2.1.33";
        }
        leaf autoclass-completed {
          type boolean;
          config false;
          description
            "Indicates whether the local PSE system has completed the
             Autoclass measurement.";
          reference
            "IEEE Std 802.3, 30.12.2.1.34";
        }
        leaf autoclass-request {
          type boolean;
          config false;
          description
            "A read-only Boolean attribute indicating whether the local
             PD system is requesting an Autoclass measurement;";
          reference
            "IEEE Std 802.3, 30.12.2.1.35";
        }
        leaf power-down-request {
          type int32;
          description
            "A read-write attribute that indicates the local PD system
             is requesting a power down when the value is  0x1D.;";
          reference
            "IEEE Std 802.3, 30.12.2.1.36";
        }
        leaf power-down-time {
          type int32;
          description
            "A read-write attribute that indicates the number of
             seconds the PD requests to stay powered off. A value of
             zero indicates an indefinite amount of time;";
          reference
            "IEEE Std 802.3, 30.12.2.1.37";
        }
        /* PoE measurement objects */
```

```
      leaf meas-voltage-support {
        type boolean;
        description
          "A read-only attribute that indicates the local device is
           capable of providing a voltage measurement.;";
        reference
          "IEEE Std 802.3, 30.12.2.1.38";
      }
      leaf meas-current-support {
        type boolean;
        description
          "A read-only attribute that indicates the local device is
           capable of providing a current measurement.;";
        reference
          "IEEE Std 802.3, 30.12.2.1.39";
      }
      leaf meas-power-support {
        type boolean;
        description
          "A read-only attribute that indicates the local device is
           capable of providing a power measurement.;";
        reference
          "IEEE Std 802.3, 30.12.2.1.40";
      }
      leaf meas-energy-support {
        type boolean;
        description
          "A read-only attribute that indicates the local device is
           capable of providing a energy measurement.;";
        reference
          "IEEE Std 802.3, 30.12.2.1.41";
      }
      /* NOTE - What do these values mean? */
      leaf measurement-source {
        type bits {
          bit bit1 {
            position 0;
            description
              "????";
          }
          bit bit2 {
            position 1;
            description
              "-";
          }
        }
        status obsolete;
        description
          "A read-write attribute value that indicates to local
           device on which Alternative or Mode the measurement
           is to be taken";
        reference
          "IEEE Std 802.3, 30.12.2.1.42";
      }
      leaf meas-voltage-request {
        type boolean;
        config false;
        description
          "A read-only attribute that indicates the local device is
```

```
          requesting a voltage measurement from the remote device.";
      reference
        "IEEE Std 802.3, 30.12.2.1.43";
    }
    leaf meas-current-request {
      type boolean;
      config false;
      description
        "A read-only attribute that indicates the local device is
         requesting a current measurement from the remote device.";
      reference
        "IEEE Std 802.3, 30.12.2.1.44";
    }
    leaf meas-power-request {
      type boolean;
      config false;
      description
        "A read-only attribute that indicates the local device is
         requesting a power measurement from the remote device.";
      reference
        "IEEE Std 802.3, 30.12.2.1.45";
    }
    leaf meas-energy-request {
      type boolean;
      config false;
      description
        "A read-only attribute that indicates the local device is
         requesting an energy measurement from the remote device.";
      reference
        "IEEE Std 802.3, 30.12.2.1.46";
    }
    leaf meas-voltage-valid {
      type boolean;
      config false;
      description
        "A read-only attribute that indicates the local deviceâ€™s
         voltage measurement is valid.";
      reference
        "IEEE Std 802.3, 30.12.2.1.47";
    }
    leaf meas-current-valid {
      type boolean;
      config false;
      description
        "A read-only attribute that indicates the local deviceâ€™s
         current measurement is valid.";
      reference
        "IEEE Std 802.3, 30.12.2.1.48";
    }
    leaf meas-power-valid {
      type boolean;
      config false;
      description
        "A read-only attribute that indicates the local deviceâ€™s
         power measurement is valid.";
      reference
        "IEEE Std 802.3, 30.12.2.1.49";
    }
    leaf meas-energy-valid {
```

```
        type boolean;
        config false;
        description
          "A read-only attribute that indicates the local deviceâ€™s
           energy measurement is valid.";
        reference
          "IEEE Std 802.3, 30.12.2.1.50";
      }
      leaf meas-voltage-uncertainty {
        type int32;
        config false;
        description
          "A read-only attribute that indicates the expanded
           uncertainty (coverage factor k = 2) for the deviceâ€™s
           voltage measurement.";
        reference
          "IEEE Std 802.3, 30.12.2.1.51";
      }
      leaf meas-current-uncertainty {
        type int32;
        config false;
        description
          "A read-only attribute that indicates the expanded
           uncertainty (coverage factor k = 2) for the deviceâ€™s
           current measurement.";
        reference
          "IEEE Std 802.3, 30.12.2.1.52";
      }
      leaf meas-power-uncertainty {
        type int32;
        config false;
        description
          "A read-only attribute that indicates the expanded
           uncertainty (coverage factor k = 2) for the deviceâ€™s
           power measurement.";
        reference
          "IEEE Std 802.3, 30.12.2.1.53";
      }
      leaf meas-energy-uncertainty {
        type int32;
        config false;
        description
          "A read-only attribute that indicates the expanded
           uncertainty (coverage factor k = 2) for the deviceâ€™s
           energy measurement.";
        reference
          "IEEE Std 802.3, 30.12.2.1.54";
      }
      leaf voltage-measurement {
        type int32;
        config false;
        description
          "A read-only attribute that returns the measured device
           voltage.";
        reference
          "IEEE Std 802.3, 30.12.2.1.55";
      }
      leaf current-measurement {
        type int32;
```

```
      config false;
      description
        "A read-only attribute that returns the measured device
         current.";
      reference
        "IEEE Std 802.3, 30.12.2.1.56";
    }
    leaf power-measurement {
      type int32;
      config false;
      description
        "A read-only attribute that returns the measured device
         power.";
      reference
        "IEEE Std 802.3, 30.12.2.1.57";
    }
    leaf energy-measurement {
      type int32;
      config false;
      description
        "A read-only attribute that returns the measured device
         energy.";
      reference
        "IEEE Std 802.3, 30.12.2.1.58";
    }
    leaf pse-power-price-index {
      type int32;
      config false;
      description
        "A read-only attribute that returns an index of the price
         of power being sourced by the PSE. For a PD, this value
         is undefined";
      reference
        "IEEE Std 802.3, 30.12.2.1.59";
    }
    leaf local-response {
      type int32;
      config false;
      description
        "The maximum time required to update
         pse-allocated-power-value";
      reference
        "IEEE Std 802.3, 30.12.2.1.60";
    }
    leaf local-system-ready {
      type boolean;
      config false;
      description
        "Initialization status of the Data Link Layer
         classification engine on the local system";
      reference
        "IEEE Std 802.3, 30.12.2.1.61";
    }
    leaf tx-system-value {
      type int32;
      config false;
      description
        "Returns the value of Tw_sys_tx that the local system can
         support in the transmit direction.";
```

```
        reference
          "IEEE Std 802.3, 30.12.2.1.62";
      }
      leaf tx-system-value-echo {
        type int32;
        config false;
        description
          "Returns the value of Tw_sys_tx that the emote system is
           advertising that it can support in the transmit direction
           and is echoed by the local system under the control of the
           EEE DLL receiver state diagram.";
        reference
          "IEEE Std 802.3, 30.12.2.1.63";
      }
      leaf rx-system-value {
        type int32;
        config false;
        description
          "Returns the value of Tw_sys_tx that the local system is
           requesting in the receive direction.";
        reference
          "IEEE Std 802.3, 30.12.2.1.64";
      }
      leaf rx-system-value-echo {
        type int32;
        config false;
        description
          "Returns the value of Tw_sys_tx that the remote system is
           advertising that it is requesting in the receive direction
           and is echoed by the local system under the control of the
           EEE DLL transmitter state diagram.";
        reference
          "IEEE Std 802.3, 30.12.2.1.65";
      }
      leaf fallback-system-value {
        type int32;
        config false;
        description
          "Returns the value of the fallback Tw_sys_tx that the local
           system is advertising to the remote system.";
        reference
          "IEEE Std 802.3, 30.12.2.1.66";
      }
      /* LDP EEE objects */
      leaf tx-dll-ready {
        type boolean;
        config false;
        description
          "Returns the initialization status of the EEE transmit Data
           Link Layer management function on the local system.";
        reference
          "IEEE Std 802.3, 30.12.2.1.67";
      }
      leaf rx-dll-ready {
        type boolean;
        config false;
        description
          "Returns the initialization status of the EEE receive Data
           Link Layer management function on the local system.";
```

```
        reference
          "IEEE Std 802.3, 30.12.2.1.68";
      }
      leaf dll-enabled {
        type boolean;
        config false;
        description
          "Returns the status of the EEE capability negotiation on
           the local system.";
        reference
          "IEEE Std 802.3, 30.12.2.1.69";
      }
      leaf tx-system-fw {
        type boolean;
        config false;
        description
          "Returns the value of LPI_FW that the local system can
           support in the transmit direction.";
        reference
          "IEEE Std 802.3, 30.12.2.1.70";
      }
      leaf tx-system-fw-echo {
        type boolean;
        config false;
        description
          "Returns the value of LPI_FW that the remote system is
           advertising that it can support in the transmit direction
           and is echoed by the local system under the control of the
           EEE DLL receiver state diagram.";
        reference
          "IEEE Std 802.3, 30.12.2.1.71";
      }
      leaf rx-system-fw {
        type boolean;
        config false;
        description
          "Returns the value of LPI_FW that the local system is
           requesting in the receive direction.";
        reference
          "IEEE Std 802.3, 30.12.2.1.72";
      }
      leaf rx-system-fw-echo {
        type boolean;
        config false;
        description
          "Returns the value of LPI_FW that the remote system is
           advertising that it is requesting in the receive direction
           and is echoed by the local system under the control of the
           EEE DLL transmitter state diagram.";
        reference
          "IEEE Std 802.3, 30.12.2.1.73";
      }
      /* LLDP preemption objects */
      leaf preemption-supported {
        type boolean;
        config false;
        description
          "Indicates whether the given port (associated with the local
           System) supports the preemption capability.";
```

```
        reference
          "IEEE Std 802.3, 30.12.2.1.74";
      }
      leaf preemption-enabled {
        type boolean;
        config false;
        description
          "Indicates whether the preemption capability is enabled on
           the given port associated with the local System.";
        reference
          "IEEE Std 802.3, 30.12.2.1.75";
      }
      leaf preemption-active {
        type boolean;
        config false;
        description
          "Indicates whether the preemption capability is active on
           the given port associated with the local System.;";
        reference
          "IEEE Std 802.3, 30.12.2.1.76";
      }
      leaf additional-fragment-size {
        type int32;
        config false;
        description
          "Indicate the minimum size of non-final fragments supported
           by the receiver on the given port associated with the local
           System. This value is expressed in units of 64 octets of
           additional fragment length.";
        reference
          "IEEE Std 802.3, 30.12.2.1.77";
      }
    }

    augment "/lldp:lldp/lldp:port" {
      description
        "Augments port with 802.3 port read-write objects";
      leaf tlvs-port-config-enable {
        type bits {
          bit mac-phy-config-status {
            position 0;
            description
              "IEEE Std 802.3, 30.12.1.1.1";
          }
          bit power-via-mdi {
            position 1;
            description
              "IEEE Std 802.3, 30.12.1.1.1";
          }
          bit unused {
            position 2;
            description
              "IEEE Std 802.3, 30.12.1.1.1";
          }
          bit max-frame-size {
            position 3;
            description
              "IEEE Std 802.3, 30.12.1.1.1";
          }
```

```
        bit eee-tlv {
          position 4;
          description
            "IEEE Std 802.3, 30.12.1.1.1";
        }
        bit eee-fast-wake-tlv {
          position 5;
          description
            "IEEE Std 802.3, 30.12.1.1.1";
        }
        bit additional-ethernet-capabilities-tlv {
          position 6;
          description
            "IEEE Std 802.3, 30.12.1.1.1";
        }
      }
      description
        "Bitmap that corresponds to an IEEE 802.3 subtype
         associated with a specific IEEE 802.3 port config TLV";
      reference
        "IEEE Std 802.3, 30.12.1.1.1";
    }
    leaf local-power-priority {
      type power-priority-type;
      description
        "Priority of a PD system. For a PSE, this is the priority
         that the PSE assigns to the PD. For a PD, this is the
         priority that the PD requests from the PSE.";
      reference
        "IEEE Std 802.3, 30.12.2.1.16";
    }
    leaf local-measurement-source {
      type measurement-source-type;
      description
        "A read-write attribute value that indicates to local
         device on which Alternative or Mode the measurement
         is to be taken.";
      reference
        "IEEE Std 802.3, 30.12.2.1.42";
    }
  }

  /* NOTE - lldp:remote-systems-data container is "config false;" */

  augment "/lldp:lldp/lldp:port/lldp:remote-systems-data" {
    description
      "Augments port with 802.3 port config tlvs";
    leaf auto-negotiation-supported {
      type boolean;
      description
        "True if the port supports Auto-negotiation";
      reference
        "IEEE Std 802.3, 30.12.3.1.1";
    }
    leaf auto-negotiation-enabled {
      type boolean;
      description
        "True if Auto-negotiation is enabled";
      reference
```

```
        "IEEE Std 802.3, 30.12.3.1.2";
    }
    /* Obsolete */
    leaf auto-negotiation-cap {
      type binary {
        length "2";
      }
      status obsolete;
      description
        "A read-only 2-octet value that contains the value (bitmap)
         of the ifMauAutoNegCapAdvertisedBits object (defined in
         IETF RFC 4836)which is associated with the given port on
         the local system.
         Supersceeded by auto-negotiation-cap-bits.";
      reference
        "IEEE Std 802.3, 30.12.3.1.3";
    }
    leaf operational-mau-type {
      type int32;
      status obsolete;
      description
        "32-bit integer value that indicates the operational MAU
         type of the given port.
         Supersceeded by operational-pmd-type.";
      reference
        "IEEE Std 802.3, 30.12.3.1.4";
    }
    leaf auto-negotiation-cap-bits {
      type ieee802-phy:auto-neg-ability-bits;
      description
        "The capabilities advertised by the remote Auto-Negotiation
         entity.";
      reference
        "IEEE Std 802.3, 30.12.3.1.3,
         aAutoNegadvertisedTechnologyAbility.";
    }
    leaf operational-pmd-type {
      type identityref {
        base ieee802-phy:pmd-type;
      }
      description
        "The operational PMD type of the given port";
      reference
        "IEEE Std 802.3, 30.12.3.1.4";
    }
    leaf power-port-class {
      type port-class-type;
      description
        "A read-only value that identifies the port Class of the
         given port";
      reference
        "IEEE Std 802.3, 30.12.3.1.5";
    }
    leaf mdi-power-supported {
      type boolean;
      description
        "True if MDI power is supported";
      reference
        "IEEE Std 802.3, 30.12.3.1.6";
```

```
    }
    leaf mdi-power-enabled {
      type boolean;
      description
        "True if MDI power is enabled";
      reference
        "IEEE Std 802.3, 30.12.3.1.7";
    }
    leaf power-pair-controlable {
      type boolean;
      description
        "True if pair selection can be controlled";
      reference
        "IEEE Std 802.3, 30.12.3.1.8";
    }
    leaf power-pairs {
      type pse-pinout-type;
      description
        "Indicates which pinout alernative is used for PD
         detection and power";
      reference
        "IEEE Std 802.3, 30.12.3.1.9";
    }
    leaf power-class {
      type pse-power-class-type;
      description
        "PD Power Class";
      reference
        "IEEE Std 802.3, 30.12.3.1.10";
    }
    /* Obsolete Link Aggregation objects
        Link aggregation has not been part of 802.3 since IEEE 802.1AX-2008
        */
    leaf link-aggregation-status {
      type bits {
        bit aggregation-capability {
          position 0;
          description
            "IEEE Std 802.3, 79.3.3.1";
        }
        bit aggregation-status {
          position 1;
          description
            "IEEE Std 802.3, 79.3.3.1";
        }
        bit bit2-reserved {
          position 2;
          description
            "IEEE Std 802.3, 79.3.3.1";
        }
        bit bit3-reserved {
          position 3;
          description
            "IEEE Std 802.3, 79.3.3.1";
        }
        bit bit4-reserved {
          position 4;
          description
            "IEEE Std 802.3, 79.3.3.1";
```

```
          }
          bit bit5-reserved {
            position 5;
            description
              "IEEE Std 802.3, 79.3.3.1";
          }
          bit bit6-reserved {
            position 6;
            description
              "IEEE Std 802.3, 79.3.3.1";
          }
          bit bit7-reserved {
            position 7;
            description
              "IEEE Std 802.3, 79.3.3.1";
          }
        }
        status obsolete;
        description
          "The bitmap value which contains the link aggregation
           capabilities and the current aggregation status
           of the link";
        reference
          "IEEE Std 802.3, 30.12.3.1.11";
      }
      leaf aggregation-port-id {
        type int32;
        status obsolete;
        description
          "The unique identifier allocated to this Aggregation Port
           by the local System.";
        reference
          "IEEE Std 802.3, 30.12.3.1.12";
      }
      leaf local-max-frame-size {
        type int32;
        status obsolete;
        description
          "An integer value indicating the maximum supported frame
           size in octets on the given port of the local system.";
        reference
          "IEEE Std 802.3, 30.12.3.1.13";
      }
      /* POE objects
          Consistency errors between PoE definitions in 30.9 (PoE),
          30.12 (LLDP), 79.3.2 (PoE TLV), 79.3.8 (PoE measurement TLV).
          Need to get these clauses reviewed against clause 145 and each other
          */
      leaf power-type {
        type bits {
          bit type1-or-greater {
            position 0;
            description
              "0-type1, 1-greater than type1";
          }
          bit pse-or-pd {
            position 1;
            description
              "0-pse, 1-pd";
```

```
          }
        }
        status obsolete;
        description
          "A read-only attribute that returns a bit string indicating
           whether the local system is a PSE or a PD and whether it
           is Type 1 or greater than Type 1. The first bit indicates
           Type 1 or greater than Type 1. The second bit indicates
           PSE or PD. A PSE sets this bit to indicate a PSE. A PD
           sets this bit to indicate a PD. See also
           aLldpXdot3LocPowerTypeExt.";
        reference
          "IEEE Std 802.3, 30.12.3.1.14";
      }
      leaf power-source {
        type power-source-type;
        description
          "Indicates the power sources of the remote system. A PSE
           indicates whether it is being powered by a primary power
           source; a backup power source; or unknown. A PD indicates
           whether it is being powered by a PSE and locally; by a PSE
           only; or unknown.;";
        reference
          "IEEE Std 802.3, 30.12.3.1.15";
      }
      leaf power-priority {
        type power-priority-type;
        description
          "The priority of the PD system received from the remote
           system";
        reference
          "IEEE Std 802.3, 30.12.3.1.16";
      }
      leaf pd-requested-power-value {
        type int32;
        description
          "PD requested power value that was used by the remote
           system to compute the power value that is has currently
           allocated to the PD.";
        reference
          "IEEE Std 802.3, 30.12.3.1.17";
      }
      leaf pd-requested-power-value-a {
        type int32;
        description
          "A read-only attribute that returns the PD requested power
           value for the Mode A pairset that was used by the remote
           system to compute the power value that it has currently
           allocated to the PD. For a PSE, it is the PD requested
           power value for the Alternative A pairset received from
           the remote system. For a PD, it is the PD requested power
           value for the Alternative A pairset that the PSE echoes
           back to the remote system. The definition and encoding of
           PD requested power value for the Mode A pairset is the same
           as described in aLldpXdot3LocPDRequestedPowerValueA";
        reference
          "IEEE Std 802.3, 30.12.3.1.18";
      }
      leaf pd-requested-power-value-b {
```

```
        type int32;
        description
          "A read-only attribute that returns the PD requested power
           value for the Mode B pairset that was used by the remote
           system to compute the power value that it has currently
           allocated to the PD. For a PSE, it is the PD requested
           power value for the Alternative B pairset received from
           the remote system. For a PD, it is the PD requested power
           value for the Alternative B pairset that the PSE echoes
           back to the remote system. The definition and encoding of
           PD requested power value for the Mode B pairset is the
           same as described in aLldpXdot3LocPDRequestedPowerValueB";
        reference
          "IEEE Std 802.3, 30.12.3.1.19";
      }
      leaf pse-allocated-power-value {
        type int32;
        description
          "PSE allocated power value. For a PSE, it is the power value
           that the PSE has currently allocated to the remote system.
           For a PD, it is the power value that the PD mirrors back to
           the remote syste";
        reference
          "IEEE Std 802.3, 30.12.3.1.20";
      }
      leaf pse-allocated-power-value-a {
        type int32;
        description
          "A read-only attribute that returns the PSE allocated power
           value for the Alternative A pairset received from the
           remote system. For a PSE, it is the PSE allocated power
           value for the Alternative A pairset that was echoed back
           by the remote PD. For a PD, it is the PSE allocated power
           value for the Mode A pairset received from the remote
           system. The definition and encoding of PSE allocated power
           value for the Alternative A pairset is the same as
           described in aLldpXdot3LocPSEAllocatedPowerValueA";
        reference
          "IEEE Std 802.3, 30.12.3.1.21";
      }
      leaf pse-allocated-power-value-b {
        type int32;
        description
          "A read-only attribute that returns the PSE allocated power
           value for the Alternative B pairset received from the
           value for the Alternative B pairset that was echoed back
           by the remote PD. For a PD, it is the PSE allocated power
           value for the Mode B pairset received from the remote
           system. The definition and encoding of PSE allocated power
           value for the Alternative B pairset is the same as
           described in aLldpXdot3LocPSEAllocatedPowerValueB";
        reference
          "IEEE Std 802.3, 30.12.3.1.22";
      }
      leaf pse-powering-status {
        type powering-status-type;
        description
          "A read only value that indicates the powering status of
           the remote PSE. For a PD, the contents of this attribute
```

```
          are undefined.";
        reference
          "IEEE Std 802.3, 30.12.3.1.23";
      }
      leaf pd-powered-status {
        type powered-status-type;
        description
          "A read only value that indicates the powering status of
           the PD. For a PSE, the contents of this attribute are
           undefined";
        reference
          "IEEE Std 802.3, 30.12.3.1.24";
      }
      leaf power-pairs-ext {
        type power-pairs-type;
        description
          "A read-only value that identifies the supported PSE
           Pinout Alternative specified in 145.2.4. For a PD, this
           attribute contains the value of the aPSEPowerPairs
           attribute (see 30.9.1.1.4). For a PSE, the contents of
           this attribute are undefined";
        reference
          "IEEE Std 802.3, 30.12.3.1.25";
      }
      leaf power-class-ext-A {
        type power-class-ext-AB-type;
        description
          "For a dual-signature PD, a read-only value that indicates
           the currently assigned Class for Mode A by the remote
           4-pair PSE. For a single-signature PD or a dual-signature
           PD connected to a 2-pair only PSE, a read-only value set
           to â€šinglesigâ€™ by the remote PSE. For a PSE connected to a
           dual-signature PD, a read-only value that indicates the
           requested Class for Mode A during Physical Layer
           classification (see 145.2.8) by the remote PD. For a PSE
           connected to a single-signature PD, a read-only value set
           to â€šinglesigâ€™ by the remote PD";
        reference
          "IEEE Std 802.3, 30.12.3.1.26";
      }
      leaf power-class-ext-B {
        type power-class-ext-AB-type;
        description
          "For a dual-signature PD, a read-only value that indicates
           the currently assigned Class for Mode B by the remote
           4-pair PSE. For a single-signature PD or a dual-signature
           PD connected to a 2-pair only PSE, a read-only value set
           to â€šinglesigâ€™ by the remote PSE. For a PSE connected to a
           dual-signature PD, a read-only value that indicates the
           requested Class for Mode B during Physical Layer
           classification (see 145.2.8) by the remote PD. For a PSE
           connected to a single-signature PD, a read-only value set
           to â€šinglesigâ€™ by the remote PD";
        reference
          "IEEE Std 802.3, 30.12.3.1.27";
      }
      leaf power-class-ext {
        type power-class-ext-type;
        description
```

```
          "For a single-signature PD or a dual-signature PD connected
           to a 2-pair only PSE, a read-only value that indicates the
           currently assigned Class by the remote PSE. For a
           dual-signature PD connected to a 4-pair capable PSE, a
           read-only value set to â€đualsigâ€™ by the remote PSE. For a
           PSE connected to a single-signature PD, a read-only value
           that indicates the requested Class during Physical Layer
           classification (see 145.2.8) by the remote PD. For a PSE
           connected to a dual-signature PD, a read-only value set to
           â€đualsigâ€™ by the remote PD.";
        reference
          "IEEE Std 802.3, 30.12.3.1.28";
      }
      leaf power-type-ext {
        type power-type;
        description
          "A read-only attribute that returns a value to indicate if
           the remote system is a Type 3 or Type 4 PSE or PD and, in
           the case of a Type 3 or Type 4 PD, if it is a
           single-signature PD or dual-signature PD.";
        reference
          "IEEE Std 802.3, 30.12.3.1.29";
      }
      leaf pd-load {
        type boolean;
        description
          "For a PSE, a read-only attribute that returns whether the
           load of the remote dual-signature PD is electrically
           isolated, as defined in 79.3.2.10.2. For a PD, this
           attribute is set to FALSE.";
        reference
          "IEEE Std 802.3, 30.12.3.1.30";
      }
      leaf pd-4pid {
        type boolean;
        description
          "A read-only Boolean attribute indicating whether the
           remote PD system supports powering of both PD Modes.";
        reference
          "IEEE Std 802.3, 30.12.3.1.31";
      }
      leaf pse-max-avail-power {
        type int32;
        description
          "A read-only attribute that returns the remote PSE maximum
           available power value in units of 0.1 W";
        reference
          "IEEE Std 802.3, 30.12.3.1.32";
      }
      leaf pse-autoclass-support {
        type boolean;
        description
          "Indicates whether the remote PSE system supports
           Autoclass.";
        reference
          "IEEE Std 802.3, 30.12.3.1.33";
      }
      leaf autoclass-completed {
        type boolean;
```

```
      description
        "Indicates whether the remote PSE system has completed the
         Autoclass measurement.";
      reference
        "IEEE Std 802.3, 30.12.3.1.34";
    }
    leaf autoclass-request {
      type boolean;
      description
        "A read-only Boolean attribute indicating whether the
         remote PD system is requesting an Autoclass measurement.";
      reference
        "IEEE Std 802.3, 30.12.3.1.35";
    }
    leaf power-down-request {
      type int32;
      description
        "A read-write attribute that indicates the remote PD system
         is requesting a power down when the value is  0x1D.;";
      reference
        "IEEE Std 802.3, 30.12.3.1.36";
    }
    leaf power-down-time {
      type int32;
      description
        "A read-only attribute that indicates the number of seconds
         the remote PD requests to stay powered off. A value of zero
         indicates an indefinite amount of time";
      reference
        "IEEE Std 802.3, 30.12.3.1.37";
    }
    /* PoE measurement objects */
    leaf meas-voltage-support {
      type boolean;
      description
        "A read-only attribute that indicates the remote device is
         capable of providing a voltage measurement.;";
      reference
        "IEEE Std 802.3, 30.12.3.1.38";
    }
    leaf meas-current-support {
      type boolean;
      description
        "A read-only attribute that indicates the remote device is
         capable of providing a current measurement.;";
      reference
        "IEEE Std 802.3, 30.12.3.1.39";
    }
    leaf meas-power-support {
      type boolean;
      description
        "A read-only attribute that indicates the remote device is
         capable of providing a power measurement.;";
      reference
        "IEEE Std 802.3, 30.12.3.1.40";
    }
    leaf meas-energy-support {
      type boolean;
      description
```

```
          "A read-only attribute that indicates the remote device is
           capable of providing a energy measurement.;";
        reference
          "IEEE Std 802.3, 30.12.3.1.41 ";
      }
      leaf measurement-source {
        type bits {
          bit bit1 {
            position 0;
            description
              "-";
          }
          bit bit2 {
            position 1;
            description
              "-";
          }
        }
        status obsolete;
        description
          "A read-write attribute value that indicates on which
           Alternative or Mode the measurement was taken by the remote
           device.
           Superseded by rem-measurement-source.";
        reference
          "IEEE Std 802.3, 30.12.3.1.42";
      }
      leaf rem-measurement-source {
        type measurement-source-type;
        description
          "A read-only attribute that indicates which Alternative or
           Mode the measurement is to be taken";
        reference
          "IEEE Std 802.3, 30.12.3.1.42";
      }
      leaf meas-voltage-request {
        type boolean;
        description
          "A read-only attribute that indicates the remote device is
           requesting a voltage measurement from the local device.";
        reference
          "IEEE Std 802.3, 30.12.3.1.43";
      }
      leaf meas-current-request {
        type boolean;
        description
          "A read-only attribute that indicates the remote device is
           requesting a current measurement from the local device.";
        reference
          "IEEE Std 802.3, 30.12.3.1.44";
      }
      leaf meas-power-request {
        type boolean;
        description
          "A read-only attribute that indicates the remote device is
           requesting a power measurement from the local device.";
        reference
          "IEEE Std 802.3, 30.12.3.1.45";
      }
```

```
      leaf meas-energy-request {
        type boolean;
        description
          "A read-only attribute that indicates the remote device is
           requesting an energy measurement from the local device.";
        reference
          "IEEE Std 802.3, 30.12.3.1.46";
      }
      leaf meas-voltage-valid {
        type boolean;
        description
          "A read-only attribute that indicates the remote deviceâ€™s
           voltage measurement is valid.";
        reference
          "IEEE Std 802.3, 30.12.3.1.47";
      }
      leaf meas-current-valid {
        type boolean;
        description
          "A read-only attribute that indicates the remote deviceâ€™s
           current measurement is valid.";
        reference
          "IEEE Std 802.3, 30.12.3.1.48";
      }
      leaf meas-power-valid {
        type boolean;
        description
          "A read-only attribute that indicates the remote deviceâ€™s
           power measurement is valid.";
        reference
          "IEEE Std 802.3, 30.12.3.1.49";
      }
      leaf meas-energy-valid {
        type boolean;
        description
          "A read-only attribute that indicates the remote deviceâ€™s
           energy measurement is valid.";
        reference
          "IEEE Std 802.3, 30.12.3.1.50";
      }
      leaf meas-voltage-uncertainty {
        type int32;
        description
          "A read-only attribute that indicates the expanded
           uncertainty (coverage factor k = 2) for the remote deviceâ€™s
           voltage measurement.";
        reference
          "IEEE Std 802.3, 30.12.3.1.51";
      }
      leaf meas-current-uncertainty {
        type int32;
        description
          "A read-only attribute that indicates the expanded
           uncertainty (coverage factor k = 2) for the remote deviceâ€™s
           current measurement.";
        reference
          "IEEE Std 802.3, 30.12.3.1.52";
      }
      leaf meas-power-uncertainty {
```

```
        type int32;
        description
          "A read-only attribute that indicates the expanded
           uncertainty (coverage factor k = 2) for the remote deviceâ€™s
           power measurement.";
        reference
          "IEEE Std 802.3, 30.12.3.1.53";
      }
      leaf meas-energy-uncertainty {
        type int32;
        description
          "A read-only attribute that indicates the expanded
           uncertainty (coverage factor k = 2) for the remote deviceâ€™s
           energy measurement.";
        reference
          "IEEE Std 802.3, 30.12.3.1.54";
      }
      leaf voltage-measurement {
        type int32;
        description
          "A read-only attribute that returns the measured remote
           device voltage.";
        reference
          "IEEE Std 802.3, 30.12.3.1.55";
      }
      leaf current-measurement {
        type int32;
        description
          "A read-only attribute that returns the measured remote
           device current.";
        reference
          "IEEE Std 802.3, 30.12.3.1.56";
      }
      leaf power-measurement {
        type int32;
        description
          "A read-only attribute that returns the measured remote
           device power.";
        reference
          "IEEE Std 802.3, 30.12.3.1.57";
      }
      leaf energy-measurement {
        type int32;
        description
          "A read-only attribute that returns the measured remote
           device energy.";
        reference
          "IEEE Std 802.3, 30.12.3.1.58";
      }
      leaf pse-power-price-index {
        type int32;
        description
          "A read-only attribute that returns an index of the price
           of power being sourced by the remote PSE. For a PSE, this
           value is undefined.";
        reference
          "IEEE Std 802.3, 30.12.3.1.59";
      }
      leaf tx-system-value {
```

```
        type int32;
        description
          "Returns the value of Tw_sys_tx that the remote system can
           support in the transmit direction.";
        reference
          "IEEE Std 802.3, 30.12.3.1.60";
      }
      leaf tx-system-value-echo {
        type int32;
        description
          "Returns the value of Tw_sys_tx that the local system is
           advertising that it can support in the transmit direction
           and is echoed by the local system under the control of the
           EEE DLL receiver state diagram.";
        reference
          "IEEE Std 802.3, 30.12.3.1.61";
      }
      leaf rx-system-value {
        type int32;
        description
          "Returns the value of Tw_sys_tx that the remote system is
           requesting in the receive direction.";
        reference
          "IEEE Std 802.3, 30.12.3.1.62";
      }
      leaf rx-system-value-echo {
        type int32;
        description
          "Returns the value of Tw_sys_tx that the local system is
           advertising that it is requesting in the receive direction
           and is echoed by the local system under the control of the
           EEE DLL transmitter state diagram.";
        reference
          "IEEE Std 802.3, 30.12.3.1.63 ";
      }
      leaf fallback-system-value {
        type int32;
        description
          "Returns the value of the fallback Tw_sys_tx that the
           remote system is advertising to the remote system.";
        reference
          "IEEE Std 802.3, 30.12.3.1.64";
      }
      /* LDP EEE objects */
      leaf tx-system-fw {
        type boolean;
        description
          "Returns the value of LPI_FW that the remote system can
           support in the transmit direction.";
        reference
          "IEEE Std 802.3, 30.12.3.1.65";
      }
      leaf tx-system-fw-echo {
        type boolean;
        description
          "Returns the value of LPI_FW that the local system is
           advertising that it can support in the transmit direction
           and is echoed by the local system under the control of the
           EEE DLL receiver state diagram.";
```

```
        reference
          "IEEE Std 802.3, 30.12.3.1.66";
      }
      leaf rx-system-fw {
        type boolean;
        description
          "Returns the value of LPI_FW that the remote system is
           requesting in the receive direction.";
        reference
          "IEEE Std 802.3, 30.12.3.1.67";
      }
      leaf rx-system-fw-echo {
        type boolean;
        description
          "Returns the value of LPI_FW that the local system is
           advertising that it is requesting in the receive direction
           and is echoed by the local system under the control of the
           EEE DLL transmitter state diagram.";
        reference
          "IEEE Std 802.3, 30.12.3.1.68";
      }
      /* LLDP preemption objects */
      leaf preemption-supported {
        type boolean;
        description
          "Indicates whether the given port (associated with the
           remote System) supports the preemption capability.";
        reference
          "IEEE Std 802.3, 30.12.3.1.69";
      }
      leaf preemption-enabled {
        type boolean;
        description
          "Indicates whether the preemption capability is enabled on
           the given port associated with the remote System.";
        reference
          "IEEE Std 802.3, 30.12.3.1.70";
      }
      leaf preemption-active {
        type boolean;
        description
          "Indicates whether the preemption capability is active on
           the given port associated with the remote System.;";
        reference
          "IEEE Std 802.3, 30.12.3.1.72";
      }
      leaf additional-fragment-size {
        type int32;
        description
          "Indicate the minimum size of non-final fragments supported
           by the receiver on the given port associated with the
           remote System. This value is expressed in units of 64
           octets of additional fragment length.";
        reference
          "IEEE Std 802.3, 30.12.3.1.72 ";
      }
    }
  }
```

5.3.2.5 Ethernet PHY

> ***Editor's Note (to be removed prior to publication):***
>
> Pretty printing of ieee802-ethernet-phy.yang file may change the appearance by adding whitespace and reformatting lines

> ***Editor's Note (to be removed prior to publication):***
>
> IEEE Std 802.3.1 and IEEE Std 802.3.2 to be updated at the publication time

```
module ieee802-ethernet-phy {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-phy";
  prefix ieee802-eth-phy;

  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991 YANG Types";
  }
  import iana-if-type {
    prefix ianaift;
    reference
      "IANA Interface Types";
  }
  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import ieee802-ethernet-phy-type {
    prefix ieee802-phy;
    reference
      "IEEE Std 802.3-2022";
  }

  organization
    "IEEE Std 802.3 Ethernet Working Group";
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
    "This module contains YANG definitions for  802.3 PHYs and PMDs.";

  revision 2025-04-17 {
    description
      "Initial revision derived from IEEE8023-MAU-MIB
       using smidump.";
    reference
      "IEEE Std 802.3-2022, unless dated explicitly";
```

```
  }

  feature fec-support {
    description
      "This object indicates support for one or more FEC modes.";
  }

  feature baset-snr-support {
    description
      "This object indicates support for SNR measurement for
       BASE-T interfaces.";
  }

  feature time-sync-support {
    description
      "This object indicates support for time syncronisation.";
  }

  feature pcs-lane-statistics-support {
    description
      "This object indicates support for PCS Lane statistics.";
  }

  feature auto-negotiation-support {
    description
      "This object indicates support for Auto-Negotiation.";
  }

  typedef snr-value {
    type int32 {
      range "-127..127";
    }
    units "0.1dB";
    description
      "SNR units of 0.1 dB to an accuracy of 0.5 dB within
       the range of -12.7 dB to 12.7 dB.";
  }

  typedef counter-1000-second {
    type yang:counter64;
    description
      "This counter increments no more than 1000 times per second.";
  }

  typedef counter-5000-second {
    type yang:counter64;
    description
      "This counter increments no more than 5000 times per second.";
  }

  typedef counter-pcs-lane {
    type yang:counter64;
    description
      "This counter has a maximum per second increment rate of:
```

```
        1.2 million for 1000 Mb/s
        5 million for 10 Gb/s and 40 Gb/s
        2.5 million for 100 Gb/s";
  }

  typedef enabled-state {
    type enumeration {
      enum unknown {
        description
          "Initializing.";
      }
      enum disabled {
        description
          "Disabled";
      }
      enum enabled {
        description
          "Enabled";
      }
    }
    description
      "Enabled, disabled or unknown.";
  }

  augment "/if:interfaces/if:interface" {
    when "derived-from-or-self(if:type,'ianaift:ethernetCsmacd')" {
      description
        "Applies to all Ethernet interfaces.";
    }
    description
      "Augment interface model with Ethernet interface
       information";
    container phy-table {
      description
        "Table of descriptive and status information
         about the Ethernet interface physical layer.";
      container control {
        description
          "Control information for the Ethernet interface
           physical layer.";
        leaf phy-reset {
          type boolean;
          description
            "Setting this to 'True' resets the interface in the
             same manner as would a power-off, power-on cycle of
             at least 0.5 s duration.";
          reference
            "IEEE Std 802.3, 30.5.1.2.1 acResetMAU";
        }
        leaf phy-admin-control {
          type enumeration {
            enum operational {
              description
                "Operational";
```

```
          }
          enum standby {
            description
              "Standby";
          }
          enum shutdown {
            description
              "Shutdown";
          }
        }
        description
          "The configured state of the interface. This object
           MAY be implemented as a read-only object by devices
           that do not implement software control of the interface
           state. Some devices may not support setting the value
           tof this object to some of the enumerated values.

           This can be set to “standby”, “operational”, and
           “shutdown” state, except that a “standby” action
           to a multidrop interface will cause the interface
           to enter the “shutdown” management state.";
        reference
          "IEEE Std 802.3, 30.5.1.2.2 acMAUAdminControl";
      }
      container fec-control {
        if-feature "fec-support";
        presence "The port supports one or more FEC modes.";
        description
          "FEC Control.";
        leaf fec-mode-control {
          type enumeration {
            enum disabled {
              description
                "FEC disabled";
            }
            enum enabled {
              description
                "FEC enabled";
            }
            enum BASE-R-enabled {
              description
                "BASE-R FEC enabled";
            }
            enum RS-FEC-enabled {
              description
                "RS-FEC enabled";
            }
            enum RS-FEC-Int-enabled {
              description
                "RS-FEC-Int enabled";
            }
          }
          description
            "The FEC mode across the MDI of the port.
```

```
              BASE-R-enabled, RS-FEC-enabled and
              RS-FEC-Int-enabled are only used for PHYs which
              support more than one type of FEC operation.
              When Clause 73 Auto-Negotiation is enabled, this
              leaf becomes read-only.";
           reference
             "IEEE Std 802.3, 30.5.1.1.16";
         }
         leaf rs-fec-bypass-control {
           type enabled-state;
           description
             "Is RS-FEC bypass enabled?";
           reference
             "IEEE Std 802.3, 30.5.1.1.31";
         }
       }
     }
     container status {
       config false;
       description
         "Status for the port.";
       leaf phy-type {
         type identityref {
           base ieee802-phy:phy-type;
         }
         description
           "This object identifies the PHY type.";
         reference
           "IEEE Std 802.3, 30.5.1.1.2, aPHYType.";
       }
       leaf pmd-type {
         type identityref {
           base ieee802-phy:pmd-type;
         }
         description
           "This object identifies the PMD type.";
         reference
           "IEEE Std 802.3, 30.5.1.1.2, aMAUType.";
       }
       leaf pmd-type-list {
         type ieee802-phy:pmd-type-list;
         description
           "PMD types this interface supports.";
       }
       leaf phy-status {
         type enumeration {
           enum unknown {
             description
               "Initializing, true state
                not yet known";
           }
           enum operational {
             description
               "Powered and connected";
```

```
            }
            enum standby {
              description
                "Inactive but on";
            }
            enum shutdown {
              description
                "Inactive and powered down";
            }
          }
          description
            "The current state of the interface.";
          reference
            "IEEE Std 802.3, 30.5.1.1.7, aMAUAdminState";
        }
        leaf phy-media-available {
          type ieee802-phy:media-available-type;
          description
            "This object reports if the media is availeable.";
          reference
            "IEEE Std 802.3, 30.5.1.1.4, aMediaAvailable.";
        }
        /* Jabber */
        leaf jabber-state {
          type enumeration {
            enum other {
              description
                "undefined";
            }
            enum unknown {
              description
                "initializing, true state
                 not yet known";
            }
            enum normal {
              description
                "state is true or normal";
            }
            enum fault {
              description
                "state is false, fault, or
                 abnormal";
            }
          }
          description
            "The JABBER state of the multidrop interface.";
          reference
            "IEEE Std 802.3, 30.5.1.1.6, aJabber.jabberFlag.";
        }
        leaf jabbering-state-enters {
          type yang:counter64;
          description
            "The number of times  this interface detected JABBER.";
          reference
```

```
            "IEEE Std 802.3,
             30.5.1.1.6, aJabber.jabberCounter.";
        }
        /* FEC */
        container fec-status {
          if-feature "fec-support";
          presence "The port supports one or more FEC modes.";
          description
            "FEC Status.";
          leaf fec-ability {
            type bits {
              bit other {
                description
                  "Other";
              }
              bit base-r {
                description
                  "BASE-R FEC supported";
              }
              bit rs-fec {
                description
                  "RS-FEC supported";
              }
              bit rs-fec-bypass {
                description
                  "RS-FEC bypass supported";
              }
              bit rs-fec-bypass-indication {
                description
                  "RS-FEC error indication bypass supported";
              }
              bit pcs-fec-bypass {
                description
                  "PCS FEC error bypass supported";
              }
              bit pcs-fec-bypass-indication {
                description
                  "PCS FEC error indication bypass supported";
              }
            }
            description
              "The FEC modes this port supports.";
            reference
              "IEEE Std 802.3, 30.5.1.1.15, 30.5.1.1.16
               30.5.1.1.29, 30.5.1.1.30, 30.5.1.1.33,
               30.5.1.1.34";
          }
          leaf fec-mode-status {
            type enumeration {
              enum unknown {
                description
                  "Initializing, true state not yet known.";
              }
              enum disabled {
```

```
                description
                  "FEC disabled";
              }
              enum enabled {
                description
                  "FEC enabled";
              }
              enum BASE-R-enabled {
                description
                  "BASE-R FEC enabled";
              }
              enum RS-FEC-enabled {
                description
                  "RS-FEC enabled";
              }
              enum RS-FEC-Int-enabled {
                description
                  "RS-FEC-Int enabled";
              }
            }
            description
              "The FEC mode of the PHY.
               BASE-R-enabled, RS-FEC-enabled and
               RS-FEC-Int-enabled are only used by PHYs which
               support more than one type of FEC operation.";
            reference
              "IEEE Std 802.3, 30.5.1.1.16";
          }
          leaf rs-fec-bypass-enabled-status {
            type enabled-state;
            description
              "Is RS-FEC bypass indication enabled?";
            reference
              "IEEE Std 802.3, 30.5.1.1.32";
          }
        }
        /* MultiGBASE-T SNR */
        container baset-snr {
          if-feature "baset-snr-support";
          presence "The port supports BASE-T SNR measurements.";
          description
            "BASE-T SNR Status.";
          leaf snr-op-margin-chnl-a {
            type snr-value;
            description
              "The current SNR operating margin on channel A
               on this port for MultiGBASE-T PMAs.";
            reference
              "IEEE Std 802.3, 30.5.1.1.19";
          }
          leaf snr-op-margin-chnl-b {
            type snr-value;
            description
              "The current SNR operating margin on channel B
```

```
             on this port for MultiGBASE-T PMAs.";
          reference
            "IEEE Std 802.3, 30.5.1.1.20";
        }
        leaf snr-op-margin-chnl-c {
          type snr-value;
          description
            "The current SNR operating margin on channel C
             on this port for MultiGBASE-T PMAs.";
          reference
            "IEEE Std 802.3, 30.5.1.1.21";
        }
        leaf snr-op-margin-chnl-d {
          type snr-value;
          description
            "The current SNR operating margin on channel CD
             on this port for MultiGBASE-T PMAs.";
          reference
            "IEEE Std 802.3, 30.5.1.1.22";
        }
      }
      /* TimeSync */
      container time-sync {
        if-feature "time-sync-support";
        presence "The port supports Time Sync.";
        description
          "Time Sync Status.";
        leaf time-sync-ability {
          type bits {
            bit transmit {
              description
                "Transmit";
            }
            bit receive {
              description
                "Receive";
            }
          }
          description
            "The TimeSync capabities of this port.";
          reference
            "IEEE Std 802.3, 30.13.1.1, 30.13.1.2";
        }
        leaf time-sync-delay-tx-max {
          type uint32;
          units "nano-seconds";
          description
            "The maximum transmit data delay between
             the xMII input to the MDI output.";
          reference
            "IEEE Std 802.3, 30.13.1.3";
        }
        leaf time-sync-delay-tx-min {
          type uint32;
```

```
            units "nano-seconds";
            description
              "The minimum transmit data delay between
               the xMII input to the MDI output.";
            reference
              "IEEE Std 802.3, 30.13.1.4";
          }
          leaf time-sync-delay-rx-max {
            type int32;
            units "nano-seconds";
            description
              "The maximum receive data delay between
               the MDI input to the xMII output.";
            reference
              "IEEE Std 802.3, 30.13.5";
          }
          leaf time-sync-delay-rx-min {
            type int32;
            units "nano-seconds";
            description
              "The minimum receive data delay between
               the MDI input to the xMII output.";
            reference
              "IEEE Std 802.3, 30.13.5";
          }
        }
      }
      container statistics {
        config false;
        description
          "Counters for the port.";
        leaf phy-media-losts {
          type yang:counter64;
          description
            "The number of times phy-media-available changed
             from 'available'.";
          reference
            "IEEE Std 802.3, 30.5.1.1.5 aLoseMediaCounter.";
        }
        leaf jabbering-state-enters {
          type yang:counter64;
          description
            "The number of times this interface detected JABBER.";
          reference
            "IEEE Std 802.3, 30.5.1.1.6, aJabber.jabberCounter.";
        }
        leaf false-carriers {
          type yang:counter64;
          description
            "The number of false carrier
             events during IDLE in 100BASE-X and
             1000BASE-X links.

             For all other interface types, this counter
```

```
              will always indicate zero.

              It can increment at a maximum rate of once
              per 100 ms for 100BASE-X and once per 10us
              for 1000BASE-X.";
           reference
             "IEEE Std 802.3, 30.5.1.1.10, aFalseCarriers.";
         }
         leaf pcs-coding-violations {
           type yang:counter64;
           description
             "The number of PCS coding violations for 100Mb/s and
              1000Mb/s ports.

              This counter increments at a maximum rate of 25 thousand
              per second for 100 Mb/s implementations and
              125 thousand per second for 1000 Mb/s
              implementations.

              The contents of this
              attribute is undefined when FEC is operating.";
           reference
             "IEEE Std 802.3, 30.5.1.1.14";
         }
         leaf fast-retrain-count-local {
           type counter-1000-second;
           description
             "The number of fast retrains initiated
              by the local device.";
           reference
             "IEEE Std 802.3, 30.5.1.1.24";
         }
         leaf fast-retrain-count-peer {
           type counter-1000-second;
           description
             "The number of fast retrains initiated
              by the link partner.";
           reference
             "IEEE Std 802.3, 30.5.1.1.25";
         }
       }
       container pcs-lane-statistics {
         if-feature "pcs-lane-statistics-support";
         config false;
         description
           "Per PCS lane counters for the port.
            These counters apply to the following:
            1000BASE-PX,
            10/25/40/50/100/200/400GBASE-R
            100GBASE-P
            10GBASE-PR
            10/1GBASE-PRX";
         list pcs-lane {
           key "pcs-lane-index";
```

```
          description
            "Per PCS lane counters for the port.";
          leaf pcs-lane-index {
            type uint32 {
              range "0..255";
            }
            description
              "PCS lane index.";
          }
          leaf fec-corrected-blocks {
            type counter-pcs-lane;
            description
              "The number blocks that FEC corrected.";
            reference
              "IEEE Std 802.3, 30.5.1.1.17";
          }
          leaf fec-uncorrected-blocks {
            type counter-pcs-lane;
            description
              "The number blocks that FEC could not correct.";
            reference
              "IEEE Std 802.3, 30.5.1.1.18";
          }
        }
      }
    }
    container auto-neg-table {
      if-feature "auto-negotiation-support";
      description
        "Configuration and status objects for Auto-Negotiation
         on Ethernet interfaces.";
      container control {
        description
          "Auto-Negotiation control.";
        leaf auto-neg-control {
          type boolean;
          default "true";
          description
            "Enable/disable Auto-Negotiation for this interface.";
          reference
            "IEEE Std 802.3, 30.6.1.1.2, acAutoNegAdminControl.";
        }
        leaf auto-neg-restart {
          type boolean;
          description
           "Setting this to true restarts Auto-Negotiation if enabled.";
          reference
            "IEEE Std 802.3,
             30.6.1.2.1, acAutoNegRestartAutoConfig.";
        }
      }
      container status {
        config false;
        description
```

```
            "Auto-Negotiation status.";
          leaf auto-neg-enabled {
            type boolean;
            description
              "Is Auto-Negotiation enabled for this interface.";
            reference
              "IEEE Std 802.3,
               30.6.1.1.2 aAutoNegAdminState.";
          }
          leaf auto-neg-remote-signaling {
            type boolean;
            description
              "Is the link partner using Auto-Negotiation?";
            reference
              "IEEE Std 802.3, 30.6.1.1.3, aAutoNegRemoteSignaling.";
          }
          leaf auto-neg-status {
            type enumeration {
              enum other {
                description
                  "Unknown state.";
              }
              enum configuring {
                description
                  "Auto-Negotiation is active.";
              }
              enum complete {
                description
                  "Auto-Negotiation is complete.";
              }
              enum disabled {
                description
                  "Auto-Negotiation s disabled.";
              }
              enum parallelDetectFail {
                description
                  "Auto-Negotiation failed because the
                   link partner does not suport, or has
                   disabled, Auto-Negotiation.";
              }
            }
            description
              "The current status of Auto-Negotiation.";
            reference
              "IEEE Std 802.3, 30.6.1.1.4, aAutoNegAutoConfig.";
          }
          leaf auto-neg-capability-bits {
            type ieee802-phy:auto-neg-ability-bits;
            description
              "The capabilities of the local Auto-Negotiation entity.";
            reference
              "IEEE Std 802.3, 30.6.1.1.5,
               aAutoNegLocalTechnologyAbility.";
          }
```

```
        leaf auto-neg-cap-advertised-bits {
          type ieee802-phy:auto-neg-ability-bits;
          description
            "The capabilities advertised by the local
             Auto-Negotiation entity.";
          reference
            "IEEE Std 802.3, 30.6.1.1.6,
             aAutoNegadvertisedTechnologyAbility.";
        }
        leaf auto-neg-cap-received-bits {
          type ieee802-phy:auto-neg-ability-bits;
          description
            "Auto-Negotiation capabilities received from
             the link partner.";
          reference
            "IEEE Std 802.3, 30.6.1.1.7,
             aAutoNegreceivedTechnologyAbility.";
        }
      }
    }
  }
}
```

5.3.2.6 Ethernet PHY Types

> ***Editor's Note (to be removed prior to publication):***
>
> Pretty printing of ieee802-ethernet-phy-type.yang file may change the appearance by adding whitespace and reformatting lines

> ***Editor's Note (to be removed prior to publication):***
>
> IEEE Std 802.3.1 and IEEE Std 802.3.2 to be updated at the publication time

```
module ieee802-ethernet-phy-type {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-phy-type";
  prefix ieee802-eth-phy-type;

  organization
    "IEEE Std 802.3 Ethernet Working Group
     Web URL: http://www.ieee802.org/3/";
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
    "This module contains YANG definitions for Physical Layer device
        (PHY) and Physical Medium Dependent(PMD) indentities and related
        types.";

  revision 2025-04-17 {
    description
      "Initial revision derived from IANA-MAU-MIB REVISION
       202309070000Z using smidump.";
```

```
    reference
      "IEEE Std 802.3-2022, unless dated explicitly";
  }

  /* PHY type identities */

  identity phy-type {
    description
      "Base type for PHY types.";
    reference
      "IEEE Std 802.3, Clause 30.3.2.1.2 aPhyType";
  }

  identity phy-type-other {
    base phy-type;
    description
      "Undefined";
  }

  identity phy-type-unknown {
    base phy-type;
    description
      "Initializing, true state or type not yet known";
  }

  identity phy-type-none {
    base phy-type;
    description
      "MII present and nothing connected";
  }

  identity phy-type-10BASE-T {
    base phy-type;
    description
      "Clause 14 10 Mb/s PAM3";
  }

  identity phy-type-10BASE-T1L {
    base phy-type;
    description
      "Clause 146 10 Mb/s PAM3";
  }

  identity phy-type-10BASE-T1S {
    base phy-type;
    description
      "Clause 147 10 Mb/s DME";
  }

  identity phy-type-100BASE-T1 {
    base phy-type;
    description
      "Clause 96 100 Mb/s PAM3";
  }
```

```
  identity phy-type-100BASE-X {
    base phy-type;
    description
      "Clause 24 or subclause 66.1 100 Mb/s 4B/5B";
  }

  identity phy-type-1000BASE-H {
    base phy-type;
    description
      "Clause 115 1000 Mb/s PAM16-THP";
  }

  identity phy-type-1000BASE-T {
    base phy-type;
    description
      "Clause 40 1000 Mb/s 4D-PAM5";
  }

  identity phy-type-1000BASE-T1 {
    base phy-type;
    description
      "Clause 97 1000 Mb/s PAM3";
  }

  identity phy-type-1000BASE-X {
    base phy-type;
    description
      "Clause 36 or subclause 66.2 1000 Mb/s 8B/10B";
  }

  identity phy-type-2.5GBASE-T {
    base phy-type;
    description
      "Clause 126 2.5 Gb/s PAM16";
  }

  identity phy-type-2.5GBASE-T1 {
    base phy-type;
    description
      "Clause 149 2.5 Gb/s PAM4";
  }

  identity phy-type-2.5GBASE-X {
    base phy-type;
    description
      "Clause 127 2.5 Gb/s 8B/10B";
  }

  identity phy-type-5GBASE-R {
    base phy-type;
    description
      "Clause 129 5 Gb/s 64/66B";
  }
```

```
  identity phy-type-5GBASE-T {
    base phy-type;
    description
      "Clause 126 5 Gb/s PAM16";
  }

  identity phy-type-5GBASE-T1 {
    base phy-type;
    description
      "Clause 149 5 Gb/s PAM4";
  }

  identity phy-type-10G-1GBASE-PRX {
    base phy-type;
    description
      "Clause 76 10/1G-EPON 10 Gb/s 64B/66B downstream and
       1 Gb/s 8B/10B upstream";
  }

  identity phy-type-10GBASE-PR {
    base phy-type;
    description
      "Clause 76 10/10G-EPON 10 Gb/s 64B/66B";
  }

  identity phy-type-10GBASE-R {
    base phy-type;
    description
      "Clause 49 10 Gb/s 64B/66B";
  }

  identity phy-type-10GBASE-T {
    base phy-type;
    description
      "Clause 55 10 Gb/s DSQ128";
  }

  identity phy-type-10GBASE-T1 {
    base phy-type;
    description
      "Clause 149 10 Gb/s PAM4";
  }

  identity phy-type-10GBASE-X {
    base phy-type;
    description
      "Clause 48 10 Gb/s 4 lane 8B/10B";
  }

  identity phy-type-10GPASS-XR {
    base phy-type;
    description
      "Clause 101 PCS up to 10 Gb/s 64B/66B OFDM downstream
```

```
       and up to 1.6 Gb/s 64B/66B OFDMA upstream";
  }

  identity phy-type-10G-2p5GGBASE-SP {
    base phy-type;
    description
      "Clause 164.3 10 Gb/s downstream and 2.5 Gb/s upstream
       256B/257B";
  }

  identity phy-type-10GBASE-SP {
    base phy-type;
    description
      "Clause 164.3 10 Gb/s 256B/257B";
  }

  identity phy-type-25G-10GBASE-PQ {
    base phy-type;
    description
      "Clause 142 25/10G-EPON 256B/257B";
  }

  identity phy-type-25GBASE-PQ {
    base phy-type;
    description
      "Clause 142 25/25G-EPON 256B/257B";
  }

  identity phy-type-25GBASE-R {
    base phy-type;
    description
      "Clause 107 25 Gb/s 64B/66B";
  }

  identity phy-type-25GBASE-T {
    base phy-type;
    description
      "Clause 113 25 Gb/s DSQ128";
  }

  identity phy-type-40GBASE-R {
    base phy-type;
    description
      "Clause 82 40 Gb/s multi-PCS lane 64B/66B";
  }

  identity phy-type-40GBASE-T {
    base phy-type;
    description
      "Clause 113 40 Gb/s DSQ128";
  }

  identity phy-type-50G-10GBASE-PQ {
    base phy-type;
```

```
    description
      "Clause 142 50/10G-EPON 256B/257B";
  }

  identity phy-type-50G-25GBASE-PQ {
    base phy-type;
    description
      "Clause 142 50/25G-EPON 256B/257B";
  }

  identity phy-type-50GBASE-PQ {
    base phy-type;
    description
      "Clause 142 50/50G-EPON 256B/257B";
  }

  identity phy-type-50GBASE-R {
    base phy-type;
    description
      "Clause 133 50 Gb/s multi-PCS lane 64B/66B";
  }

  identity phy-type-100GBASE-P {
    base phy-type;
    description
      "Clause 82 100 Gb/s multi-PCS lane using >2-level PAM";
  }

  identity phy-type-100GBASE-R {
    base phy-type;
    description
      "Clause 82 100 Gb/s multi-PCS lane using 2-level PAM";
  }

  identity phy-type-200GBASE-R {
    base phy-type;
    description
      "Clause 119 200 Gb/s multi-PCS lane 64B/66B";
  }

  identity phy-type-400GBASE-R {
    base phy-type;
    description
      "Clause 119 400 Gb/s multi-PCS lane 64B/66B";
  }

  identity phy-type-800GBASE-R {
    base phy-type;
    description
      "Clause 172 800 Gb/s multi-PCS lane 64B/66B";
  }

  /* PMD  identities */
```

```
  identity pmd-type {
    description
      "Base type for Physical Medium Dependent(PMD)sublayer types.";
    reference
      "IEEE Std 802.3, Clause 30.5.1.1.2 aMAUType";
  }

  identity pmd-type-global {
    base pmd-type;
    description
      "Undefined";
  }

  identity pmd-type-other {
    base pmd-type;
    description
      "Nonconformant PMD.";
  }

  identity pmd-type-unknown {
    base pmd-type;
    description
      "Initializing, true state or type not yet known.";
  }

  identity pmd-type-10BASE-T {
    base pmd-type;
    description
      "Twisted-pair cabling MAU as specified in Clause 14";
  }

  identity pmd-type-10BASE-T1L {
    base pmd-type;
    description
      "Single balanced pair PHY as specified in Clause 146";
  }

  identity pmd-type-10BASE-T1S {
    base pmd-type;
    description
      "Single balanced pair PHY as specified in Clause 147, full
       duplex mode";
  }

  identity pmd-type-10BASE-T1SMD {
    base pmd-type;
    description
      "Single balanced pair PHY as specified in Clause 147,
       multidrop mode";
  }

  identity pmd-type-100BASE-BX10D {
    base pmd-type;
    description
```

```
      "One single-mode fiber OLT PHY as specified in Clause 58";
  }

  identity pmd-type-100BASE-BX10U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY as specified in Clause 58";
  }

  identity pmd-type-100BASE-FX {
    base pmd-type;
    description
      "100BASE-FX PMD as specified in Clause 26";
  }

  identity pmd-type-100BASE-LX10 {
    base pmd-type;
    description
      "Two fiber PHY as specified in Clause 58";
  }

  identity pmd-type-100BASE-T1 {
    base pmd-type;
    description
      "Single balanced twisted-pair copper cabling PHY as specified
       in Clause 96";
  }

  identity pmd-type-100BASE-TX {
    base pmd-type;
    description
      "Two-pair Category 5 twisted-pair cabling as specified in
       Clause 25";
  }

  identity pmd-type-1000BASE-BX10D {
    base pmd-type;
    description
      "One single-mode fiber OLT PHY as specified in Clause 59";
  }

  identity pmd-type-1000BASE-BX10U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY as specified in Clause 59";
  }

  identity pmd-type-1000BASE-KX {
    base pmd-type;
    description
      "1000BASE-KX PCS/PMA over an electrical backplane PMD as
       specified in Clause 70";
  }
```

```
  identity pmd-type-1000BASE-LX {
    base pmd-type;
    description
      "1000BASE-LX fiber over long-wavelength laser PMD as specified
       in Clause 38";
  }

  identity pmd-type-1000BASE-LX10 {
    base pmd-type;
    description
      "Two fiber 10 km PHY as specified in Clause 59";
  }

  identity pmd-type-1000BASE-PX10-D {
    base pmd-type;
    description
      "One single-mode fiber OMP OLT PHY, as specified in Clause 60,
       supporting a distance of at least 10 km, and a split of at
       least 1:16";
  }

  identity pmd-type-1000BASE-PX10-U {
    base pmd-type;
    description
      "One single-mode fiber OMP ONU PHY, as specified in Clause 60,
       supporting a distance of at least 10 km, and a split of at
       least 1:16";
  }

  identity pmd-type-1000BASE-PX20-D {
    base pmd-type;
    description
      "One single-mode fiber OMP OLT PHY, as specified in Clause 60,
       supporting a distance of at least 20 km, and a split of at
       least 1:16";
  }

  identity pmd-type-1000BASE-PX20-U {
    base pmd-type;
    description
      "One single-mode fiber OMP ONU PHY, as specified in Clause 60,
       supporting a distance of at least 20 km, and a split of at
       least 1:16";
  }

  identity pmd-type-1000BASE-PX30-D {
    base pmd-type;
    description
      "One single-mode fiber OMP OLT PHY, as specified in Clause 60,
       supporting a distance of at least 20 km, and a split of at
       least 1:32";
  }

  identity pmd-type-1000BASE-PX30-U {
```

```
    base pmd-type;
    description
      "One single-mode fiber OMP ONU PHY, as specified in Clause 60,
       supporting a distance of at least 20 km, and a split of at
       least 1:32";
  }

  identity pmd-type-1000BASE-PX40-D {
    base pmd-type;
    description
      "One single-mode fiber OMP OLT PHY as specified in Clause 60,
       supporting a distance of at least 20 km, and a split of at
       least 1:64";
  }

  identity pmd-type-1000BASE-PX40-U {
    base pmd-type;
    description
      "One single-mode fiber OMP ONU PHY as specified in Clause 60,
       supporting a distance of at least 20 km, and a split of at
       least 1:64";
  }

  identity pmd-type-1000BASE-RHA {
    base pmd-type;
    description
      "Plastic optical fiber PHY as specified in Clause 115.";
  }

  identity pmd-type-1000BASE-RHB {
    base pmd-type;
    description
      "Plastic optical fiber PHY as specified in Clause 115.";
  }

  identity pmd-type-1000BASE-RHC {
    base pmd-type;
    description
      "Plastic optical fiber PHY as specified in Clause 115.";
  }

  identity pmd-type-1000BASE-SX {
    base pmd-type;
    description
      "1000BASE-SX fiber over short-wavelength laser PMD as
       specified in Clause 38";
  }

  identity pmd-type-1000BASE-T {
    base pmd-type;
    description
      "Four-pair Category 5 twisted-pair cabling PHY as specified in
       Clause 40";
  }
```

```
  identity pmd-type-1000BASE-T1 {
    base pmd-type;
    description
      "Single twisted-pair copper cable PHY as specified in Clause
       97";
  }

  identity pmd-type-1000BASE-X {
    base pmd-type;
    description
      "1000BASE-X PCS/PMA as specified in Clause 36 over undefined
       PMD";
  }

  identity pmd-type-2.5GBASE-KX {
    base pmd-type;
    description
      "2.5GBASE-X PMD as specified in Clause 128 over an electrical
       backplane as specified in Clause 128";
  }

  identity pmd-type-2.5GBASE-T {
    base pmd-type;
    description
      "Four-pair twisted-pair balanced copper cabling PHY as
       specified in Clause 126";
  }

  identity pmd-type-2.5GBASE-T1 {
    base pmd-type;
    description
      "Single balanced pair of conductors PHY as specified in Clause
       149";
  }

  identity pmd-type-2.5GBASE-X {
    base pmd-type;
    description
      "2.5GBASE-X PCS/PMA as specified in Clause 127 over undefined
       PMD";
  }

  identity pmd-type-5GBASE-KR {
    base pmd-type;
    description
      "5GBASE-KR PMD as specified in Clause 130 over an electrical
       backplane as specified in Clause 130";
  }

  identity pmd-type-5GBASE-R {
    base pmd-type;
    description
      "5GBASE-R PCS/PMA as specified in Clause 129 over undefined
```

```
        PMD";
  }

  identity pmd-type-5GBASE-T {
    base pmd-type;
    description
      "Four-pair twisted-pair balanced copper cabling PHY as
       specified in Clause 126";
  }

  identity pmd-type-5GBASE-T1 {
    base pmd-type;
    description
      "Single balanced pair of conductors PHY as specified in Clause
       149";
  }

  identity pmd-type-10G-1GBASE-PRX-D1 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
  }

  identity pmd-type-10G-1GBASE-PRX-D2 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
  }

  identity pmd-type-10G-1GBASE-PRX-D3 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
  }

  identity pmd-type-10G-1GBASE-PRX-D4 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
  }

  identity pmd-type-10G-1GBASE-PRX-U1 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
  }

  identity pmd-type-10G-1GBASE-PRX-U2 {
    base pmd-type;
```

```
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
  }

  identity pmd-type-10G-1GBASE-PRX-U3 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
  }

  identity pmd-type-10G-1GBASE-PRX-U4 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
  }

  identity pmd-type-10GBASE-BR10-D {
    base pmd-type;
    description
      "One single-mode fiber OLT PHY supporting a distance of at
       least 10 km as specified in Clause 158";
  }

  identity pmd-type-10GBASE-BR10-U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY supporting a distance of at
       least 10 km as specified in Clause 158";
  }

  identity pmd-type-10GBASE-BR20-D {
    base pmd-type;
    description
      "One single-mode fiber OLT PHY supporting a distance of at
       least 20 km as specified in Clause 158";
  }

  identity pmd-type-10GBASE-BR20-U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY supporting a distance of at
       least 20 km as specified in Clause 158";
  }

  identity pmd-type-10GBASE-BR40-D {
    base pmd-type;
    description
      "One single-mode fiber OLT PHY upporting a distance of at
       least 40 km as specified in Clause 158";
  }
```

```
  identity pmd-type-10GBASE-BR40-U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY supporting a distance of at
       least 40 km as specified in Clause 158";
  }

  identity pmd-type-10GBASE-CX4 {
    base pmd-type;
    description
      "10GBASE-CX4 copper over 8 pair 100-Ohm balanced cable as
       specified in Clause 54";
  }

  identity pmd-type-10GBASE-ER {
    base pmd-type;
    description
      "10GBASE-ER fiber over 1550nm optics as specified in Clause 52";
  }

  identity pmd-type-10GBASE-EW {
    base pmd-type;
    description
      "10GBASE-EW fiber over 1550nm optics as specified in Clause 52";
  }

  identity pmd-type-10GBASE-KR {
    base pmd-type;
    description
      "10GBASE-KR PCS/PMA over an electrical backplane PMD as
       specified in Clause 72";
  }

  identity pmd-type-10GBASE-KX4 {
    base pmd-type;
    description
      "10GBASE-KX4 PCS/PMA over an electrical backplane PMD as
       specified in Clause 71";
  }

  identity pmd-type-10GBASE-LR {
    base pmd-type;
    description
      "10GBASE-LR fiber over 1310nm optics as specified in Clause 52";
  }

  identity pmd-type-10GBASE-LRM {
    base pmd-type;
    description
      "10GBASE-LRM fiber over 1310 nm optics as specified in Clause
       68";
  }

  identity pmd-type-10GBASE-LW {
```

```
    base pmd-type;
    description
      "10GBASE-LW fiber over 1310nm optics as specified in Clause 52";
  }

  identity pmd-type-10GBASE-LX4 {
    base pmd-type;
    description
      "10GBASE-LX4 fiber over 4 lane 1310nm optics as specified in
       Clause 53";
  }

  identity pmd-type-10GBASE-PR-D1 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
  }

  identity pmd-type-10GBASE-PR-D2 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
  }

  identity pmd-type-10GBASE-PR-D3 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
  }

  identity pmd-type-10GBASE-PR-D4 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
  }

  identity pmd-type-10GBASE-PR-U1 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
  }

  identity pmd-type-10GBASE-PR-U3 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
  }
```

```
identity pmd-type-10GBASE-PR-U4 {
  base pmd-type;
  description
    "One single-mode fiber 10.3125 GBd continuous downstream /
     1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
}

identity pmd-type-10GBASE-R {
  base pmd-type;
  description
    "10GBASE-R PCS/PMA as specified in Clause 49 over undefined
     PMD";
}

identity pmd-type-10GBASE-SR {
  base pmd-type;
  description
    "10GBASE-SR fiber over 850nm optics as specified in Clause 52";
}

identity pmd-type-10GBASE-SW {
  base pmd-type;
  description
    "10GBASE-SW fiber over 850nm optics as specified in Clause 52";
}

identity pmd-type-10GBASE-T {
  base pmd-type;
  description
    "Four-pair twisted-pair balanced copper cabling PHY as
     specified in Clause 55";
}

identity pmd-type-10GBASE-T1 {
  base pmd-type;
  description
    "Single balanced pair of conductors PHY as specified in Clause
     149";
}

identity pmd-type-10GBASE-X {
  base pmd-type;
  description
    "10GBASE-X PCS/PMA as specified in Clause 48 over undefined
     PMD";
}

identity pmd-type-10GPASS-XR {
  base pmd-type;
  description
    "Coax cable distribution network PHY continuous
     downstream/burst mode upstream PHY as specified in Clause 100
     and Clause 101";
}
```

```
identity pmd-type-10G-2p5GGBASE-SP1-Dx {
  base pmd-type;
  description
    "One single-mode fiber 10.3125 GBd continuous downstream /
     2.578125 GBd burst mode upstream OLT PHY supporting
     channel x of FSR set 1, as specified in 164.2,";
}

identity pmd-type-10G-2p5GGBASE-SP1-Uxy {
  base pmd-type;
  description
    "One single-mode fiber 10.3125 GBd continuous downstream /
     2.578125 GBd burst mode upstream ONU PHY supporting
     channels x to y of FSR set 1, as specified in 164.2";
}

identity pmd-type-10GBASE-SP {
  base pmd-type;
  description
    "Clause 164.3 10 Gb/s 256B/257B";
}

identity pmd-type-25G-10GBASE-PQG-D2 {
  base pmd-type;
  description
    "One single mode fiber, 1 Ã— 25.78125 GBd continuous
     transmission / 1 Ã— 10.3125 GBd burst mode reception, medium
     power class, as specified in Clause 141";
}

identity pmd-type-25G-10GBASE-PQG-D3 {
  base pmd-type;
  description
    "One single mode fiber, 1 Ã— 25.78125 GBd continuous
     transmission / 1 Ã— 10.3125 GBd burst mode reception, high
     power class, as specified in Clause 141";
}

identity pmd-type-25G-10GBASE-PQG-U2 {
  base pmd-type;
  description
    "One single mode fiber, 1 Ã— 25.78125 GBd continuous reception
     / 1 Ã— 10.3125 GBd burst mode transmission, medium power class,
     as specified in Clause 141";
}

identity pmd-type-25G-10GBASE-PQG-U3 {
  base pmd-type;
  description
    "One single mode fiber, 1 Ã— 25.78125 GBd continuous reception
     / 1 Ã— 10.3125 GBd burst mode transmission, high power class,
     as specified in Clause 141";
}
```

```
  identity pmd-type-25G-10GBASE-PQX-D2 {
    base pmd-type;
    description
      "One single mode fiber, 1 Ã— 25.78125 GBd continuous
       transmission / 1 Ã— 10.3125 GBd burst mode reception, medium
       power class, as specified in Clause 141";
  }

  identity pmd-type-25G-10GBASE-PQX-D3 {
    base pmd-type;
    description
      "One single mode fiber, 1 Ã— 25.78125 GBd continuous
       transmission / 1 Ã— 10.3125 GBd burst mode reception, high
       power class, as specified in Clause 141";
  }

  identity pmd-type-25G-10GBASE-PQX-U2 {
    base pmd-type;
    description
      "One single mode fiber, 1 Ã— 25.78125 GBd continuous reception
       / 1 Ã— 10.3125 GBd burst mode transmission, medium power class,
       as specified in Clause 141";
  }

  identity pmd-type-25G-10GBASE-PQX-U3 {
    base pmd-type;
    description
      "One single mode fiber, 1 Ã— 25.78125 GBd continuous reception
       / 1 Ã— 10.3125 GBd burst mode transmission, high power class,
       as specified in Clause 141";
  }

  identity pmd-type-25GBASE-BR10-D {
    base pmd-type;
    description
      "One single-mode fiber OLT PHY supporting a distance of at
       least 10 km as specified in Clause 159";
  }

  identity pmd-type-25GBASE-BR10-U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY supporting a distance of at
       least 10 km as specified in Clause 159";
  }

  identity pmd-type-25GBASE-BR20-D {
    base pmd-type;
    description
      "One single-mode fiber OLT PHY supporting a distance of at
       least 20 km as specified in Clause 159";
  }
```

```
  identity pmd-type-25GBASE-BR20-U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY supporting a distance of at
       least 20 km as specified in Clause 159";
  }

  identity pmd-type-25GBASE-BR40-D {
    base pmd-type;
    description
      "One single-mode fiber OLT PHY supporting a distance of at
       least 40 km as specified in Clause 159";
  }

  identity pmd-type-25GBASE-BR40-U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY supporting a distance of at
       least 40 km as specified in Clause 159";
  }

  identity pmd-type-25GBASE-CR {
    base pmd-type;
    description
      "25GBASE-R PCS/PMA over shielded balanced copper cable PMD as
       specified in Clause 110";
  }

  identity pmd-type-25GBASE-CR-S {
    base pmd-type;
    description
      "25GBASE-R PCS/PMA over shielded balanced copper cable PMD as
       specified in Clause 110 without support for RS-FEC ";
  }

  identity pmd-type-25GBASE-ER {
    base pmd-type;
    description
      "25GBASE-R PCS/PMA over single-mode fiber PMD, with extended
       reach, as specified in Clause 114";
  }

  identity pmd-type-25GBASE-KR {
    base pmd-type;
    description
      "25GBASE-R PCS/PMA over an electrical backplane PMD as
       specified in Clause 111";
  }

  identity pmd-type-25GBASE-KR-S {
    base pmd-type;
    description
      "25GBASE-R PCS/PMA over an electrical backplane PMD as
       specified in Clause 111 without support for RS-FEC";
```

```
  }

  identity pmd-type-25GBASE-LR {
    base pmd-type;
    description
      "25GBASE-R PCS/PMA over single-mode fiber PMD, with long
       reach, as specified in Clause 114";
  }

  identity pmd-type-25GBASE-PQG-D2 {
    base pmd-type;
    description
      "One single mode fiber, 1 Ã— 25.78125 GBd continuous
       transmission / 1 Ã— 25.78125 GBd burst mode reception, medium
       power class,  as specified in Clause 141";
  }

  identity pmd-type-25GBASE-PQG-D3 {
    base pmd-type;
    description
      "One single mode fiber, 1 Ã— 25.78125 GBd continuous
       transmission / 1 Ã— 25.78125 GBd burst mode reception, high
       power class, as specified in Clause 141";
  }

  identity pmd-type-25GBASE-PQG-U2 {
    base pmd-type;
    description
      "One single mode fiber, 1 Ã— 25.78125 GBd continuous reception
       / 1 Ã— 25.78125 GBd burst mode transmission, medium power
       class, as specified in Clause 141";
  }

  identity pmd-type-25GBASE-PQG-U3 {
    base pmd-type;
    description
      "One single mode fiber, 1 Ã— 25.78125 GBd continuous reception
       / 1 Ã— 25.78125 GBd burst mode transmission, high power class,
       as specified in Clause 141";
  }

  identity pmd-type-25GBASE-PQX-D2 {
    base pmd-type;
    description
      "One single mode fiber, 1 Ã— 25.78125 GBd continuous
       transmission / 1 Ã— 25.78125 GBd burst mode reception, medium
       power class, as specified in Clause 141";
  }

  identity pmd-type-25GBASE-PQX-D3 {
    base pmd-type;
    description
      "One single mode fiber, 1 Ã— 25.78125 GBd continuous
       transmission / 1 Ã— 25.78125 GBd burst mode reception, high
```

```
        power class, as specified in Clause 141";
  }

  identity pmd-type-25GBASE-PQX-U2 {
    base pmd-type;
    description
      "One single mode fiber, 1 Ã— 25.78125 GBd continuous reception
       / 1 Ã— 25.78125 GBd burst mode transmission, medium power
       class, as specified in Clause 141";
  }

  identity pmd-type-25GBASE-PQX-U3 {
    base pmd-type;
    description
      "One single mode fiber, 1 Ã— 25.78125 GBd continuous reception
       / 1 Ã— 25.78125 GBd burst mode transmission, high power class,
       as specified in Clause 141";
  }

  identity pmd-type-25GBASE-R {
    base pmd-type;
    description
      "PCS as specified in Clause 107 with PMA as specified in
       Clause 109 over undefined PMD";
  }

  identity pmd-type-25GBASE-SR {
    base pmd-type;
    description
      "25GBASE-R PCS/PMA over multimode fiber PMD as specified in
       Clause 112";
  }

  identity pmd-type-25GBASE-T {
    base pmd-type;
    description
      "Four-pair twisted-pair balanced copper cabling PHY as
       specified in Clause 113";
  }

  identity pmd-type-40GBASE-CR4 {
    base pmd-type;
    description
      "40GBASE-R PCS/PMA over 4 lane shielded copper balanced cable
       PMD as specified in Clause 85";
  }

  identity pmd-type-40GBASE-ER4 {
    base pmd-type;
    description
      "40GBASE-R PCS/PMA over 4 WDM lane single mode fiber PMD, with
       extended reach, as specified in Clause 87";
  }
```

```
  identity pmd-type-40GBASE-FR {
    base pmd-type;
    description
      "40GBASE-R PCS/PMA over single mode fiber PMD as specified in
       Clause 89";
  }

  identity pmd-type-40GBASE-KR4 {
    base pmd-type;
    description
      "40GBASE-R PCS/PMA over an electrical backplane PMD as
       specified in Clause 84";
  }

  identity pmd-type-40GBASE-LR4 {
    base pmd-type;
    description
      "40GBASE-R PCS/PMA over 4 WDM lane single mode fiber PMD, with
       long reach, as specified in Clause 87";
  }

  identity pmd-type-40GBASE-R {
    base pmd-type;
    description
      "Multi-lane PCS as specified in Clause 82 over undefined
       PMA/PMD";
  }

  identity pmd-type-40GBASE-SR4 {
    base pmd-type;
    description
      "40GBASE-R PCS/PMA over 4 lane multimode fiber PMD as
       specified in Clause 86";
  }

  identity pmd-type-40GBASE-T {
    base pmd-type;
    description
      "Four-pair twisted-pair balanced copper cabling PHY as
       specified in Clause 113";
  }

  identity pmd-type-50G-10GBASE-PQG-D2 {
    base pmd-type;
    description
      "One single mode fiber, 2 Ã— 25.78125 GBd continuous
       transmission / 1 Ã— 10.3125 GBd burst mode reception, medium
       power class, as specified in Clause 141";
  }

  identity pmd-type-50G-10GBASE-PQG-D3 {
    base pmd-type;
    description
      "One single mode fiber, 2 Ã— 25.78125 GBd continuous
```

```
        transmission / 1 Ã— 10.3125 GBd burst mode reception, high
        power class, as specified in Clause 141";
  }

  identity pmd-type-50G-10GBASE-PQG-U2 {
    base pmd-type;
    description
      "One single mode fiber, 2 Ã— 25.78125 GBd continuous reception
       / 1 Ã— 10.3125 GBd burst mode transmission, medium power class,
       as specified in Clause 141";
  }

  identity pmd-type-50G-10GBASE-PQG-U3 {
    base pmd-type;
    description
      "One single mode fiber, 2 Ã— 25.78125 GBd continuous reception
       / 1 Ã— 10.3125 GBd burst mode transmission, high power class,
       as specified in Clause 141";
  }

  identity pmd-type-50G-10GBASE-PQX-D2 {
    base pmd-type;
    description
      "One single mode fiber, 2 Ã— 25.78125 GBd continuous
       transmission / 1 Ã— 10.3125 GBd burst mode reception, medium
       power class, as specified in Clause 141";
  }

  identity pmd-type-50G-10GBASE-PQX-D3 {
    base pmd-type;
    description
      "One single mode fiber, 2 Ã— 25.78125 GBd continuous
       transmission / 1 Ã— 10.3125 GBd burst mode reception, high
       power class, as specified in Clause 141";
  }

  identity pmd-type-50G-10GBASE-PQX-U2 {
    base pmd-type;
    description
      "One single mode fiber, 2 Ã— 25.78125 GBd continuous reception
       / 1 Ã— 10.3125 GBd burst mode transmission, medium power class,
       as specified in Clause 141";
  }

  identity pmd-type-50G-10GBASE-PQX-U3 {
    base pmd-type;
    description
      "One single mode fiber, 2 Ã— 25.78125 GBd continuous reception
       / 1 Ã— 10.3125 GBd burst mode transmission, high power class,
       as specified in Clause 141";
  }

  identity pmd-type-50G-25GBASE-PQG-D2 {
    base pmd-type;
```

```
    description
      "One single mode fiber, 2 Ã— 25.78125 GBd continuous
       transmission / 1 Ã— 25.78125 GBd burst mode reception, medium
       power class, as specified in Clause 141";
  }

  identity pmd-type-50G-25GBASE-PQG-D3 {
    base pmd-type;
    description
      "One single mode fiber, 2 Ã— 25.78125 GBd continuous
       transmission / 1 Ã— 25.78125 GBd burst mode reception, high
       power class, as specified in Clause 141";
  }

  identity pmd-type-50G-25GBASE-PQG-U2 {
    base pmd-type;
    description
      "One single mode fiber, 2 Ã— 25.78125 GBd continuous reception
       / 1 Ã— 25.78125 GBd burst mode transmission, medium power
       class, as specified in Clause 141";
  }

  identity pmd-type-50G-25GBASE-PQG-U3 {
    base pmd-type;
    description
      "One single mode fiber, 2 Ã— 25.78125 GBd continuous reception
       / 1 Ã— 25.78125 GBd burst mode transmission, high power class,
       as specified in Clause 141";
  }

  identity pmd-type-50G-25GBASE-PQX-D2 {
    base pmd-type;
    description
      "One single mode fiber, 2 Ã— 25.78125 GBd continuous
       transmission / 1 Ã— 25.78125 GBd burst mode reception, medium
       power class, as specified in Clause 141";
  }

  identity pmd-type-50G-25GBASE-PQX-D3 {
    base pmd-type;
    description
      "One single mode fiber, 2 Ã— 25.78125 GBd continuous
       transmission / 1 Ã— 25.78125 GBd burst mode reception, high
       power class, as specified in Clause 141";
  }

  identity pmd-type-50G-25GBASE-PQX-U2 {
    base pmd-type;
    description
      "One single mode fiber, 2 Ã— 25.78125 GBd continuous reception
       / 1 Ã— 25.78125 GBd burst mode transmission, medium power
       class, as specified in Clause 141";
  }
```

```
  identity pmd-type-50G-25GBASE-PQX-U3 {
    base pmd-type;
    description
      "One single mode fiber, 2 Ã— 25.78125 GBd continuous reception
       / 1 Ã— 25.78125 GBd burst mode transmission, high power class,
       as specified in Clause 141";
  }

  identity pmd-type-50GBASE-BR10-D {
    base pmd-type;
    description
      "One single-mode fiber OLT PHY supporting a distance of at
       least 10 km as specified in Clause 160";
  }

  identity pmd-type-50GBASE-BR10-U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY supporting a distance of at
       least 10 km as specified in Clause 160";
  }

  identity pmd-type-50GBASE-BR20-D {
    base pmd-type;
    description
      "One single-mode fiber OLT PHY supporting a distance of at
       least 20 km as specified in Clause 160";
  }

  identity pmd-type-50GBASE-BR20-U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY supporting a distance of at
       least 20 km as specified in Clause 160";
  }

  identity pmd-type-50GBASE-BR40-D {
    base pmd-type;
    description
      "One single-mode fiber OLT PHY supporting a distance of at
       least 40 km as specified in Clause 160";
  }

  identity pmd-type-50GBASE-BR40-U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY supporting a distance of at
       least 40 km as specified in Clause 160";
  }

  identity pmd-type-50GBASE-CR {
    base pmd-type;
    description
      "50GBASE-R PCS/PMA over shielded copper balanced cable PMD as
```

```
       specified in Clause 136 ";
  }

  identity pmd-type-50GBASE-ER {
    base pmd-type;
    description
      "50GBASE-R PCS/PMA over single-mode fiber PMD with reach up to
       at least 40 km as specified in Clause 139";
  }

  identity pmd-type-50GBASE-FR {
    base pmd-type;
    description
      "50GBASE-R PCS/PMA over single mode fiber PMD as specified in
       Clause 139";
  }

  identity pmd-type-50GBASE-KR {
    base pmd-type;
    description
      "50GBASE-R PCS/PMA over an electrical backplane PMD as
       specified in Clause 137";
  }

  identity pmd-type-50GBASE-LR {
    base pmd-type;
    description
      "50GBASE-R PCS/PMA over single mode fiber PMD as specified in
       Clause 139";
  }

  identity pmd-type-50GBASE-PQG-D2 {
    base pmd-type;
    description
      "One single mode fiber, 2 Ã— 25.78125 GBd continuous
       transmission / 2 Ã— 25.78125 GBd burst mode reception, medium
       power class, as specified in Clause 141";
  }

  identity pmd-type-50GBASE-PQG-D3 {
    base pmd-type;
    description
      "One single mode fiber, 2 Ã— 25.78125 GBd continuous
       transmission / 2 Ã— 25.78125 GBd burst mode reception, high
       power class, as specified in Clause 141";
  }

  identity pmd-type-50GBASE-PQG-U2 {
    base pmd-type;
    description
      "One single mode fiber, 2 Ã— 25.78125 GBd continuous reception
       / 2 Ã— 25.78125 GBd burst mode transmission, medium power
       class, as specified in Clause 141";
  }
```

```
identity pmd-type-50GBASE-PQG-U3 {
  base pmd-type;
  description
    "One single mode fiber, 2 Ã— 25.78125 GBd continuous reception
     / 2 Ã— 25.78125 GBd burst mode transmission, high power class,
     as specified in Clause 141";
}

identity pmd-type-50GBASE-PQX-D2 {
  base pmd-type;
  description
    "One single mode fiber, 2 Ã— 25.78125 GBd continuous
     transmission / 2 Ã— 25.78125 GBd burst mode reception, medium
     power class, as specified in Clause 141";
}

identity pmd-type-50GBASE-PQX-D3 {
  base pmd-type;
  description
    "One single mode fiber, 2 Ã— 25.78125 GBd continuous
     transmission / 2 Ã— 25.78125 GBd burst mode reception, high
     power class, as specified in Clause 141";
}

identity pmd-type-50GBASE-PQX-U2 {
  base pmd-type;
  description
    "One single mode fiber, 2 Ã— 25.78125 GBd continuous reception
     / 2 Ã— 25.78125 GBd burst mode transmission, medium power
     class, as specified in Clause 141";
}

identity pmd-type-50GBASE-PQX-U3 {
  base pmd-type;
  description
    "One single mode fiber, 2 Ã— 25.78125 GBd continuous reception
     / 2 Ã— 25.78125 GBd burst mode transmission, high power class,
     as specified in Clause 141";
}

identity pmd-type-50GBASE-R {
  base pmd-type;
  description
    "Multi-lane PCS as specified in Clause 133 with PMA as
     specified in Clause 135 over undefined PMD";
}

identity pmd-type-50GBASE-SR {
  base pmd-type;
  description
    "50GBASE-R PCS/PMA over multimode fiber PMD as specified in
     Clause 138";
}
```

```
  identity pmd-type-100CR1 {
    base pmd-type;
    description
      "100GBASE-CR1 as specified in Clause 162 or 100GBASE-KR1 as
       specified in Clause 163";
  }

  identity pmd-type-100GBASE-CR2 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over 2 lane shielded copper balanced cable
       PMD as specified in Clause 136";
  }

  identity pmd-type-100GBASE-CR4 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over 4 lane shielded copper balanced cable
       PMD as specified in Clause 92";
  }

  identity pmd-type-100GBASE-CR10 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over 10 lane shielded copper balanced
       cable PMD as specified in Clause 85";
  }

  identity pmd-type-100GBASE-DR {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over single mode fiber PMD as specified in
       Clause 140";
  }

  identity pmd-type-100GBASE-ER4 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over 4 WDM lane single mode fiber PMD,
       with extended reach, as specified in Clause 88";
  }

  identity pmd-type-100GBASE-FR1 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over single-mode fiber PMD with reach up
       to at least 2 km as specified in Clause 140";
  }

  identity pmd-type-100GBASE-KP4 {
    base pmd-type;
    description
      "100GBASE-P PCS/PMA over an electrical backplane PMD as
```

```
       specified in Clause 94";
  }

  identity pmd-type-100GBASE-KR2 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over an electrical backplane PMD as
       specified in Clause 137";
  }

  identity pmd-type-100GBASE-KR4 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over an electrical backplane PMD as
       specified in Clause 93";
  }

  identity pmd-type-100GBASE-LR1 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over single-mode fiber PMD with reach up
       to at least 10 km as specified in Clause 140";
  }

  identity pmd-type-100GBASE-LR4 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over 4 WDM lane single mode fiber PMD,
       with long reach, as specified in Clause 88";
  }

  identity pmd-type-100GBASE-R {
    base pmd-type;
    description
      "Multi-lane PCS as specified in Clause 82 over undefined
       100GBASE-R or 100GBASE-P PMA/PMD";
  }

  identity pmd-type-100GBASE-SR1 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over multimode fiber PMD with
       reach up to at least 100 m as specified in Clause 167";
  }

  identity pmd-type-100GBASE-SR2 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over 2 lane multimode fiber PMD as
       specified in Clause 138";
  }

  identity pmd-type-100GBASE-SR4 {
    base pmd-type;
```

```
    description
      "100GBASE-R PCS/PMA over 4 lane multimode fiber PMD as
       specified in Clause 95";
  }

  identity pmd-type-100GBASE-SR10 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over 10 lane multimode fiber PMD as
       specified in Clause 86";
  }

  identity pmd-type-100GBASE-VR1 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over multimode fiber PMD with
       reach up to at least 50 m as specified in Clause 167";
  }

  identity pmd-type-100GBASE-ZR {
    base pmd-type;
    description
      "100GBASE-R PCS/100GBASE-ZR PMA over a PMD with reach up to at
       least 80 km as specified in Clause 154";
  }

  identity pmd-type-200GBASE-CR2 {
    base pmd-type;
    description
      "200GBASE-CR2 as specified in Clause 162 or 200GBASE-KR2 as
       specified in Clause 163";
  }

  identity pmd-type-200GBASE-CR4 {
    base pmd-type;
    description
      "200GBASE-R PCS/PMA over 4 lane shielded copper balanced cable
       PMD as specified in Clause 136 ";
  }

  identity pmd-type-200GBASE-DR4 {
    base pmd-type;
    description
      "200GBASE-R PCS/PMA over 4-lane single-mode fiber PMD as
       specified in Clause 121";
  }

  identity pmd-type-200GBASE-ER4 {
    base pmd-type;
    description
      "200GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD with
       reach up to at least 40 km as specified in Clause 122";
  }
```

```
identity pmd-type-200GBASE-FR4 {
  base pmd-type;
  description
    "200GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD with
     reach up to at least 2 km as specified in Clause 122";
}

identity pmd-type-200GBASE-KR2 {
  base pmd-type;
  description
    "200GBASE-R PCS/PMA over an electrical backplane PMD
     as specified in Clause 163";
}

identity pmd-type-200GBASE-KR4 {
  base pmd-type;
  description
    "200GBASE-R PCS/PMA over an electrical backplane PMD as
     specified in Clause 137";
}

identity pmd-type-200GBASE-LR4 {
  base pmd-type;
  description
    "200GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD with
     reach up to at least 10 km as specified in Clause 122";
}

identity pmd-type-200GBASE-R {
  base pmd-type;
  description
    "Multi-lane PCS as specified in Clause 119 over undefined
     PMA/PMD";
}

identity pmd-type-200GBASE-SR2 {
  base pmd-type;
  description
    "200GBASE-R PCS/PMA over 2 lane multimode fiber PMD with
     reach up to at least 100 m as specified in Clause 167";
}

identity pmd-type-200GBASE-SR4 {
  base pmd-type;
  description
    "200GBASE-R PCS/PMA over 4 lane multimode fiber PMD as
     specified in Clause 138";
}

identity pmd-type-200GBASE-VR2 {
  base pmd-type;
  description
    "200GBASE-R PCS/PMA over 2 lane multimode fiber PMD with
     reach up to at least 50 m as specified in Clause 167";
```

```
  }

  identity pmd-type-400GBASE-DR4 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 4-lane single-mode fiber PMD as
       specified in Clause 124";
  }

  identity pmd-type-400GBASE-DR4-2 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 4 single-mode fibers in each
       direction PMD with reach up to at least 2 km as specified
       in Clause 124";
  }

  identity pmd-type-400GBASE-ER8 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 8 WDM lane single-mode fiber PMD with
       reach up to at least 40 km as specified in Clause 122";
  }

  identity pmd-type-400GBASE-FR4 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD with
       reach up to at least 2 km as specified in Clause 151";
  }

  identity pmd-type-400GBASE-FR8 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 8 WDM lane single-mode fiber PMD with
       reach up to at least 2 km as specified in Clause 122";
  }

  identity pmd-type-400GBASE-LR4-6 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD with
       reach up to at least 6 km as specified in Clause 151";
  }

  identity pmd-type-400GBASE-LR8 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 8 WDM lane single-mode fiber PMD with
       reach up to at least 10 km as specified in Clause 122";
  }

  identity pmd-type-400GBASE-R {
    base pmd-type;
```

```
    description
      "Multi-lane PCS as specified in Clause 119 over undefined
       PMA/PMD";
  }

  identity pmd-type-400GBASE-SR4 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 4 lane multimode fiber PMD with
       reach up to at least 100 m as specified in Clause 167";
  }

  identity pmd-type-400GBASE-SR4d2 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 8-lane multimode fiber PMD as
       specified in Clause 150";
  }

  identity pmd-type-400GBASE-SR8 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 8-lane multimode fiber PMD as
       specified in Clause 138";
  }

  identity pmd-type-400GBASE-SR16 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 16-lane multimode fiber PMD as
       specified in Clause 123";
  }

  identity pmd-type-400GBASE-VR4 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 4 lane multimode fiber PMD with
       reach up to at least 50 m as specified in Clause 167";
  }

  identity pmd-type-800GBASE-CR8 {
    base pmd-type;
    description
      "800GBASE-R PCS/PMA over 8-lane shielded balanced copper cable
       PMD as specified in Clause 162";
  }

  identity pmd-type-800GBASE-DR8 {
    base pmd-type;
    description
      "800GBASE-R PCS/PMA over 8 single-mode fibers in each
          direction PMD with reach up to at least 500 m as specified
          in Clause 124";
  }
```

```
  identity pmd-type-800GBASE-DR8-2 {
    base pmd-type;
    description
      "800GBASE-R PCS/PMA over 8 single-mode fibers in each
       direction PMD with reach up to at least 2 km as specified
       in Clause 124";
  }

  identity pmd-type-800GBASE-KR8 {
    base pmd-type;
    description
      "800GBASE-R PCS/PMA over an electrical backplane PMD as
       specified in Clause 163";
  }

  identity pmd-type-800GBASE-R {
    base pmd-type;
    description
      "Multi-lane PCS as specified in Clause 172 over
       undefined PMA/PMDï€ ";
  }

  identity pmd-type-800GBASE-SR8 {
    base pmd-type;
    description
      "800GBASE-R PCS/PMA over 8 multimode fibers in each
       direction PMD with reach up to at least 100 m as specified
       in Clause 167";
  }

  identity pmd-type-800GBASE-VR8 {
    base pmd-type;
    description
      "800GBASE-R PCS/PMA over 8 multimode fibers in each
       direction PMD with reach up to at least 50 m as specified
       in Clause 167";
  }

  typedef pmd-type-list {
    type bits {
      /* 802.3-2022 */
      bit bOther {
        description
          "Other";
      }
      bit b10baseT {
        description
          "10BASE-T";
      }
      bit b10baseFL {
        description
          "10BASE-FL";
      }
```

```
        bit b100baseTX {
          description
            "100BASE-TX";
        }
        bit b100baseFX {
          description
            "100BASE-FX";
        }
        bit b1000baseX {
          description
            "1000BASE-X";
        }
        bit b1000baseLX {
          description
            "1000BASE-LX";
        }
        bit b1000baseSX {
          description
            "1000BASE-SX";
        }
        bit b1000baseCX {
          description
            "1000BASE-CX";
        }
        bit b1000baseT {
          description
            "1000BASE-T";
        }
        bit b10GbaseX {
          description
            "10GBASE-X";
        }
        bit b10GbaseLX4 {
          description
            "10GBASE-LX4";
        }
        bit b10GbaseR {
          description
            "10GBASE-R";
        }
        bit b10GbaseER {
          description
            "10GBASE-ER";
        }
        bit b10GbaseLR {
          description
            "10GBASE-LR";
        }
        bit b10GbaseSR {
          description
            "10GBASE-SR";
        }
        bit b10GbaseSW {
          description
```

```
            "10GBASE-SW";
        }
        bit b10GbaseCX4 {
          description
            "10GBASE-CX4";
        }
        bit b100BaseBX10D {
          description
            "100BASE-BX10D";
        }
        bit b100BaseBX10U {
          description
            "100BASE-BX10U";
        }
        bit b100BaseLX10 {
          description
            "100BASE-LX10";
        }
        bit b1000BaseBX10D {
          description
            "1000BASE-BX10D";
        }
        bit b1000BaseBX10U {
          description
            "1000BASE-BX10U";
        }
        bit b1000BaseLX10 {
          description
            "1000BASE-LX10";
        }
        bit b1000BasePX10D {
          description
            "1000BASE-PX10D";
        }
        bit b1000BasePX10U {
          description
            "1000BASE-PX10U";
        }
        bit b1000BasePX20D {
          description
            "1000BASE-PX20D";
        }
        bit b1000BasePX20U {
          description
            "1000BASE-PX20U";
        }
        bit b10GbaseT {
          description
            "10GBASE-T";
        }
        bit b10GbaseLRM {
          description
            "10GBASE-LRM";
        }
```

```
        bit b1000baseKX {
          description
            "1000BASE-KX";
        }
        bit b10GbaseKX4 {
          description
            "10GBASE-KX4";
        }
        bit b10GbaseKR {
          description
            "10GBASE-KR";
        }
        bit b10G1GbasePRXD1 {
          description
            "10/1GBASE-PRXD1";
        }
        bit b10G1GbasePRXD2 {
          description
            "10/1GBASE-PRXD2";
        }
        bit b10G1GbasePRXD3 {
          description
            "10/1GBASE-PRXD3";
        }
        bit b10G1GbasePRXU1 {
          description
            "10/1GBASE-PRXU1";
        }
        bit b10G1GbasePRXU2 {
          description
            "10/1GBASE-PRXU2";
        }
        bit b10G1GbasePRXU3 {
          description
            "10/1GBASE-PRXU3";
        }
        bit b10GbasePRD1 {
          description
            "10GBASE-PRD1";
        }
        bit b10GbasePRD2 {
          description
            "10GBASE-PRD2";
        }
        bit b10GbasePRD3 {
          description
            "10GBASE-PRD3";
        }
        bit b10GbasePRU1 {
          description
            "10GBASE-PRU1";
        }
        bit b10GbasePRU3 {
          description
```

```
            "10GBASE-PRU3";
        }
        bit b40GbaseKR4 {
          description
            "40GBASE-KR4";
        }
        bit b40GbaseCR4 {
          description
            "40GBASE-CR4";
        }
        bit b40GbaseSR4 {
          description
            "40GBASE-SR4";
        }
        bit b40GbaseFR {
          description
            "40GBASE-FR";
        }
        bit b40GbaseLR4 {
          description
            "40GBASE-LR4";
        }
        bit b100GbaseCR10 {
          description
            "100GBASE-CR10";
        }
        bit b100GbaseSR10 {
          description
            "100GBASE-SR10";
        }
        bit b100GbaseLR4 {
          description
            "100GBASE-LR4";
        }
        bit b100GbaseER4 {
          description
            "100GBASE-ER4";
        }
        bit b1000baseT1 {
          description
            "1000BASE-T1";
        }
        bit b1000basePX30D {
          description
            "1000BASE-PX30D";
        }
        bit b1000basePX30U {
          description
            "1000BASE-PX30U";
        }
        bit b1000basePX40D {
          description
            "1000BASE-PX40D";
        }
```

```
        bit b1000basePX40U {
          description
            "1000BASE-PX40U";
        }
        bit b10G1GbasePRXD4 {
          description
            "10/1GBASE-PRXD4";
        }
        bit b10G1GbasePRXU4 {
          description
            "10/1GBASE-PRXU4";
        }
        bit b10GbasePRD4 {
          description
            "10GBASE-PRD4";
        }
        bit b10GbasePRU4 {
          description
            "10GBASE-PRU4";
        }
        bit b25GbaseCR {
          description
            "25GBASE-CR";
        }
        bit b25GbaseCRS {
          description
            "25GBASE-CRS";
        }
        bit b25GbaseKR {
          description
            "25GBASE-KR";
        }
        bit b25GbaseKRS {
          description
            "25GBASE-KRS";
        }
        bit b25GbaseR {
          description
            "25GBASE-R";
        }
        bit b25GbaseSR {
          description
            "25GBASE-SR";
        }
        bit b25GbaseT {
          description
            "25GBASE-T";
        }
        bit b40GbaseER4 {
          description
            "40GBASE-ER4";
        }
        bit b40GbaseR {
          description
```

```
            "40GBASE-R";
        }
        bit b40GbaseT {
          description
            "40GBASE-T";
        }
        bit b100GbaseCR4 {
          description
            "100GBASE-CR4";
        }
        bit b100GbaseKR4 {
          description
            "100GBASE-KR4";
        }
        bit b100GbaseKP4 {
          description
            "100GBASE-KP4";
        }
        bit b100GbaseR {
          description
            "100GBASE-R";
        }
        bit b100GbaseSR4 {
          description
            "100GBASE-SR4";
        }
        bit b2p5GbaseT {
          description
            "2.5GBASE-T";
        }
        bit b5GbaseT {
          description
            "5GBASE-T";
        }
        bit b100baseT1 {
          description
            "100BASE-T1";
        }
        bit b1000baseRHA {
          description
            "1000BASE-RHA";
        }
        bit b1000baseRHB {
          description
            "1000BASE-RHB";
        }
        bit b1000baseRHC {
          description
            "1000BASE-RHC";
        }
        bit b2p5GbaseKX {
          description
            "2.5GBASE-KX";
        }
```

```
      bit b2p5GbaseX {
        description
          "2.5GBASE-X";
      }
      bit b5GbaseKR {
        description
          "5GBASE-KR";
      }
      bit b5GbaseR {
        description
          "5GBASE-R";
      }
      bit b10GpassXR {
        description
          "10GPASS-XR";
      }
      bit b25GbaseLR {
        description
          "25GBASE-LR";
      }
      bit b25GbaseER {
        description
          "25GBASE-ER";
      }
      bit b50GbaseR {
        description
          "50GBASE-R";
      }
      bit b50GbaseCR {
        description
          "50GBASE-CR";
      }
      bit b50GbaseKR {
        description
          "50GBASE-KR";
      }
      bit b50GbaseSR {
        description
          "50GBASE-SR";
      }
      bit b50GbaseFR {
        description
          "50GBASE-FR";
      }
      bit b50GbaseLR {
        description
          "50GBASE-LR";
      }
      bit b50GbaseER {
        description
          "50GBASE-ER";
      }
      bit b100GbaseCR2 {
        description
```

```
            "100GBASE-CR2";
        }
        bit b100GbaseKR2 {
          description
            "100GBASE-KR2";
        }
        bit b100GbaseSR2 {
          description
            "100GBASE-SR2";
        }
        bit b100GbaseDR {
          description
            "100GBASE-DR";
        }
        bit b200GbaseR {
          description
            "200GBASE-R";
        }
        bit b200GbaseDR4 {
          description
            "200GBASE-DR4";
        }
        bit b200GbaseFR4 {
          description
            "200GBASE-FR4";
        }
        bit b200GbaseLR4 {
          description
            "200GBASE-LR4";
        }
        bit b200GbaseCR4 {
          description
            "200GBASE-CR4";
        }
        bit b200GbaseKR4 {
          description
            "200GBASE-KR4";
        }
        bit b200GbaseSR4 {
          description
            "200GBASE-SR4";
        }
        bit b200GbaseER4 {
          description
            "200GBASE-ER4";
        }
        bit b400GbaseR {
          description
            "400GBASE-R";
        }
        bit b400GbaseSR16 {
          description
            "400GBASE-SR16";
        }
```

```
        bit b400GbaseDR4 {
          description
            "400GBASE-DR4";
        }
        bit b400GbaseFR8 {
          description
            "400GBASE-FR8";
        }
        bit b400GbaseLR8 {
          description
            "400GBASE-LR8";
        }
        bit b400GbaseER8 {
          description
            "400GBASE-ER8";
        }
        bit b10baseT1L {
          description
            "10BASE-T1L";
        }
        bit b10baseT1S {
          description
            "10BASE-T1S";
        }
        bit b10baseT1SMD {
          description
            "10BASE-T1SMD";
        }
        bit b100GbaseFR1 {
          description
            "100GBASE-FR1";
        }
        bit b100GbaseLR1 {
          description
            "100GBASE-LR1";
        }
        bit b100GBASE-CR1 {
          description
            "100GBASE-LR1";
        }
        bit b400GbaseFR4 {
          description
            "400GBASE-FR4";
        }
        bit b400GbaseLR46 {
          description
            "400GBASE-LR46";
        }
        bit b400GbaseSR8 {
          description
            "400GBASE-SR8";
        }
        bit b400GbaseSR4p2 {
          description
```

```
            "400GBASE-SR4p2";
        }
        bit b2p5GbaseT1 {
          description
            "2.5GBASE-T1";
        }
        bit b5GbaseT1 {
          description
            "5GBASE-T1";
        }
        bit b10GbaseT1 {
          description
            "10GBASE-T1";
        }
        bit b25G10GbasePQGD2 {
          description
            "25/10GBASE-PQGD2";
        }
        bit b25G10GbasePQGD3 {
          description
            "25/10GBASE-PQGD3";
        }
        bit b25G10GbasePQGU2 {
          description
            "25/10GBASE-PQGU2";
        }
        bit b25G10GbasePQGU3 {
          description
            "25/10GBASE-PQGU3";
        }
        bit b25G10GbasePQXD2 {
          description
            "25/10GBASE-PQXD2";
        }
        bit b25G10GbasePQXD3 {
          description
            "25/10GBASE-PQXD3";
        }
        bit b25G10GbasePQXU2 {
          description
            "25/10GBASE-PQXU2";
        }
        bit b25G10GbasePQXU3 {
          description
            "25/10GBASE-PQXU3";
        }
        bit b25GbasePQGD2 {
          description
            "25GBASE-PQGD2";
        }
        bit b25GbasePQGD3 {
          description
            "25GBASE-PQGD3";
        }
```

```
      bit b25GbasePQGU2 {
        description
          "25GBASE-PQGU2";
      }
      bit b25GbasePQGU3 {
        description
          "25GBASE-PQGU3";
      }
      bit b25GbasePQXD2 {
        description
          "25GBASE-PQXD2";
      }
      bit b25GbasePQXD3 {
        description
          "25GBASE-PQXD3";
      }
      bit b25GbasePQXU2 {
        description
          "25GBASE-PQXU2";
      }
      bit b25GbasePQXU3 {
        description
          "25GBASE-PQXU3";
      }
      bit b50G10GbasePQGD2 {
        description
          "50/10GBASE-PQGD2";
      }
      bit b50G10GbasePQGD3 {
        description
          "50/10GBASE-PQGD3";
      }
      bit b50G10GbasePQGU2 {
        description
          "50/10GBASE-PQGU2";
      }
      bit b50G10GbasePQGU3 {
        description
          "50/10GBASE-PQGU3";
      }
      bit b50G10GbasePQXD2 {
        description
          "50/10GBASE-PQXD2";
      }
      bit b50G10GbasePQXD3 {
        description
          "50/10GBASE-PQXD3";
      }
      bit b50G10GbasePQXU2 {
        description
          "50/10GBASE-PQXU2";
      }
      bit b50G10GbasePQXU3 {
        description
```

```
            "50/10GBASE-PQXU3";
        }
        bit b50G25GbasePQGD2 {
          description
            "50/25GBASE-PQGD2";
        }
        bit b50G25GbasePQGD3 {
          description
            "50/25GBASE-PQGD3";
        }
        bit b50G25GbasePQGU2 {
          description
            "50/25GBASE-PQGU2";
        }
        bit b50G25GbasePQGU3 {
          description
            "50/25GBASE-PQGU3";
        }
        bit b50G25GbasePQXD2 {
          description
            "50/25GBASE-PQXD2";
        }
        bit b50G25GbasePQXD3 {
          description
            "50/25GBASE-PQXD3";
        }
        bit b50G25GbasePQXU2 {
          description
            "50/25GBASE-PQXU2";
        }
        bit b50G25GbasePQXU3 {
          description
            "50/25GBASE-PQXU3";
        }
        bit b50GbasePQGD2 {
          description
            "50GBASE-PQGD2";
        }
        bit b50GbasePQGD3 {
          description
            "50GBASE-PQGD3";
        }
        bit b50GbasePQGU2 {
          description
            "50GBASE-PQGU2";
        }
        bit b50GbasePQGU3 {
          description
            "50GBASE-PQGU3";
        }
        bit b50GbasePQXD2 {
          description
            "50GBASE-PQXD2";
        }
```

```
        bit b50GbasePQXD3 {
          description
            "50GBASE-PQXD3";
        }
        bit b50GbasePQXU2 {
          description
            "50GBASE-PQXU2";
        }
        bit b50GbasePQXU3 {
          description
            "50GBASE-PQXU3";
        }
        bit b100GbaseZR {
          description
            "100GBASE-ZR";
        }
        bit b10GbaseBR10D {
          description
            "10GBASE-BR10D";
        }
        bit b10GbaseBR10U {
          description
            "10GBASE-BR10U";
        }
        bit b10GbaseBR20D {
          description
            "10GBASE-BR20D";
        }
        bit b10GbaseBR20U {
          description
            "10GBASE-BR20U";
        }
        bit b10GbaseBR40D {
          description
            "10GBASE-BR40D";
        }
        bit b10GbaseBR40U {
          description
            "10GBASE-BR40U";
        }
        bit b25GbaseBR10D {
          description
            "25GBASE-BR10D";
        }
        bit b25GbaseBR10U {
          description
            "25GBASE-BR10U";
        }
        bit b25GbaseBR20D {
          description
            "25GBASE-BR20D";
        }
        bit b25GbaseBR20U {
          description
```

```
            "25GBASE-BR20U";
        }
        bit b25GbaseBR40D {
          description
            "25GBASE-BR40D";
        }
        bit b25GbaseBR40U {
          description
            "25GBASE-BR40U";
        }
        bit b50GbaseBR10D {
          description
            "50GBASE-BR10D";
        }
        bit b50GbaseBR10U {
          description
            "50GBASE-BR10U";
        }
        bit b50GbaseBR20D {
          description
            "50GBASE-BR20D";
        }
        bit b50GbaseBR20U {
          description
            "50GBASE-BR20U";
        }
        bit b50GbaseBR40D {
          description
            "50GBASE-BR40D";
        }
        bit b50GbaseBR40U {
          description
            "50GBASE-BR40U";
        }
        /* 802.3ck-2022 */
        bit b100GBaseCR1 {
          description
            "100GBASE-CR1";
        }
        bit b100GBaseKR1 {
          description
            "100GBASE-KR1";
        }
        bit b200GBaseCR2 {
          description
            "200GBASE-CR2";
        }
        bit b200GBaseKR2 {
          description
            "200GBASE-KR2";
        }
        bit b400GBaseCR4 {
          description
            "400GBASE-CR4";
```

```
      }
      bit b400GBaseKR4 {
        description
          "400GBASE-KR4";
      }
      /* 802.3cs-2022 */
      bit b10G2p5GBaseSP1Dx {
        description
          "10/2.5GBASE-SP1-Dx";
      }
      bit b10G2p5GBaseSP1Uxy {
        description
          "10/2.5GBASE-SP1-Uxy";
      }
      bit b10GBaseSP1Dx {
        description
          "10GBASE-SP1-Dx";
      }
      bit b10GBaseSP1Uxy {
        description
          "10GBASE-SP1-Uxy";
      }
      /* 802.3db-2022 */
      bit b100GBaseSR1 {
        description
          "100GBASE-SR1";
      }
      bit b100GBaseVR1 {
        description
          "100GBASE-VR1";
      }
      bit bit200GBaseSR2 {
        description
          "200GBASE-SR2";
      }
      bit bit200GBaseVR2 {
        description
          "200GBASE-VR2";
      }
      bit bit400GBaseSR4 {
        description
          "400GBASE-SR4";
      }
      bit bit400GBaseVR4 {
        description
          "400GBASE-VR4";
      }
      /* 802.3cz-2023 */
      bit b2p5GbaseAU {
        description
          "2.5GBASE-AU";
      }
      bit b5GbaseAU {
        description
```

```
            "5GBASE-AU";
        }
        bit b10GbaseAU {
          description
            "10GBASE-AU";
        }
        bit b25GbaseAU {
          description
            "25GBASE-AU";
        }
        bit b50GbaseAU {
          description
            "50GBASE-AU";
        }
        /* 802.3cy-2023 */
        bit b25GbaseT1 {
          description
            "25GBASE-T1";
        }
        /* 802.3ca-2023 */
        bit bit400GBaseDR4d2 {
          description
            "400GBASE-DR4-2";
        }
        bit bit800GBaseCR8 {
          description
            "800GBASE-CR8";
        }
        bit bit800GBaseDR8 {
          description
            "800GBASE-DR8";
        }
        bit bit800GBaseDR8d2 {
          description
            "800GBASE-DR8-2";
        }
        bit bit800GBaseKR8 {
          description
            "800GBASE-KR8";
        }
        bit bit800GBaseR {
          description
            "800GBASE-R";
        }
        bit bit800GBaseSR8 {
          description
            "800GBASE-SR8";
        }
        bit bit800GBaseVR8 {
          description
            "800GBASE-VR8";
        }
      }
      description
```

```
      "This data type is used to report the set of pmd-type values
       an interface can implement.";
    reference
      "IEEE Std 802.3, Clause 30.5.1.1.3 aMAUTypeList";
  }

  typedef media-available-type {
    type enumeration {
      /* 802.3-2022 */
      enum other {
        description
          "Undefined";
      }
      enum unknown {
        description
          "initializing, true state not yet known";
      }
      enum available {
        description
          "link or light normal, loopback normal";
      }
      enum notAvailable {
        description
          "link loss or low light, no loopback";
      }
      enum remoteFault {
        description
          "remote fault with no detail";
      }
      enum remoteJabber {
        description
          "remote fault, reason known to be jabber";
      }
      enum remoteLinkLoss {
        description
          "remote fault, reason known to be far-end link loss";
      }
      enum remoteTest {
        description
          "remote fault, reason known to be test";
      }
      enum offline {
        description
          "offline, applies only to Clause 37 Auto-Negotiation";
      }
      enum autoNegError {
        description
          "Auto-Negotiation Error, applies only to Clause 37
                auto-Negotiation";
      }
      enum pmdLinkFault {
        description
          "PMD/PMA receive link fault";
      }
```

```
        enum pcsLinkFault {
          description
            "PCS receive link fault";
        }
        enum excessiveBER {
          description
            "PCS Bit Error Ratio monitor reporting excessive error
             ratio";
        }
        enum dxsLinkFault {
          description
            "DTE XGXS receive link fault, applies only to XAUI ";
        }
        enum pxsLinkFault {
          description
            "PHY XGXS transmit link fault, applies only to XAUI";
        }
      }
      description
        "This data type is used to report media availability.";
      reference
        "IEEE Std 802.3, Clause 30.5.1.1.4 aMediaAvailable";
    }

    typedef auto-neg-ability-bits {
      type bits {
        /* Not PHY specific */
        bit bOther {
          description
            "Other abilities beyond the scope of this module.";
        }
        bit bHalfDuplex {
          description
            "Half duplex";
        }
        bit bFullDuplex {
          description
            "Full duplex";
        }
        bit bPause {
          description
            "PAUSE operation as defined in Annex 31B";
        }
        bit bAPause {
          description
            "Asymmetric PAUSE operation as defined in
             Clause 37, Annex 28B, and Annex 31B";
        }
        bit bSPause {
          description
            "Symmetric PAUSE operation as defined in
             Clause 37, Annex 28B, and Annex 31B";
        }
        bit bBPause {
```

```
          description
            "Asymmetric and Symmetric PAUSE operation
             as defined in Clause 37, Annex 28B, and Annex 31B";
        }
        bit bRemFault {
          description
            "Remote fault bit (RF) as specified in Clause 73
             and Clause 98";
        }
        bit bRemFault1 {
          description
            "Remote fault bit 1 (RF1) as specified in Clause 37";
        }
        bit bRemFault2 {
          description
            "Remote fault bit 2 (RF2) as specified in Clause 37";
        }
        bit bFECCapable {
          description
            "FEC ability as specified in Clause 73 (see 73.6.5)
             and Clause 74";
        }
        bit bFECRequested {
          description
            "FEC requested as specified in Clause 73 (see 73.6.5)
             and Clause 74";
        }
        bit bRSFEC25Greq {
          description
            "25G RS-FEC requested as specified in Clause 73 (see 73.6.5)
             and Clause 108";
        }
        bit bBaseFEC25Greq {
          description
            "25Gb/s BASE-R FEC ";
        }
        bit bForceMS {
          description
            "Force MASTER-SLAVE as specified in Clause 98 (see
             98.2.1.2.5)";
        }
        /* PHY/PMD  */
        bit b10baseT {
          description
            "10BASE-T as defined in Clause 14";
        }
        bit b10baseT1L {
          description
            "10BASE-T1L as specified in Clause 146";
        }
        bit b10baseT1S {
          description
            "10BASE-T1S as specified in Clause 147";
        }
```

```
      bit b100baseTX {
        description
          "100BASE-TX as defined in Clause 25";
      }
      bit b1000baseX {
        description
          "1000BASE-X half duplex as specified in Clause 36";
      }
      bit b1000baseT {
        description
          "1000BASE-T PHY as specified in Clause 40";
      }
      bit b1000baseT1 {
        description
          "1000BASE-T1 as specified in Clause 97";
      }
      bit b2p5GbaseT {
        description
          "2.5GBASE-T PHY as specified in Clause 126";
      }
      bit b2p5GbaseT1 {
        description
          "2.5GBASE-T1 as specified in Clause 149";
      }
      bit b2p5GbaseKX {
        description
          "2.5GBASE-KX as specified in Clause 128";
      }
      bit b5GBaseT {
        description
          "5GBASE-T PHY as specified in Clause 126";
      }
      bit b5GbaseT1 {
        description
          "5GBASE-T1 as specified in Clause 149";
      }
      bit b5GbaseKR {
        description
          "5GBASE-KR as specified in Clause 130";
      }
      bit b10GbaseT {
        description
          "10GBASE-T PHY as specified in Clause 55";
      }
      bit b1000baseKX {
        description
          "1000BASE-KX as specified in Clause 70";
      }
      bit b10GbaseKX4 {
        description
          "10GBASE-KX4 as specified in Clause 71";
      }
      bit b10GbaseKR {
        description
```

```
            "10GBASE-KR as specified in Clause 72";
        }
        bit b25GbaseT {
          description
            "25GBASE-T as specified in Clause 113";
        }
        bit b25GbaseRS {
          description
            "25GBASE-CR-S as specified in Clause 110 or
             25GBASE-KR-S as specified in Clause 111";
        }
        bit b25GbaseR {
          description
            "25GBASE-CR as specified in Clause 110 or
             25GBASE-KR as specified in Clause 111";
        }
        bit b25GbaseT1 {
          description
            "25GBASE-T1 as specified in Clause 165";
        }
        bit b40GbaseKR4 {
          description
            "40GBASE-KR4 as specified in Clause 84";
        }
        bit b40GbaseCR4 {
          description
            "40GBASE-CR4 as specified in Clause 85";
        }
        bit b40GbaseT {
          description
            "40GBASE-T as specified in Clause 113";
        }
        bit b50GbaseR {
          description
            "50GBASE-CR as specified in Clause 136 or
             50GBASE-KR as specified in Clause 137";
        }
        bit b100GbaseCR10 {
          description
            "100GBASE-CR10 as specified in Clause 85";
        }
        bit b100GbaseCR4 {
          description
            "100GBASE-CR4 as specified in Clause 92";
        }
        bit b100GbaseKR4 {
          description
            "100GBASE-KR4 as specified in Clause 93";
        }
        bit b100GbaseKP4 {
          description
            "100GBASE-KP4 as specified in Clause 94";
        }
        bit b100GbaseR2 {
```

```
        description
          "100GBASE-CR2 as specified in Clause 136 or
           100GBASE-KR2 as specified in Clause 137";
      }
      bit b200GbaseR4 {
        description
          "200GBASE-CR4 as specified in Clause 136
           or 200GBASE-KR4 as specified in Clause 137";
      }
    }
    description
      "This data type is used as the syntax of the
       ieee802-eth-phy-auto-neg-capability-bits,
       ieee802-eth-phy-auto-neg-cap-advertised-bits,
       and ieee802-eth-phy-auto-neg-cap-received-bits
       objects in ieee802-eth-phy-auto-neg-table.";
    reference
      "IEEE Std 802.3, Clause 30.6.1.1.5";
  }
}
```

6. YANG module for Ethernet data terminal equipment (DTE) power via medium dependent interface (MDI) and Power over Data Lines (PoDL)

6.1 Introduction

This clause defines a YANG module to manage power via MDI Power Sourcing Equipment (PSE) and Power over Data Line (PoDL) PSE.

IEEE Std 802.3 defines the hardware registers that allow management interfaces to be built for a DTE Power via MDI and Power over Data Line device. The YANG module defined in this clause extends the Ethernet-interface YANG data modules defined in Clause 5 with the management objects required for the management of PoE and PoDL devices and ports.

6.2 YANG module structure

The deprecated *ieee802-ethernet-pse superseded by ieee802-ethernet-pse-2* YANG module of this clause is focused on the configuration and monitoring of the Power over Ethernet (PoE) function defined in IEEE Std 802.3, including power via MDI, as well as Power over Data Line which can also be considered as the single pair PoE. The module augments the *ieee802-ethernet-interface* YANG module with attributes for the PoE function. The module is partitioned into two major containers.

The PoE PSE container describes a multi-pair PSE, while the PoDL PSE describes a single-pair PSE.

6.3 Security considerations for Ethernet data terminal equipment (DTE) power via medium dependent interface (MDI) and Power over Data Line Module

There are a number of data nodes defined in this YANG module that are configurable as read-write. Such data nodes can be considered sensitive or vulnerable in some network environments. The support for configuration operations in a non-secure environment without proper protection can have a negative effect on network operations.

Setting the following data nodes to incorrect values can result in improper operation of the PSE, including the possibility that the Powered Device (PD) does not receive power from the PSE port:

— pse-enable
— powering-pairs

Some of the readable operational states in this module can be considered sensitive or vulnerable in some network environments. These are as follows:

— pairs-control-ability
— classifications
— pd-power-class
— pse-type
— detected-pd-type

It is thus important to control GET access to these data nodes and to possibly encrypt their values when sending them over the network.

6.4 Mapping of IEEE Std 802.3, Clause 30 managed objects

This subclause contains the mapping between YANG data nodes included in *ieee802-ethernet-pse* (see Table 6–1) YANG module, managed objects, and attributes defined in IEEE Std 802.3, Clause 30.

Table 6–1—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-pse* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-pse* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
N/A	N/A		interfaces/interface/ethernet/pse	supported-pse-type	R
oPSE	aPSEAdminState	30.9.1	interfaces/interface/ethernet/pse/multi-pair	pse-enable	R
	aPSEPowerPairs			powering-pairs	R/W
	aPSEPowerPairsControlAbility			pairs-control-ability	R
	aPSEPowerDetectionStatus			detection-status	R
	aPSEPowerClassification			classifications	R
	aPSEActualPower			actual-power	R
	aPSEPowerAccuracy			power-accuracy	R
	aPSEInvalidSignatureCounter		interfaces/interface/ethernet/pse/multi-pair/statistics	invalid-signature	R
	aPSEPowerDeniedCounter			power-denied	R
	aPSEOverLoadCounter			overload	R
	aPSEShortCounter			short	R
	aPSEMPSAbsentCounter			mps-absent	R
	aPSECumulativeEnergy			cumulative-energy	R

Table 6–1—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-pse* YANG data nodes *(continued)*

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-pse* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
oPoDLPSE	aPoDLPSEAdminState	30.15	interfaces/interface/ethernet/pse/single-pair	pse-enable	R
	aPoDLPSEPowerDetectionStatus			detection-status	R
	aPoDLPSEType			podl-type	R
	aPoDLPSEDetectedPDType			detected-pd-type	R
	aPoDLPSEDetectedPDPowerClass			pd-power-class	R
	aPoDLPSEActualPower			actual-power	R
	aPoDLPSEPowerAccuracy			power-accuracy	R
	aPoDLPSEInvalidSignatureCounter		interfaces/interface/ethernet/pse/single-pair/statistics	invalid-signature	R
	aPoDLPSEInvalidClassCounter			invalid-class	R
	aPoDLPSEPowerDeniedCounter			power-denied	R
	aPoDLPSEOverLoadCounter			overload	R
	aPoDLPSEMaintainFullVoltageSignatureAbsentCounter			fvs-absent	R
	aPoDLPSECumulativeEnergy			cumulative-energy	R

6.5 YANG module definition[i]

The YANG module tree hierarchy uses terms defined in IETF RFC 8407.

6.5.1 Tree hierarchy

6.5.1.1 ieee802-ethernet-pse

```
module: ieee802-ethernet-pse

  augment /if:interfaces/if:interface/ieee802-eth-if:ethernet:
    +--rw pse
       +--ro supported-pse-type?   identityref
       x--rw multi-pair!
       |  +--rw pse-enable?              boolean
       |  +--rw powering-pairs?          identityref
       |  +--ro pairs-control-ability?   boolean
       |  x--ro detection-status?        multi-pair-detection-state
       |  x--ro classifications?         power-class
       |  x--ro statistics
       |  |  +--ro power-denied?        yang:counter64
       |  |  +--ro invalid-signature?   yang:counter64
       |  |  +--ro mps-absent?          yang:counter64
       |  |  x--ro overload?            yang:counter64
       |  |  x--ro short?               yang:counter64
       |  |  +--ro cumulative-energy?   yang:counter64
       |  +--ro actual-power?            decimal64
       |  +--ro power-accuracy?          int64
       x--rw single-pair!
          x--rw pse-enable?         boolean
          x--ro detection-status?   single-pair-detection-state
          +--ro podl-type?          enumeration
          +--ro detected-pd-type?   enumeration
          x--ro pd-power-class?     power-class
          x--ro statistics
          |  +--ro power-denied?        yang:counter64
```

[i]Copyright release for YANG modules: Users of this standard may freely reproduce the YANG module contained in this subclause so that it can be used for its intended purpose.

```
          |  +--ro invalid-signature?   yang:counter64
          |  +--ro invalid-class?       yang:counter64
          |  +--ro overload?            yang:counter64
          |  +--ro fvs-absence?         yang:counter64
          |  +--ro cumulative-energy?   yang:counter64
          +--ro actual-power?       decimal64
          +--ro power-accuracy?     int64
```

6.5.1.2 ieee802-ethernet-pse-2

```
module: ieee802-ethernet-pse-2

  augment /if:interfaces/if:interface/ieee802-eth-if:ethernet:
    +--rw pse-2
       +--ro supported-pse-type?   pse-support
       +--rw multi-pair! {multi-pair-pse}?
       |  +--rw pse-enable?                 boolean
       |  +--ro pse-state?                  boolean
       |  +--rw multi-pair-powering-pairs?   multi-pair-powering-pairs
       |  +--ro pairs-control-ability?      boolean
       |  +--ro detection-status?           multi-pair-detection-state
       |  +--ro classifications?            multi-pair-power-class
       |  +--ro statistics
       |  |  +--ro power-denied?        yang:counter64
       |  |  +--ro invalid-signature?   yang:counter64
       |  |  +--ro mps-absent?          yang:counter64
       |  |  x--ro overload?            yang:counter64
       |  |  x--ro short?               yang:counter64
       |  |  +--ro cumulative-energy?   yang:counter64
       |  +--ro actual-power?               uint32
       |  +--ro power-accuracy?             int64
       +--rw single-pair! {single-pair-pse}?
          +--rw pse-enable?         boolean
          +--ro detection-status?   single-pair-detection-state
          +--ro single-pair-type?   single-pair-pse-type
          +--ro detected-pd-type?   single-pair-pd-type
          +--ro pd-power-class?     single-pair-power-class
```

```
+--ro statistics
|  +--ro power-denied?         yang:counter64
|  +--ro invalid-signature?    yang:counter64
|  +--ro invalid-class?        yang:counter64
|  +--ro overload?             yang:counter64
|  +--ro fvs-absence?          yang:counter64
|  +--ro cumulative-energy?    yang:counter64
+--ro actual-power?        uint32
+--ro power-accuracy?      int64
```

6.5.2 YANG module

In the YANG module definition, should any discrepancy between the text of the description for individual YANG nodes in the YANG modules of this clause and the corresponding definition in 6.2 through 6.5 of this clause occur, the definitions in the YANG modules attached to this standard shall take precedence.

An ASCII text version of the YANG module can be found at the following URL:[j]

> ***Editor's Note (to be removed prior to publication):***
>
> *Yang files contained in https://github.com/YangModels/yang/tree/main/standard/ieee/published/802.3 are IEEE 802.3.2-2019 version and will be updated at the publication time.*

https://github.com/YangModels/yang/tree/master/standard/ieee/published/802.3

6.5.2.1 Ethernet PSE Module

> ***Editor's Note (to be removed prior to publication):***
>
> *Pretty printing of ieee802-ethernet-pse.yang file may change the appearance by adding whitespace and reformatting lines*

> ***Editor's Note (to be removed prior to publication):***
>
> IEEE Std 802.3.1 and IEEE Std 802.3.2 to be updated at the publication time

```
module ieee802-ethernet-pse {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-pse";
  prefix ieee802-pse;

  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
  }
  import ieee802-ethernet-interface {
    prefix ieee802-eth-if;
  }

  organization
    "IEEE 802.3 Ethernet Working Group
     Web URL: http://www.ieee802.org/3/";
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
    "This module is deprecated and superceeded by
         ieee802-ethernet-pse-2.
```

[j]Copyright release for YANG modules: Users of this standard may freely reproduce the YANG module contained in this subclause so that it can be used for its intended purpose.

```
        This module contains YANG definitions for configuring and
    managing ports with Power Over Ethernet feature defined by
    IEEE 802.3. It provides functionality roughly equivalent to
    that of the POWER-ETHERNET-MIB defined in IETF RFC 3621.";

  revision 2025-04-17 {
    description
      "Deprecated in IEEE Std 802.3.2-202x, Draft D3.3, superceeded by
          ieee802-ethernet-pse-2";
    reference
      "IEEE Std IEEE Std 802.3.2-202x, unless dated explicitly";
  }

  identity pse-type {
    description
      "Base type for PSE.";
  }

  identity all {
    base powering-pairs;
    description
      "All pairs are in use.";
  }

  identity four-pair {
    base pse-type;
    description
      "PSE support IEEE Std 802.3, Clause 145.";
  }

  identity two-pair {
    base pse-type;
    description
      "PSE supports IEEE Std 802.3, Clause 33.";
  }

  identity single-pair {
    base pse-type;
    description
      "PSE support IEEE Std 802.3, Clause 104.";
  }

  identity powering-pairs {
    description
      "Base type for powering pairs.";
  }

  identity signal {
    base powering-pairs;
    description
      "The signal pairs are in use.";
  }

  identity spare {
    base powering-pairs;
    description
      "The spare pairs are in use.";
  }
```

```
  typedef multi-pair-detection-state {
    type enumeration {
      enum disabled {
        value 1;
        description
          "PSE disabled.";
      }
      enum searching {
        value 2;
        description
          "PSE is searching.";
      }
      enum deliveringPower {
        value 3;
        description
          "PSE is delivering power.";
      }
      enum fault {
        value 4;
        description
          "PSE fault detected.";
      }
      enum test {
        value 5;
        description
          "PSE test mode.";
      }
      enum otherFault {
        value 6;
        description
          "PSE implementation specific fault detected.";
      }
    }
    status deprecated;
    description
      "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded by
           ieee802-ethernet-pse-2
           Detection state of a multi-pair PSE.";
    reference
      "IEEE Std 802.3, 30.9.1.1.5";
  }

  typedef single-pair-detection-state {
    type enumeration {
      enum unknown {
        value 1;
        description
          "True detection state unknown.";
      }
      enum disabled {
        value 2;
        description
          "PoDL PSE is disabled.";
      }
      enum searching {
        value 3;
        description
          "PoDL PSE is searching.";
      }
```

```
      enum deliveringPower {
        value 4;
        description
          "PoDL PSE is delivering power.";
      }
      enum sleep {
        value 5;
        description
          "PoDL PSE is in sleep state.";
      }
      enum idle {
        value 6;
        description
          "PoDL PSE is idle.";
      }
      enum error {
        value 7;
        description
          "PoDL PSE error.";
      }
    }
    status deprecated;
    description
      "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded by
          ieee802-ethernet-pse-2
       Detection state of a PoDL PSE.";
    reference
      "IEEE Std 802.3, 30.15.1.1.3";
  }

  typedef power-class {
    type enumeration {
      enum class0 {
        value 1;
        description
          "Class 0";
      }
      enum class1 {
        value 2;
        description
          "Class 1";
      }
      enum class2 {
        value 3;
        description
          "Class 2";
      }
      enum class3 {
        value 4;
        description
          "Class 3";
      }
      enum class4 {
        value 5;
        description
          "Class 4";
      }
      enum class5 {
        value 6;
```

```
          description
            "Class 5 (for PoDL-only)";
        }
        enum class6 {
          value 7;
          description
            "Class 6 (for PoDL-only)";
        }
        enum class7 {
          value 8;
          description
            "Class 7 (for PoDL-only)";
        }
        enum class8 {
          value 9;
          description
            "Class 8 (for PoDL-only)";
        }
        enum class9 {
          value 10;
          description
            "Class 9 (for PoDL-only)";
        }
        enum unknown {
          value 11;
          description
            "Initializing, true Power Class not yet known
             (only for PoDL PSE).";
        }
      }
      status deprecated;
      description
        "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded by
            ieee802-ethernet-pse-2
            The power class.";
      reference
        "IEEE Std 802.3, 30.9.1.1.6 aPSEPowerClassification and
         IEEE Std 802.3, 30.15.1.1.6 aPoDLPSEDetectedPDPowerClass.";
    }

    augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet" {
      status deprecated;
      description
        "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded by
             ieee802-ethernet-pse-2
             Augments ethernet interface configuration model with
         nodes specific to DTE Power via MDI devices and ports";
      container pse {
        description
          "DTE Power via MDI port configuration";
        reference
          "IEEE Std 802.3, 30.9.1 PoE PSE & IEEE Std 802.3, 30.15.1
           PoDL PSE";
        leaf supported-pse-type {
          type identityref {
            base ieee802-pse:pse-type;
          }
          config false;
          description
```

```
          "PSE supports one or more of IEEE Std 802.3 Clause 33,
           Clause 104, or Clause 145.";
      }
      container multi-pair {
        presence "PSE port supports IEEE Std 802.3, Clause 33.";
        status deprecated;
        description
          "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded by
           ieee802-ethernet-pse-2PSE port configuration in IEEE Std 802.3,
           30.9.1.";
        leaf pse-enable {
          type boolean;
          default "false";
          description
            "When true enables the PSE function on the interface,
             when false disables the PSE function on the
             interface.";
          reference
            "IEEE Std 802.3, 30.9.1.1.2 aPSEAdminState";
        }
        leaf powering-pairs {
          type identityref {
            base powering-pairs;
          }
          description
            "Describes or controls the PSE pairs in use. If the
             value of pairs-control-ability is true, this object
             is writeable.";
          reference
            "IEEE Std 802.3, 30.9.1.1.4 aPSEPowerPairs";
        }
        leaf pairs-control-ability {
          type boolean;
          default "true";
          config false;
          description
            "Describes the ability to control switching the
             power sourcing pins of the PSE.";
          reference
            "IEEE Std 802.3, 30.9.1.1.3
             aPSEPowerPairsControlAbility";
        }
        leaf detection-status {
          type multi-pair-detection-state;
          config false;
          status deprecated;
          description
            "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded by
             ieee802-ethernet-pse-2
             Describes the operational status of the port
             PD detection.";
          reference
            "IEEE Std 802.3, 30.9.1.1.5 aPSEPowerDetectionStatus";
        }
        leaf classifications {
          when "../detection-status = 'deliveringPower'" {
            description
              "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded by
               ieee802-ethernet-pse-2
```

```
              This node only applies when the detection status is
              delivering power.";
          }
          type power-class;
          config false;
          status deprecated;
          description
            "The power class of the PSE port.";
          reference
            "IEEE Std 802.3, 30.9.1.1.6 aPSEPowerClassfication";
        }
        container statistics {
          config false;
          status deprecated;
          description
            "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded by
             ieee802-ethernet-pse-2statistics information of the multi-pair
             port.";
          leaf power-denied {
            type yang:counter64;
            description
              "This counter is incremented when the PSE state
               diagram enters the POWER_DENIED state, per
               IEEE Std 802.3, Figure 33-9.";
            reference
              "IEEE Std 802.3, 30.9.1.1.14";
          }
          leaf invalid-signature {
            type yang:counter64;
            description
              "This counter is incremented when the PSE state
               diagram enters the SIGNATURE_INVALID state per
               IEEE Std 802.3, Figure 33-9.";
            reference
              "IEEE Std 802.3, 30.9.1.1.11";
          }
          leaf mps-absent {
            type yang:counter64;
            description
              "This counter is incremented when the PSE
               transitions directly from the POWER_ON state to the
               IDLE state due to tmpdo_timer_done being asserted,
               per IEEE Std 802.3, Figure 33-9.";
            reference
              "IEEE Std 802.3, 30.9.1.1.20";
          }
          leaf overload {
            type yang:counter64;
            status deprecated;
            description
              "This counter is incremented when the PSE state
               diagram enters the ERROR_DELAY state due to the
               ovld_detected variable being TRUE, per
               IEEE Std 802.3, Figure 33-9.";
            reference
              "IEEE Std 802.3, 30.9.1.1.17";
          }
          leaf short {
            type yang:counter64;
```

```
            status deprecated;
            description
              "This Yang object is deprecated as its not defined in
               base standard.
               This counter is incremented when the PSE state
               diagram enters the ERROR_DELAY state due to the
               short_detected variable being TRUE, per
               IEEE Std 802.3, Figure 33-9.";
            reference
              "IEEE Std 802.3, 30.9.1.1.10 aPSEShortCounter";
          }
          leaf cumulative-energy {
            type yang:counter64;
            units "millijoules";
            description
              "The cumulative energy supplied by the PSE as
               measured at the MDI in millijoules.";
            reference
              "IEEE Std 802.3, 30.9.1.1.25";
          }
        }
        leaf actual-power {
          type decimal64 {
            fraction-digits 4;
          }
          units "milliwatts";
          config false;
          description
            "The actual power drawn by a PD over the port.";
          reference
            "IEEE Std 802.3, 30.9.1.1.23";
        }
        leaf power-accuracy {
          type int64;
          units "milliwatts";
          config false;
          description
            "An integer value indicating the accuracy
             associated with power-accuracy in +/- milliwatts.";
          reference
            "IEEE Std 802.3, 30.9.1.1.24";
        }
      }
      container single-pair {
        presence "PSE port working in PoDL.";
        status deprecated;
        description
          "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded by
           ieee802-ethernet-pse-2
           PoDL PSE configuration as defined in
           IEEE Std 802.3, 30.15.1.";
        leaf pse-enable {
          type boolean;
          default "false";
          status deprecated;
          description
            "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded by
             ieee802-ethernet-pse-2
             When true enables the PSE function on the interface,
```

```
              when false disables the PSE function on the
              interface.";
          reference
            "IEEE Std 802.3, 30.15.1.1.2 aPoDLPSEAdminState";
        }
        leaf detection-status {
          type single-pair-detection-state;
          config false;
          status deprecated;
          description
            "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded by
              ieee802-ethernet-pse-2
              Indicates the current status of the PoDL PSE.";
          reference
            "IEEE Std 802.3, 30.15.1.1.3
              aPoDLPSEPowerDetectionStatus";
        }
        leaf podl-type {
          type enumeration {
            enum unknown {
              description
                "Unknown PSE type.";
            }
            enum typeA {
              description
                "TypeA PSE";
            }
            enum typeB {
              description
                "TypeB PSE";
            }
            enum typeC {
              description
                "Type PSEC";
            }
            enum typeD {
              description
                "TypeD PSE";
            }
            enum typeE {
              description
                "TypeE PSE";
            }
            enum typeF {
              description
                "TypeF PSE";
            }
          }
          config false;
          description
            "PSE type specified in and
              IEEE Std 802.3,  30.15.1.1.4.";
        }
        leaf detected-pd-type {
          when "../detection-status = 'deliveringPower'" {
            description
              "This node only applies when the detection status is
                delivering power.";
          }
```

```
          type enumeration {
            enum unknown {
              description
                "Unknown PD type";
            }
            enum typeA {
              description
                "TypeA PD";
            }
            enum typeB {
              description
                "TypeB PD";
            }
            enum typeC {
              description
                "TypeC PD";
            }
            enum typeD {
              description
                "TypeD PD";
            }
            enum typeE {
              description
                "TypeE PD";
            }
            enum typeF {
              description
                "TypeF PD";
            }
          }
          config false;
          description
            "Indicates the Type of the detected PoDL PD as
             specified in IEEE Std 802.3, 104.5.1.";
          reference
            "IEEE Std 802.3, 30.15.1.1.5 aPoDLPSEDetectedPDType";
        }
        leaf pd-power-class {
          when "../detection-status = 'deliveringPower'" {
            description
              "This node only applies when the detection status is
               delivering power.";
          }
          type power-class;
          config false;
          status deprecated;
          description
            "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded by
             ieee802-ethernet-pse-2
             Power class of the PD detected on the PSE port.";
          reference
            "IEEE Std 802.3, 30.15.1.1.6
             aPoDLPSEDetectedPDPowerClass";
        }
        container statistics {
          config false;
          status deprecated;
          description
            "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded by
```

```
      ieee802-ethernet-pse-2
      PSE Discontinuities in the values of counters in this
      container can occur at re-initialization of the
      management system, and at other times as indicated by the
      value of the 'discontinuity-time' leaf  defined in the
      ietf-interfaces YANG module (IETF RFC 8343).";
   leaf power-denied {
     type yang:counter64;
     description
       "This counter is incremented when the PoDL PSE state
        diagram variable power_available transitions from
        true to false (see IEEE Std 802.3, 104.4.3.3).";
     reference
       "IEEE Std 802.3, 30.15.1.1.9
        aPoDLPSEPowerDeniedCounter";
   }
   leaf invalid-signature {
     type yang:counter64;
     description
       "This counter is incremented when the PSE state
        diagram enters the SIGNATURE_INVALID state per
        IEEE Std 802.3, Figure 33-9.";
     reference
       "IEEE Std 802.3, 30.15.1.1.7
        aPoDLPSEInvalidSignatureCounter";
   }
   leaf invalid-class {
     type yang:counter64;
     description
       "This counter is incremented when the PoDL PSE state
        diagram variable tclass_timer_done transitions from
        false to true or when the valid_class variable
        transitions from true to false
        (see IEEE Std 802.3, 104.4.3.3).";
     reference
       "IEEE Std 802.3, 30.15.1.1.8
        aPoDLPSEInvalidClassCounter";
   }
   leaf overload {
     type yang:counter64;
     description
       "This counter is incremented when the PSE state
        diagram variable overload_held transitions from
        false to true (see IEEE Std 802.3, 104.4.3.3).";
     reference
       "IEEE Std 802.3, 30.15.1.1.10
        aPoDLPSEOverLoadCounter";
   }
   leaf fvs-absence {
     type yang:counter64;
     description
       "Maintain Full Voltage Signature absent counter.
        This counter is incremented when the PoDL PSE state
        diagram variable mfvs_timeout transitions from false
        to true (see IEEE Std 802.3, 104.4.3.3).";
     reference
       "IEEE Std 802.3, 30.15.1.1.11
        aPoDLPSEMaintainFullVoltageSignatureAbsentCounter";
   }
```

```
          leaf cumulative-energy {
            type yang:counter64;
            units "millijoules";
            description
              "A count of the cumulative energy supplied by the
               PoDL PSE, measured at the MDI, and expressed in
               units of millijoules.";
            reference
              "IEEE Std 802.3, 30.15.1.1.14
               aPoDLPSECumulativeEnergy";
          }
        }
        leaf actual-power {
          type decimal64 {
            fraction-digits 4;
          }
          units "milliwatts";
          config false;
          description
            "An integer value indicating present (actual) power
             being supplied by the PoDL PSE as measured at the MDI
             in milliwatts.";
          reference
            "IEEE Std 802.3, 30.15.1.1.12 aPoDLPSEActualPower";
        }
        leaf power-accuracy {
          type int64;
          units "milliwatts";
          config false;
          description
            "A signed integer value indicating the accuracy
             associated
             with power-accuracy in milliwatts.";
          reference
            "IEEE Std 802.3, 30.15.1.1.13 aPoDLPSEPowerAccuracy";
        }
      }
    }
  }
}
```

59.5.2.2 Ethernet PSE-2 Module

> ***Editor's Note (to be removed prior to publication):***
>
> *Pretty printing of ieee802-ethernet-interface.yang file may change the appearance by adding whitespace and reformatting lines*

> ***Editor's Note (to be removed prior to publication):***
>
> IEEE Std 802.3.1 and IEEE Std 802.3.2 to be updated at the publication time

```
module ieee802-ethernet-pse-2 {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-pse-2";
  prefix ieee802-pse-2024;
```

```
  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
  }
  import ieee802-ethernet-interface {
    prefix ieee802-eth-if;
  }

  organization
    "IEEE 802.3 Ethernet Working Group
     Web URL: http://www.ieee802.org/3/";
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
    "This module contains YANG definitions for configuring and
     managing ports with Power Over Ethernet features defined by
     IEEE Std 802.3-2022.
     This module is based on,and superscedes ieee802-ethernet-pse.yang
     from IEEE Std 802.3.2-2019.";

  revision 2025-04-17 {
    description
      "Updates under IEEE Std 802.3.2-202x, Draft D3.3";
    reference
      "IEEE Std IEEE Std 802.3.2-202x, unless dated explicitly";
  }

  feature multi-pair-pse {
    description
      "This object indicates the support for IEEE Std 802.3
          Clause 33 and/or 145.";
  }

  feature single-pair-pse {
    description
      "This object indicates the support for IEEE Std 802.3
          Clause 104.";
  }

  typedef multi-pair-detection-state {
    type enumeration {
      enum disabled {
        description
          "PSE disabled.";
      }
      enum searching {
        description
          "PSE is searching.";
      }
      enum deliveringPower {
        description
          "PSE is delivering power.";
```

```
      }
      enum fault {
        description
          "PSE fault detected.";
      }
      enum test {
        description
          "PSE test mode.";
      }
      enum otherFault {
        description
          "PSE implementation specific fault detected.";
      }
    }
    description
      "Detection state of a multi-pair PSE.";
    reference
      "IEEE Std 802.3, 30.9.1.1.5";
  }

  typedef single-pair-detection-state {
    type enumeration {
      enum unknown {
        description
          "True detection state unknown.";
      }
      enum disabled {
        description
          "PSE is disabled.";
      }
      enum searching {
        description
          "PSE is searching.";
      }
      enum deliveringPower {
        description
          "PSE is delivering power.";
      }
      enum sleep {
        description
          "PSE is in sleep state.";
      }
      enum idle {
        description
          "PSE is idle.";
      }
      enum error {
        description
          "PSE error.";
      }
    }
    description
      "Detection state of a single-pair PSE.";
    reference
      "IEEE Std 802.3, 30.15.1.1.3";
  }

  typedef multi-pair-power-class {
    type enumeration {
```

```
      enum class0 {
        description
          "Class 0";
      }
      enum class1 {
        description
          "Class 1";
      }
      enum class2 {
        description
          "Class 2";
      }
      enum class3 {
        description
          "Class 3";
      }
      enum class4 {
        description
          "Class 4";
      }
      enum class5 {
        description
          "Class 5";
      }
      enum class6 {
        description
          "Class 6";
      }
      enum class7 {
        description
          "Class 7";
      }
      enum class8 {
        description
          "Class 8";
      }
    }
    description
      "Multi-pair PoE power class.";
    reference
      "IEEE Std 802.3, 30.9.1.1.6 aPSEPowerClassification ";
  }

  typedef pse-support {
    type enumeration {
      enum two-pair {
        description
          "PSE port supports IEEE Std 802.3, Clause 33.";
      }
      enum four-pair {
        description
          "PSE port supports IEEE Std 802.3, Clause 33 and 145.";
      }
      enum single-pair {
        description
          "PSE port supports IEEE Std 802.3, Clause 104.";
      }
    }
    description
```

```
      "PSE support.";
  }

  typedef multi-pair-powering-pairs {
    type enumeration {
      enum signal {
        description
          "Using signal pairs.";
      }
      enum spare {
        description
          "Using spare pairs.";
      }
      enum both {
        description
          "Using signal and spare pairs.";
      }
    }
    description
      "Powering pairs.";
  }

  typedef single-pair-pd-type {
    type enumeration {
      enum unknown {
        description
          "Unknown PD type";
      }
      enum typeA {
        description
          "TypeA PD";
      }
      enum typeB {
        description
          "TypeB PD";
      }
      enum typeC {
        description
          "TypeC PD";
      }
      enum typeD {
        description
          "TypeD PD";
      }
      enum typeE {
        description
          "TypeE PD";
      }
      enum typeF {
        description
          "TypeF PD";
      }
    }
    description
      "Single-pair PoE PD type.";
    reference
      "IEEE Std 802.3, 30.15.1.1.5 aPoDLPSEDetectedPDType";
  }
```

```
  typedef single-pair-pse-type {
    type enumeration {
      enum unknown {
        description
          "Unknown PSE type.";
      }
      enum typeA {
        description
          "TypeA PSE";
      }
      enum typeB {
        description
          "TypeB PSE";
      }
      enum typeC {
        description
          "TypeC PSE";
      }
      enum typeD {
        description
          "TypeD PSE";
      }
      enum typeE {
        description
          "TypeE PSE";
      }
      enum typeF {
        description
          "TypeF PSE";
      }
    }
    description
      "Single-pair PoE PSE type.";
    reference
      "IEEE Std 802.3, 30.15.1.1.5 aPoDLPSEDetectedPDType";
  }

  typedef single-pair-power-class {
    type enumeration {
      enum class0 {
        description
          "Class 0";
      }
      enum class1 {
        description
          "Class 1";
      }
      enum class2 {
        description
          "Class 2";
      }
      enum class3 {
        description
          "Class 3";
      }
      enum class4 {
        description
          "Class 4";
      }
```

```
        enum class5 {
          description
            "Class 5";
        }
        enum class6 {
          description
            "Class 6";
        }
        enum class7 {
          description
            "Class 7";
        }
        enum class8 {
          description
            "Class 8";
        }
        enum class9 {
          description
            "Class 9";
        }
        enum class10 {
          description
            "Class 10";
        }
        enum class11 {
          description
            "Class 11";
        }
        enum class12 {
          description
            "Class 12";
        }
        enum class13 {
          description
            "Class 13";
        }
        enum class14 {
          description
            "Class 14";
        }
        enum class15 {
          description
            "Class 15";
        }
        enum unknown {
          description
            "Initializing, true Power Class not yet known.";
        }
      }
      description
        "Single-pair PoE power class.";
      reference
        "IEEE Std 802.3, 30.15.1.1.6 aPoDLPSEDetectedPDPowerClass";
    }

    augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet" {
      description
        "Augments ethernet interface configuration model with
         nodes specific to Multi-pair and Single-pair PoE.";
```

```
    container pse-2 {
      description
        "Multi-pair and Single-pair PoE port configuration.

         Discontinuities in the values of counters in this container
         can occur at re-initialization of the  management system,
         and at other times as indicated by the value of the
         'discontinuity-time' leaf  defined in the ietf-interfaces
          YANG module (IETF RFC 8343).";
      reference
        "IEEE Std 802.3, 30.9.1 multi-pair PoE PSE and
                IEEE Std 802.3, 30.15.1 single-pair PoE PSE";
      leaf supported-pse-type {
        type pse-support;
        config false;
        description
          "The PSE types are supported.";
      }
      /* Multi-pair objects */
      container multi-pair {
        if-feature "multi-pair-pse";
        presence "PSE port supports IEEE Std 802.3 Clause 33
                          and/or Clause 145.";
        description
          "PSE port configuration in IEEE Std 802.3, 30.9.1.";
        leaf pse-enable {
          type boolean;
          default "false";
          description
            "When true enables the PSE function on the port,
             when false disables the PSE function on the port.";
          reference
            "IEEE Std 802.3, 30.9.1.2.1 acPSEAdminControl";
        }
        leaf pse-state {
          type boolean;
          config false;
          description
            "The PSE enabled state on the port.";
          reference
            "IEEE Std 802.3, 30.9.1.1.2 aPSEAdminState";
        }
        leaf multi-pair-powering-pairs {
          type multi-pair-powering-pairs;
          description
            "The PSE pairs in use. If the value of
             pairs-control-ability is true, this object is
             writeable.";
          reference
            "IEEE Std 802.3, 30.9.1.1.4 aPSEPowerPairs";
        }
        leaf pairs-control-ability {
          type boolean;
          default "true";
          config false;
          description
            "Can the PSE Port control switching the
             power sourcing pins.";
          reference
```

```
      "IEEE Std 802.3, 30.9.1.1.3 aPSEPowerPairsControlAbility";
  }
  leaf detection-status {
    type multi-pair-detection-state;
    config false;
    description
      "The operational status of the port PD detection.";
    reference
      "IEEE Std 802.3, 30.9.1.1.5 aPSEPowerDetectionStatus";
  }
  leaf classifications {
    when "../detection-status  = 'deliveringPower'" {
      description
        "This node only applies when the detection status is
         deliveringPower.";
    }
    type multi-pair-power-class;
    config false;
    description
      "The power class of the detected PD.";
    reference
      "IEEE Std 802.3, 30.9.1.1.6 aPSEPowerClassfication";
  }
  /* Multi-pair statistics */
  container statistics {
    config false;
    description
      "Multi-pair PoE Statistics information for the port.";
    leaf power-denied {
      type yang:counter64;
      description
        "The number of times the PSE denied power.";
      reference
        "IEEE Std 802.3, 30.9.1.1.14";
    }
    leaf invalid-signature {
      type yang:counter64;
      description
        "The number of times the PSE detected an
         invalid signature.";
      reference
        "IEEE Std 802.3, 30.9.1.1.11";
    }
    leaf mps-absent {
      type yang:counter64;
      description
        "The number of times the PSE
         lost the 'Maintain Power Signature'(MPS).";
      reference
        "IEEE Std 802.3, 30.9.1.1.20";
    }
    leaf overload {
      type yang:counter64;
      status deprecated;
      description
        "The number of times the PSE detected
         an output current overload.";
      reference
        "IEEE Std 802.3, 30.9.1.1.17";
```

```
          }
          leaf short {
            type yang:counter64;
            status deprecated;
            description
              "This object is deprecated as it's not defined in base
               standard.";
          }
          leaf cumulative-energy {
            type yang:counter64;
            units "millijoules";
            description
              "The cumulative energy supplied by the PSE.";
            reference
              "IEEE Std 802.3, 30.9.1.1.25";
          }
        }
        /* Multi-pair measurement */
        leaf actual-power {
          type uint32;
          units "milliwatts";
          config false;
          description
            "The actual power drawn by a PD over the port.";
          reference
            "IEEE Std 802.3, 30.9.1.1.23";
        }
        leaf power-accuracy {
          type int64;
          units "milliwatts";
          config false;
          description
            "The power accuracy of the port.";
          reference
            "IEEE Std 802.3, 30.9.1.1.24";
        }
      }
      /* Single-pair objects */
      container single-pair {
        if-feature "single-pair-pse";
        presence "PSE port supports IEEE Std 802.3 Clause 104.";
        description
          "Single-pair PSE configuration as defined in
           IEEE Std 802.3, 30.15.1.";
        leaf pse-enable {
          type boolean;
          default "false";
          description
            "When true enables the PSE function on the port,
             when false disables the PSE function on the port.";
          reference
            "IEEE Std 802.3, 30.15.1.1.2 aPoDLPSEAdminState";
        }
        leaf detection-status {
          type single-pair-detection-state;
          config false;
          description
            "Indicates the current status of the Single-pair PSE.";
          reference
```

```
            "IEEE Std 802.3,
             30.15.1.1.3 aPoDLPSEPowerDetectionStatus";
        }
        leaf single-pair-type {
          type single-pair-pse-type;
          config false;
          description
            "Single-pair PSE type.";
          reference
            "IEEE Std 802.3, 30.15.1.1.4 aPoDLPSEType";
        }
        leaf detected-pd-type {
          when "../detection-status = 'deliveringPower'" {
            description
              "This node only applies when the detection status is
               delivering power.";
          }
          type single-pair-pd-type;
          config false;
          description
            "Indicates the detected type of the single-pair PD.";
          reference
            "IEEE Std 802.3, 30.15.1.1.5 aPoDLPSEDetectedPDType";
        }
        leaf pd-power-class {
          when "../detection-status = 'deliveringPower'" {
            description
              "This node only applies when the detection status is
               delivering power.";
          }
          type single-pair-power-class;
          config false;
          description
            "The power class of the detected PD.";
          reference
            "IEEE Std 802.3,
             30.15.1.1.6 aPoDLPSEDetectedPDPowerClass";
        }
        /* Single-pair statistics */
        container statistics {
          config false;
          description
            "Single-pair PoE Statistics information for the port.";
          leaf power-denied {
            type yang:counter64;
            description
              "The number of times the PSE denied power.";
            reference
              "IEEE Std 802.3,
               30.15.1.1.9 aPoDLPSEPowerDeniedCounter";
          }
          leaf invalid-signature {
            type yang:counter64;
            description
              "The number of times the PSE detected an
               invalid signature.";
            reference
              "IEEE Std 802.3, 30.15.1.1.7
               aPoDLPSEInvalidSignatureCounter";
```

```
          }
          leaf invalid-class {
            type yang:counter64;
            description
              "The number of times PD classification timed out
               or did not get valid class information from SCCP.";
            reference
              "IEEE Std 802.3,
               30.15.1.1.8 aPoDLPSEInvalidClassCounter";
          }
          leaf overload {
            type yang:counter64;
            description
              "The number of times the PSE detected
               an output current overload.";
            reference
              "IEEE Std 802.3, 30.15.1.1.10 aPoDLPSEOverLoadCounter";
          }
          leaf fvs-absence {
            type yang:counter64;
            description
              "The number of times the PSE
               lost the 'Maintain Full Voltage Signature'(MFVS).";
            reference
              "IEEE Std 802.3, 30.15.1.1.11
               aPoDLPSEMaintainFullVoltageSignatureAbsentCounter";
          }
          leaf cumulative-energy {
            type yang:counter64;
            units "millijoules";
            description
              "The cumulative energy supplied by the PSE as
               measured at the MDI.";
            reference
              "IEEE Std 802.3, 30.15.1.1.14 aPoDLPSECumulativeEnergy";
          }
        }
        /* Single-pair measurement */
        leaf actual-power {
          type uint32;
          units "milliwatts";
          config false;
          description
            "The actual power drawn by a PD over the port.";
          reference
            "IEEE Std 802.3, 30.15.1.1.12 aPoDLPSEActualPower";
        }
        leaf power-accuracy {
          type int64;
          units "milliwatts";
          config false;
          description
            "The power accuracy of the port.";
          reference
            "IEEE Std 802.3, 30.15.1.1.13 aPoDLPSEPowerAccuracy";
        }
      }
    }
  }
```

```
}
```

7. YANG module for Ethernet Passive Optical Network (EPON)

7.1 Introduction

This clause defines a YANG module to manage Ethernet Passive Optical Network (EPON).

EPON is defined in IEEE Std 802.3, covering Physical Layers and Media Access Control sublayers. The Passive Optical Network (PON) is comprised of sections of single-mode fiber connected with passive optical splitter/coupler devices, forming a passive optical tree, as shown in Figure 7–1. Individual branches of the PON are terminated with the Optical Line Terminal (OLT) in the Central Office or at remote optical nodes, and Optical Network Units (ONUs) near the subscribers. ONUs can be located either in some remote location (e.g., basement in a multi-dwelling unit) or directly at the subscriber premises. Various types of Customer Premises Equipment (CPE) can be connected to ONUs or even integrated with such devices.

7.2 YANG module structure

The *ieee802-ethernet-pon* YANG module of this clause is focused on the configuration and monitoring of EPON.

7.3 Mapping of IEEE Std 802.3, Clause 30 managed objects

This sub-clause contains the mapping between YANG data nodes included in *ieee802-ethernet-pon* (see Table 7–1) YANG module, managed objects, and attributes defined in IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB.

Table 7–1—Mapping between IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB managed objects and *ieee802-ethernet-pon* YANG data nodes

IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB		Corresponding *ieee802-ethernet-pon* YANG data nodes		
Managed object(s)	**Attribute(s)**	**Container(s)**	**Data node(s)**	**R/W**
dot3EponFecTable	dot3EponFecMode		fec-mode	R/W
	dot3EponFecPCSCodingViolation	statistics-pon-fec	fec-code-group-violations	R
	dot3EponFecAbility		fec-capability	R
	dot3EponFecCorrectedBlocks	statistics-pon-fec	fec-code-word-corrected-errors	R
	dot3EponFecUncorrectableBlocks	statistics-pon-fec	fec-code-word-uncorrected-errors	R
	dot3EponFecBufferHeadCodingViolation	statistics-pon-fec	fec-buffer-head-coding-violation	R
dot3MpcpControl Table	dot3MpcpAdminState		mpcp-admin-state	R/W
	dot3MpcpMode		mpcp-mode	R
	dot3MpcpLinkID		mpcp-logical-link-id	R
	dot3MpcpRemoteMACAddress		mpcp-remote-mac-address	R
	dot3MpcpRegistrationState		mpcp-logical-link-state	R
	dot3MpcpSyncTime		mpcp-sync-time	R
	dot3MpcpTransmitElapsed		mpcp-elapsed-time-out	R
	dot3MpcpReceiveElapsed		mpcp-elapsed-time-in	R
	dot3MpcpRoundTripTime		mpcp-round-trip-time	R
	dot3MpcpMaximumPendingGrants		mpcp-maximum-grant-count	R

Table 7–1—Mapping between IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB managed objects and *ieee802-ethernet-pon* YANG data nodes *(continued)*

IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB		Corresponding *ieee802-ethernet-pon* YANG data nodes		
Managed object(s)	**Attribute(s)**	**Container(s)**	**Data node(s)**	**R/W**
dot3ExtPkgQueueSets Table	dot3QueueSetIndex	mpcp-queue-thresholds	mpcp-queue-set-index	R/W
	dot3ExtPkgObjectReportThreshold		mpcp-queue-set-threshold	R/W
	dot3QueueIndex	mpcp-queues	mpcp-queue-index	R/W
	dot3ExtPkgObjectReportNumThreshold		mpcp-queue-threshold-count	R/W
	dot3ExtPkgObjectReportMaximumNumThreshold		mpcp-queue-threshold-count-max	R
	dot3ExtPkgStatTxFramesQueue		in-mpcp-queue-frames	R
	dot3ExtPkgStatRxFramesQueue		out-mpcp-queue-frames	R
	dot3ExtPkgStatDroppedFramesQueue		mpcp-queue-frames-drop	R
dot3ExtPkgControl Table	dot3ExtPkgObjectReset dot3MpcpOperStatus		mpcp-logical-link-admin-state	R/W
	dot3ExtPkgObjectNumberOfLLIDs		mpcp-logical-link-count	R
	dot3ExtPkgObjectReportMaximumNumQueues		mpcp-maximum-queue-count-per-report	R
dot3RecognizedMulticast-IDs Table	dot3RecognizedMulticastID	multicast-IDs	multicast-ID	R/W

Table 7–1—Mapping between IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB managed objects and *ieee802-ethernet-pon* YANG data nodes *(continued)*

IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB		Corresponding *ieee802-ethernet-pon* YANG data nodes		
Managed object(s)	**Attribute(s)**	**Container(s)**	**Data node(s)**	**R/W**
dot3OmpEmulation Table	dot3OmpEmulationType	statistics-ompe	ompe-mode	R
	dot3OmpEmulationSLDErrors		in-ompe-frames-errored-sld	R
	dot3OmpEmulationCRC8Errors		in-ompe-frames-errored-crc8	R
	dot3OmpEmulationBadLLID		in-ompe-frames-with-bad-llid	R
	dot3OmpEmulationGoodLLID		in-ompe-frames-with-good-llid	R
	dot3OmpEmulationBroadcastBitNotOnuLlid		in-ompe-frames-not-match-onu-llid-broadcast	R
	dot3OmpEmulationOnuLLIDNotBroadcast		in-ompe-frames-match-onu-llid-not-broadcast	R
	dot3OmpEmulationBroadcastBitPlusOnuLlid		in-ompe-frames-match-onu-llid-broadcast	R
	dot3OmpEmulationNotBroadcastBitNotOnuLlid		in-ompe-frames-not-match-onu-llid-not-broad-cast	R
			in-ompe-frames	R
			ompe-onu-frames-with-good-llid-good-crc8	
			ompe-olt-frames-with-good-llid-good-crc8	

Table 7–1—Mapping between IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB managed objects and *ieee802-ethernet-pon* YANG data nodes *(continued)*

IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB		Corresponding *ieee802-ethernet-pon* YANG data nodes		
Managed object(s)	**Attribute(s)**	**Container(s)**	**Data node(s)**	**R/W**
dot3MpcpStat Table	dot3MpcpMACCtrlFramesTransmitted	statistics-mpcp	out-mpcp-mac-ctrl-frames	R
	dot3MpcpMACCtrlFramesReceived		in-mpcp-mac-ctrl-frames	R
	dot3MpcpDiscoveryWindowsSent		mpcp-discovery-window-count	R
	dot3MpcpDiscoveryTimeout		mpcp-discovery-timeout-count	R
	dot3MpcpTxRegRequest		out-mpcp-register-req	R
	dot3MpcpRxRegRequest		in-mpcp-register-req	R
	dot3MpcpTxRegAck		out-mpcp-register-ack	R
	dot3MpcpRxRegAck		in-mpcp-register-ack	R
	dot3MpcpTxReport		out-mpcp-report	R
	dot3MpcpRxReport		in-mpcp-report	R
	dot3MpcpTxGate		out-mpcp-gate	R
	dot3MpcpRxGate		in-mpcp-gate	R
	dot3MpcpTxRegister		out-mpcp-register	R
	dot3MpcpRxRegister		in-mpcp-register	R

Table 7–1—Mapping between IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB managed objects and *ieee802-ethernet-pon* YANG data nodes *(continued)*

IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB		Corresponding *ieee802-ethernet-pon* YANG data nodes		
Managed object(s)	**Attribute(s)**	**Container(s)**	**Data node(s)**	**R/W**
dot3ExtPkgOptIf Table	dot3ExtPkgOptIfLowerInputPowerThreshold	thresholds-trx	in-trx-power-low-threshold	R/W
	dot3ExtPkgOptIfUpperInputPowerThreshold		in-trx-power-high-threshold	R/W
	dot3ExtPkgOptIfLowerOutputPowerThreshold		out-trx-power-low-threshold	R/W
	dot3ExtPkgOptIfUpperOutputPowerThreshold		out-trx-power-high-threshold	R/W
	dot3ExtPkgOptIfSignalDetect		in-trx-power-signal-detect	R
	dot3ExtPkgOptIfInputPower		in-trx-power	R
	dot3ExtPkgOptIfLowInputPower		in-trx-power-low-15-minutes-bin	R
	dot3ExtPkgOptIfHighInputPower		in-trx-power-high-15-minutes-bin	R
	dot3ExtPkgOptIfTransmitEnable		out-trx-power-signal-detect	R/W
	dot3ExtPkgOptIfOutputPower		out-trx-power	R
	dot3ExtPkgOptIfLowOutputPower		out-trx-power-low-15-minutes-bin	R
	dot3ExtPkgOptIfHighOutputPower		out-trx-power-high-15-minutes-bin	R
	dot3ExtPkgOptIfSuspectedFlag		trx-data-reliable	R

7.4 YANG module definition[k]

The YANG module tree hierarchy uses terms defined in IETF RFC 8407.

7.4.1 Tree hierarchy

7.4.1.1 ieee802-ethernet-pon

```
module: ieee802-ethernet-pon

  augment /if:interfaces/if:interface/ieee802-eth-if:ethernet:
    +--rw fec-mode?                               fec-mode {fec-supported}?
    +--rw mpcp-admin-state?                       mpcp-admin-state
    +--ro mpcp-logical-link-admin-state?          mpcp-logical-link-admin-state
    +--rw trx-transmit-admin-state?               trx-admin-state {trx-power-level-reporting-supported}?
    +--ro capabilities
    |  +--ro mpcp-supported?   mpcp-supported
    +--ro statistics-mpcp
    |  +--ro out-mpcp-mac-ctrl-frames?       yang:counter64
    |  +--ro in-mpcp-mac-ctrl-frames?        yang:counter64
    |  +--ro mpcp-discovery-window-count?    yang:counter64
    |  +--ro mpcp-discovery-timeout-count?   yang:counter64
    |  +--ro out-mpcp-register-req?          yang:counter64
    |  +--ro in-mpcp-register-req?           yang:counter64
    |  +--ro out-mpcp-register-ack?          yang:counter64
    |  +--ro in-mpcp-register-ack?           yang:counter64
    |  +--ro out-mpcp-report?                yang:counter64
    |  +--ro in-mpcp-report?                 yang:counter64
    |  +--ro out-mpcp-gate?                  yang:counter64
    |  +--ro in-mpcp-gate?                   yang:counter64
    |  +--ro out-mpcp-register?              yang:counter64
    |  +--ro in-mpcp-register?               yang:counter64
    +--rw statistics-ompe
    |  +--ro in-ompe-frames-errored-sld?                       yang:counter64
    |  +--ro in-ompe-frames-errored-crc8?                      yang:counter64
    |  +--ro ompe-onu-frames-with-good-llid-good-crc8?         yang:counter64
```

[k]Copyright release for YANG modules: Users of this standard may freely reproduce the YANG module contained in this subclause so that it can be used for its intended purpose.

```
|  +--ro ompe-olt-frames-with-good-llid-good-crc8?          yang:counter64
|  +--ro in-ompe-frames-with-bad-llid?                      yang:counter64
|  +--ro in-ompe-frames-with-good-llid?                     yang:counter64
|  +--ro in-ompe-frames?                                    yang:counter64
|  +--ro in-ompe-frames-not-match-onu-llid-broadcast?       yang:counter64
|  +--ro in-ompe-frames-match-onu-llid-not-broadcast?       yang:counter64
|  +--ro in-ompe-frames-match-onu-llid-broadcast?           yang:counter64
|  +--ro in-ompe-frames-not-match-onu-llid-not-broadcast?   yang:counter64
+--rw thresholds-trx {trx-power-level-reporting-supported}?
|  +--rw in-trx-power-low-threshold?     power-level {trx-power-level-reporting-supported}?
|  +--rw in-trx-power-high-threshold?    power-level {trx-power-level-reporting-supported}?
|  +--rw out-trx-power-low-threshold?    power-level {trx-power-level-reporting-supported}?
|  +--rw out-trx-power-high-threshold?   power-level {trx-power-level-reporting-supported}?
x--rw statistics-trx {trx-power-level-reporting-supported}?
|  +--ro in-trx-power-signal-detect?          boolean
|  +--ro in-trx-power?                        power-level
|  +--ro in-trx-power-low-15-minutes-bin?     power-level
|  +--ro in-trx-power-high-15-minutes-bin?    power-level
|  +--ro out-trx-power-signal-detect?         boolean
|  +--ro out-trx-power?                       power-level
|  +--ro out-trx-power-low-15-minutes-bin?    power-level
|  +--ro out-trx-power-high-15-minutes-bin?   power-level
|  +--ro trx-data-reliable?                   boolean {trx-power-level-reporting-supported}?
+--rw monitoring-trx {trx-power-level-reporting-supported}?
|  +--ro in-trx-power-signal-detect?          boolean
|  +--ro in-trx-power?                        power-level
|  +--ro in-trx-power-low-15-minutes-bin?     power-level
|  +--ro in-trx-power-high-15-minutes-bin?    power-level
|  +--ro out-trx-power-signal-detect?         boolean
|  +--ro out-trx-power?                       power-level
|  +--ro out-trx-power-low-15-minutes-bin?    power-level
|  +--ro out-trx-power-high-15-minutes-bin?   power-level
|  +--ro trx-data-reliable?                   boolean {trx-power-level-reporting-supported}?
+--ro statistics-pon-fec {fec-supported}?
|  +--ro fec-code-group-violations?           yang:counter64
|  +--ro fec-buffer-head-coding-violations?   yang:counter64
|  +--ro fec-code-word-corrected-errors?      yang:counter64
|  +--ro fec-code-word-uncorrected-errors?    yang:counter64
+--rw mpcp-logical-link-admin-actions
```

```
|  +---x state-change-action-type
|  |  +---w input
|  |     +---w state-change-action-type?   identityref
|  +---x reset-action-type
|  |  +---w input
|  |     +---w reset-action-type?   identityref
|  +---x register-type
|     +---w input
|        +---w register-type?   identityref
+--rw mpcp-queues* [mpcp-queue-index]
|  +--rw mpcp-queue-index                  uint8
|  +--rw mpcp-queue-threshold-count?       uint8
|  +--ro mpcp-queue-threshold-count-max?   uint8
|  +--rw mpcp-queue-thresholds* [mpcp-queue-set-index]
|  |  +--rw mpcp-queue-set-index        uint8
|  |  +--rw mpcp-queue-set-threshold?   uint64
|  +--ro in-mpcp-queue-frames?             yang:counter64
|  +--ro out-mpcp-queue-frames?            yang:counter64
|  +--ro mpcp-queue-frames-drop?           yang:counter64
+--rw multicast-IDs* [multicast-ID]
|  +--rw multicast-ID    uint32
+--ro fec-capability?                       fec-capability
+--ro mpcp-mode?                            mpcp-mode
+--ro mpcp-sync-time?                       uint64
+--ro mpcp-logical-link-id?                 mpcp-supported
+--ro mpcp-remote-mac-address?              ieee:mac-address
+--ro mpcp-logical-link-state?              mpcp-logical-link-state
+--ro mpcp-elapsed-time-out?                uint64
+--ro mpcp-elapsed-time-in?                 uint64
+--ro mpcp-round-trip-time?                 uint16
+--ro mpcp-maximum-grant-count?             uint8
+--ro mpcp-logical-link-count?              mpcp-llid-count
+--ro mpcp-maximum-queue-count-per-report?  mpcp-maximum-queue-count-per-report
+--ro ompe-mode?                            ompe-mode
```

7.4.2 YANG module

In the YANG module definition, should any discrepancy between the text of the description for individual YANG nodes in the YANG modules of this clause and the corresponding definition in 7.2 through 7.4 of this clause occur, the definitions in the YANG modules attached to this standard shall take precedence.

An ASCII text version of the YANG module can be found at the following URL:[1]

> ***Editor's Note (to be removed prior to publication):***
>
> *Yang files contained in https://github.com/YangModels/yang/tree/main/standard/ieee/published/802.3 are IEEE 802.3.2-2019 version and will be updated at the publication time.*

https://github.com/YangModels/yang/tree/master/standard/ieee/published/802.3

> ***Editor's Note (to be removed prior to publication):***
>
> *Pretty printing of ieee802-ethernet-pon.yang file may change the appearance by adding whitespace and reformatting lines*

> ***Editor's Note (to be removed prior to publication):***
>
> IEEE Std 802.3.1 and IEEE Std 802.3.2 to be updated at the publication time

```
module ieee802-ethernet-pon {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-pon";
  prefix ieee802-eth-pon;

  import ieee802-types {
    prefix ieee;
    reference
      "IEEE 802 types";
  }
  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
  }
  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import ieee802-ethernet-interface {
    prefix ieee802-eth-if;
  }

  organization
    "IEEE 802.3 Ethernet Working Group
     Web URL: http://www.ieee802.org/3/";
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
```

[1]Copyright release for YANG modules: Users of this standard may freely reproduce the YANG module contained in this subclause so that it can be used for its intended purpose.

```
    "This module contains a collection of YANG definitions for
     managing the Multi Point Control Protocol for
     Ethernet PON (EPON),as defined in IEEE Std 802.3, Clause 64
     and Clause 77.

     This YANG module augments the 'ethernet' module.";

  revision 2025-04-17 {
    description
      "Updates under IEEE Std 802.3.2-202x, Draft D3.3";
    reference
      "IEEE Std 802.3-2022, unless dated explicitly";
  }

  feature trx-power-level-reporting-supported {
    description
      "This object indicates the support for optical transceiver
       power level monitoring and reporting capability.
       When 'true', the given interface supports the optical power
       level monitoring and reporting function. Otherwise,
       the value is 'false'.";
  }

  feature fec-supported {
    description
      "This object indicates the support of operation of the
       optional FEC sublayer of the 1G-EPON PHY specified in
       IEEE Std 802.3, 65.2. The value of 'unknown' is reported in
       the initialization, for non FEC support state or type not
       yet known. The value of 'not supported' is reported when the
       sublayer is not supported. The value of 'supported' is
       reported when the sublayer is supported. This object is
       applicable for an OLT, with the same value for all logical
       links, and for an ONU.";
    reference
      "IEEE Std 802.3, 30.5.1.1.15";
  }

  identity state-change-action-type {
    description
      "Type of interface state change requested.";
  }

  identity power-down {
    base state-change-action-type;
    description
      "Power down the EPON logical interface.
       Power-down actions are applicable for the OLT and ONU.  A
       power down of a specific logical interface affects only
       the logical interface (and not the physical interface).
       the logical interface will be unavailable while the
       power-down occurs and data may be lost. Other logical
       interface are unaffected by power-down.

       This action is relevant when the admin state is active.";
  }

  identity power-up {
    base state-change-action-type;
```

```
      description
        "Exit EPON logical interface power-down state.";
    }

    identity reset-action-type {
      description
        "Type of reset action requested.";
    }

    identity reset-interface {
      base reset-action-type;
      description
        "Reset the EPON logical interface. Resetting an interface
         can lead an interruption of service for the users connected
         to the respective EPON interface.

         This object is applicable for an OLT and an ONU. At the
         OLT, it has a distinct value for each logical interface.
         A reset for a specific logical interface resets only
         this logical interface and not the physical interface.

         Thus, a logical link that is malfunctioning can be
         reset without affecting the operation of other logical
         interfaces.

         The reset can cause Discontinuities in the values of the
         counters of the interface, similar to re-initialization
         of the management system.";
    }

    identity register-type {
      description
        "Type of registration requested.";
    }

    identity register {
      base register-type;
      description
        "Register indicates a request to register an LLID.
         This action applies to an OLT or ONU logical interface.";
    }

    identity reregister {
      base register-type;
      description
        "Re-register indicates an request to re-register an LLID.
         This action applies to an OLT or ONU logical interface.";
    }

    identity deregister {
      base register-type;
      description
        "De-register indicates an request to de-register an LLID.
         This action applies to an OLT or ONU logical interface.
         Deregister may result in an interruption of service to
         users connected to the respective EPON interface.";
    }

    typedef mpcp-supported {
```

```
    type boolean;
    description
      "This object indicates that the given interface supports
       MPCP, i.e., it is an Ethernet PON (EPON) interface.";
  }

  typedef mpcp-llid {
    type uint64 {
      range "0 .. 32767";
    }
    description
      "Logical Link Identifiers (LLIDs) are used to identify a
       single MAC from a number of MACs which may be present in the
       EPON OLT or ONU. LLIDs between the value of 0x07FFE
       and 0x7FFF are assigned for ONU discovery and registration.
       Other LLIDs are dynamically assigned by the OLT during the
       registration process. For a complete description of how the
       LLID is used in an EPON device, see IEEE Std 802.3,
       Clause 65 for 1G-EPON and Clause 76 for 10G-EPON.";
    reference
      "IEEE Std 802.3, 65.1.3.3 for 1G-EPON and
       76.2.6.1.3 for 10G-EPON";
  }

  typedef mpcp-maximum-queue-count-per-report {
    type uint8 {
      range "0..7";
    }
    default "0";
    description
      "Defines the maximum number of queues (0-7) in the REPORT
       MPCPDU as defined in IEEE Std 802.3, Clause 64 and
       Clause 77.";
  }

  typedef mpcp-llid-count {
    type uint64 {
      range "0 .. 32767";
    }
    description
      "Indicates the number of registered LLIDs. The initialization
       value is 0. This is applicable for an OLT with the same
       value for all logical interfaces and for an ONU.";
    reference
      "IEEE Std 802.3, 65.1.3.3 for 1G-EPON and
          76.2.6.1.3 for 10G-EPON";
  }

  typedef mpcp-admin-state {
    type enumeration {
      enum enabled {
        description
          "When selecting the value of 'enabled', the MultiPoint
           Control Protocol sublayer on the OLT / ONU is enabled.";
      }
      enum disabled {
        description
          "When selecting the value of 'disabled', the MultiPoint
           Control Protocol sublayer on the OLT / ONU is
```

```
          disabled.";
      }
    }
    description
      "Enumeration of valid administrative states for a
       MultiPoint MAC Control sublayer on the OLT or ONU.";
    reference
      "IEEE Std 802.3, 30.3.5.2.1";
  }

  typedef mpcp-mode {
    type enumeration {
      enum olt {
        description
          "MPCP mode: olt";
      }
      enum onu {
        description
          "MPCP mode: onu";
      }
    }
    description
      "Enumeration of valid MPCP modes for EPON interfaces.";
    reference
      "IEEE Std 802.3, 30.3.5.1.3";
  }

  typedef mpcp-logical-link-state {
    type enumeration {
      enum unregistered {
        description
          "MPCP registration state: logical link is
           NOT registered.";
      }
      enum registering {
        description
          "MPCP registration state: logical link is currently in
           the process of registering.";
      }
      enum registered {
        description
          "MPCP registration state: logical link is currently
           registered.";
      }
    }
    description
      "Enumeration of valid MPCP registration states for EPON
       interfaces.";
    reference
      "IEEE Std 802.3, 30.3.5.1.6";
  }

  typedef mpcp-logical-link-admin-state {
    type enumeration {
      enum reset {
        description
          "When read, the value of 'reset' indicates that the given
           logical link on the OLT / ONU has been reset.";
      }
```

```
      enum operate {
        description
          "When read, the value of 'operate' indicates that the
           given logical link on the OLT / ONU has moved into
           operating mode.";
      }
      enum unknown {
        description
          "When read, the value of 'unknown' indicates that the
           status of the given logical link on the OLT / ONU is
           currently not known.";
      }
      enum registered {
        description
          "When read, the value of 'registered' indicates that the
           given logical link on the OLT / ONU has been
           registered.";
      }
      enum deregistered {
        description
          "When read, the value of 'deregistered' indicates that the
           given logical link on the OLT / ONU has been
           deregistered.";
      }
      enum reregistered {
        description
          "When read, the value of 'reregistered' indicates that the
           given logical link on the OLT / ONU has been
           reregistered.";
      }
    }
    description
      "Enumeration of valid administrative states for a logical
       link on the OLT or ONU.";
  }

  typedef ompe-mode {
    type enumeration {
      enum unknown {
        description
          "omp-emulation mode: unknown = system is initializing";
      }
      enum olt {
        description
          "omp-emulation mode: olt";
      }
      enum onu {
        description
          "omp-emulation mode: onu";
      }
    }
    description
      "Enumeration of valid OMP-Emulation modes for EPON
       interfaces.";
    reference
      "IEEE Std 802.3, 30.3.7.1.2";
  }

  typedef fec-capability {
```

```
      type enumeration {
        enum unknown {
          description
            "FEC capability: unknown = system is initializing.";
        }
        enum supported {
          description
            "FEC capability: supported.";
        }
        enum NotSupported {
          description
            "FEC capability: not supported.";
        }
      }
      description
        "Enumeration of valid FEC capability values for EPON
         interfaces with enabled MPCP.";
      reference
        "IEEE Std 802.3, 30.5.1.1.15";
    }

    typedef fec-mode {
      type enumeration {
        enum unknown {
          description
            "FEC mode: unknown = system is initializing.";
        }
        enum disabled {
          description
            "FEC mode: disabled = FEC is disabled for the given
             logical link (both Tx and Rx directions).";
        }
        enum enabled-Tx-Rx {
          description
            "FEC mode: enabled-Tx-Rx = FEC is enabled for the given
             logical link in both Tx and Rx directions.";
        }
        enum enabled-Tx-only {
          description
            "FEC mode: enabled-Tx-only = FEC is enabled for
             the given logical link but only in Tx direction.";
        }
        enum enabled-Rx-only {
          description
            "FEC mode: enabled-Rx-only = FEC is enabled for
             the given logical link but only in Rx direction.";
        }
      }
      description
        "Enumeration of valid FEC modes for EPON interfaces.";
      reference
        "IEEE Std 802.3, 30.5.1.1.16";
    }

    typedef power-level {
      type int32;
      units "0.1 dBm";
      description
        "Power level reflects the value of power, as measured at the
```

```
       optical transceiver, expressed in units of 0.1 dBm.";
  }

  typedef trx-admin-state {
    type enumeration {
      enum enabled {
        description
          "When read as 'enabled', the transmitter is enabled and
           operating under the control of the logical control
           protocol. When set to 'enabled', the transmitter is
           enabled to operate under the control of the logical
           control protocol.";
      }
      enum disabled {
        description
          "When read as 'disabled', the transmitter is currently
           disabled (not transmitting). When set to 'disabled',
           the transmitter is expected to be disabled
           (to stop transmitting).";
      }
    }
    description
      "Enumeration of valid administrative states for an optical
       transceiver.";
    reference
      "IEEE Std 802.3.1, dot3ExtPkgOptIfTransmitEnable";
  }

  augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet" {
    description
      "Augments the definition of Ethernet interface (
          /if:interfaces/if:interface/ieee802-eth-if:ethernet)
          module with nodes specific to Ethernet PON (EPON).";
    leaf fec-mode {
      if-feature "fec-supported";
      type fec-mode;
      description
        "This object reflects the current administrative state of
         the FEC function for the given logical link on an ONU or
         OLT.

         When reading the value of 'disabled', the FEC function on
         the given logical link is disabled.

         When reading the value of 'enabled-Tx-Rx', the FEC function
         on the given logical link is enabled in both Tx and Rx
         directions.

         When reading the value of 'enabled-Tx-only', the FEC
         function on the given logical link is enabled in Tx
         direction only.

         When reading the value of 'enabled-Rx-only', the FEC
         function on the given logical link is enabled in Rx
         direction only.

         When reading the value of 'unknown', the state of the FEC
         function on the given logical link is unknown or the FEC
         function is currently initializing.
```

```
          This object is applicable for an OLT and an ONU. This
          object has the same value for each logical link.";
      reference
        "IEEE Std 802.3, 30.5.1.1.16";
    }
    leaf mpcp-admin-state {
      type mpcp-admin-state;
      description
        "This object reflects the current administrative state of
         the MultiPoint MAC Control sublayer, as defined in
         IEEE Std 802.3, Clause 64 and Clause 77, for the
         OLT / ONU.

         When reading the value of 'enabled', the MultiPoint
         Control Protocol on the OLT / ONU is enabled.

         When reading the value of 'disabled', the MultiPoint
         Control Protocol on the OLT / ONU is disabled.

         This object is applicable for an OLT and an ONU. It has
         the same value for all logical links.";
      reference
        "IEEE Std 802.3, 30.3.5.1.2";
    }
    leaf mpcp-logical-link-admin-state {
      type mpcp-logical-link-admin-state;
      config false;
      description
        "This object reflects the current administrative state of a
         logical link on an ONU or OLT.

         When reading the value of 'reset', the given logical link
         is undergoing a reset.

         When reading the value of 'unknown', the current status of
         the given logical link is unknown and the link might be
         undergoing initialization.

         When reading the value of 'operate', the given logical
         link is operating  normally.

         When reading the value of 'registered', the given logical
         link was requested to perform registration.

         When reading the value of 'deregistered', the given
         logical link was requested to perform deregistration.

         When reading the value of 'reregistered', the given
         logical link was requested to perform reregistration.

         This object is applicable for an OLT and an ONU. It has a
         distinct value for each logical link.";
      reference
        "IEEE Std 802.3.1, dot3ExtPkgObjectRegisterAction";
    }
    leaf trx-transmit-admin-state {
      when "../../ieee802-eth-if:ethernet/
               ieee802-eth-pon:mpcp-admin-state = 'enabled'";
```

```
        if-feature "trx-power-level-reporting-supported";
        type trx-admin-state;
        description
          "This object reflects the current status of the transmitter
           in the optical transceiver.

           When read as 'enabled', the optical transmitter is enabled
           and operating under the control of the logical control
           protocol.

           When read as 'disabled', the optical transmitter is
           disabled.

           This object is applicable for an OLT and an ONU.
           At the OLT, this object has a distinct value for each
           logical link.

           The value of this object is only reliable when
           /if:interfaces-state/if:interface/ieee802-eth-if:ethernet/
           'mpcp-admin-state' is equal to 'enabled'.";
        reference
          "IEEE Std 802.3.1, dot3ExtPkgOptIfTransmitEnable";
      }
      container capabilities {
        config false;
        description
          "This container includes all EPON interface-specific
           capabilities.";
        leaf mpcp-supported {
          type mpcp-supported;
          default "true";
          description
            "This object indicates that the given interface supports
             MPCP, i.e., it is an Ethernet PON (EPON) interface.";
        }
      }
      container statistics-mpcp {
        config false;
        description
          "This container defines a set of MPCP-related statistics
           counters of an EPON interface, as defined in
           IEEE Std 802.3, Clause 64 and Clause 77.

           Discontinuities in the values of counters in this
           container can occur at re-initialization of the management
           system, and at other times as indicated by the value of
           the 'discontinuity-time' leaf defined in the ietf-interfaces
           YANG module (IETF RFC 8343).";
        leaf out-mpcp-mac-ctrl-frames {
          type yang:counter64;
          units "frames";
          config false;
          description
            "A count of MPCP frames passed to the MAC sublayer for
             transmission.

             This counter is incremented when a MA_CONTROL.request
             service primitive is generated within the MAC control
             sublayer with an opcode indicating an MPCP frame.
```

```
            This object is applicable for an OLT and an ONU. It has
            a distinct value for each logical link.";
         reference
           "IEEE Std 802.3, 30.3.5.1.7";
       }
       leaf in-mpcp-mac-ctrl-frames {
         type yang:counter64;
         units "frames";
         config false;
         description
           "A count of MPCP frames passed by the MAC sublayer to the
            MAC Control sublayer.

            This counter is incremented when a frame is received at
            the interface which is an MPCP frame or has a
            Length/Type Ethernet header field value equal to the
            Type assigned for 802.3_MAC_Control as specified in
            IEEE Std 802.3, 31.4.1.3.

            This object is applicable for an OLT and an ONU. It has
            a distinct value for each logical link.";
         reference
           "IEEE Std 802.3, 30.3.5.1.8";
       }
       leaf mpcp-discovery-window-count {
         when "../../ompe-mode = 'olt'";
         type yang:counter64;
         units "discovery windows";
         config false;
         description
           "A count of discovery windows generated by the OLT.

            The counter is incremented by one for each generated
            discovery window.

            This object is applicable for an OLT and has the same
            value for each logical link.";
         reference
           "IEEE Std 802.3, 30.3.5.1.22";
       }
       leaf mpcp-discovery-timeout-count {
         when "../../ompe-mode = 'olt'";
         type yang:counter64;
         units "discovery timeouts";
         config false;
         description
           "A count of the number of times a discovery timeout
            occurs.

            This counter is incremented by one for each discovery
            processing state-machine reset resulting from timeout
            waiting for message arrival.

            This object is applicable for an OLT and has the same
            value for each logical link.";
         reference
           "IEEE Std 802.3, 30.3.5.1.23";
       }
```

```
      leaf out-mpcp-register-req {
        when "../../ompe-mode = 'onu'";
        type yang:counter64;
        units "frames";
        config false;
        description
          "A count of the number of times a REGISTER_REQ MPCP frame
           transmission occurs.

           This counter is incremented by one for each
           REGISTER_REQ MPCP frame transmitted as defined in
           IEEE Std 802.3, Clause 64 and Clause 77.

           This object is applicable for an ONU and has the same
           value for each logical link.";
        reference
          "IEEE Std 802.3, 30.3.5.1.12";
      }
      leaf in-mpcp-register-req {
        when "../../ompe-mode = 'olt'";
        type yang:counter64;
        units "frames";
        config false;
        description
          "A count of the number of times a REGISTER_REQ MPCP frame
           reception occurs.

           This counter is incremented by one for each
           REGISTER_REQ MPCP frame received as defined in
           IEEE Std 802.3, Clause 64 and Clause 77.

           This object is applicable for an OLT and has the same
           value for each logical link.";
        reference
          "IEEE Std 802.3, 30.3.5.1.17";
      }
      leaf out-mpcp-register-ack {
        when "../../ompe-mode = 'onu'";
        type yang:counter64;
        units "frames";
        config false;
        description
          "A count of the number of times a REGISTER_ACK MPCP frame
           transmission occurs.

           This counter is incremented by one for each
           REGISTER_ACK MPCP frame transmitted as defined in
           IEEE Std 802.3, Clause 64 and Clause 77.

           This object is applicable for an ONU and has a distinct
           value for each logical link.";
        reference
          "IEEE Std 802.3, 30.3.5.1.10";
      }
      leaf in-mpcp-register-ack {
        when "../../ompe-mode = 'olt'";
        type yang:counter64;
        units "frames";
        config false;
```

```
        description
          "A count of the number of times a REGISTER_ACK MPCP frame
           reception occurs.

           This counter is incremented by one for each
           REGISTER_ACK MPCP frame received as defined in
           IEEE Std 802.3, Clause 64 and Clause 77.

           This object is applicable for an OLT and has a distinct
           value for each logical link.";
        reference
          "IEEE Std 802.3, 30.3.5.1.15";
      }
      leaf out-mpcp-report {
        when "../../ompe-mode = 'onu'";
        type yang:counter64;
        units "frames";
        config false;
        description
          "A count of the number of times a REPORT MPCP frame
           transmission occurs.

           This counter is incremented by one for each REPORT MPCP
           frame transmitted as defined in IEEE Std 802.3,
           Clause 64 and Clause 77.

           This object is applicable for an ONU and has a distinct
           value for each logical link.";
        reference
          "IEEE Std 802.3, 30.3.5.1.13";
      }
      leaf in-mpcp-report {
        when "../../ompe-mode = 'olt'";
        type yang:counter64;
        units "frames";
        config false;
        description
          "A count of the number of times a REPORT MPCP frame
           reception occurs.

           This counter is incremented by one for each REPORT MPCP
           frame received as defined in IEEE Std 802.3,
           Clause 64 and Clause 77.

           This object is applicable for an OLT and has a distinct
           value for each logical link.";
        reference
          "IEEE Std 802.3, 30.3.5.1.18";
      }
      leaf out-mpcp-gate {
        when "../../ompe-mode = 'olt'";
        type yang:counter64;
        units "frames";
        config false;
        description
          "A count of the number of times a GATE MPCP frame
           transmission occurs.

           This counter is incremented by one for each GATE MPCP
```

```
            frame transmitted as defined in IEEE Std 802.3,
            Clause 64 and Clause 77.

            This object is applicable for an OLT and has a distinct
            value for each logical link.";
         reference
           "IEEE Std 802.3, 30.3.5.1.9";
       }
       leaf in-mpcp-gate {
         when "../../ompe-mode = 'onu'";
         type yang:counter64;
         units "frames";
         config false;
         description
           "A count of the number of times a GATE MPCP frame
            reception occurs.

            This counter is incremented by one for each GATE MPCP
            frame received as defined in IEEE Std 802.3,
            Clause 64 and Clause 77.

            This object is applicable for an ONU and has a distinct
            value for each logical link.";
         reference
           "IEEE Std 802.3, 30.3.5.1.14";
       }
       leaf out-mpcp-register {
         when "../../ompe-mode = 'olt'";
         type yang:counter64;
         units "frames";
         config false;
         description
           "A count of the number of times a REGISTER MPCP frame
            transmission occurs.

            This counter is incremented by one for each
            REGISTER MPCP frame transmitted as defined in
            IEEE Std 802.3, Clause 64 and Clause 77.

            This object is applicable for an OLT and has a distinct
            value for each logical link.";
         reference
           "IEEE Std 802.3, 30.3.5.1.11";
       }
       leaf in-mpcp-register {
         when "../../ompe-mode = 'onu'";
         type yang:counter64;
         units "frames";
         config false;
         description
           "A count of the number of times a REGISTER MPCP frame
            reception occurs.

            This counter is incremented by one for each
            REGISTER MPCP frame received as defined in
            IEEE Std 802.3, Clause 64 and Clause 77.

            This object is applicable for an ONU and has a distinct
            value for each logical link.";
```

```
        reference
          "IEEE Std 802.3, 30.3.5.1.16";
      }
    }
    container statistics-ompe {
      description
        "This container defines a set of OMP-Emulation-related
         statistics counters of an EPON interface, as defined in
         IEEE Std 802.3, Clause 65 and Clause 76.

         Discontinuities in the values of counters in this
         container can occur at re-initialization of the management
         system, and at other times as indicated by the value of
         the 'discontinuity-time' leaf defined in the
         ietf-interfaces YANG module (IETF RFC 8343).";
      reference
        "IEEE Std 802.3.1, dot3OmpEmulationStatEntry";
      leaf in-ompe-frames-errored-sld {
        type yang:counter64;
        units "frames";
        config false;
        description
          "A count of frames received that do not contain a valid
           SLD field as defined in IEEE Std 802.3, 65.1.3.3.1 or
           76.2.6.1.3.1, as appropriate.

           This object is applicable for an OLT and an ONU.
           It has a distinct value for each logical link.";
        reference
          "IEEE Std 802.3, 30.3.7.1.3";
      }
      leaf in-ompe-frames-errored-crc8 {
        type yang:counter64;
        units "frames";
        config false;
        description
          "A count of frames received that contain a valid SLD
           field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
           76.2.6.1.3.1 as appropriate, but do not pass the CRC-8
           check as defined in IEEE Std 802.3, 65.1.3.3.3 or
           76.2.6.1.3.3 as appropriate.

           This object is applicable for an OLT and an ONU.
           It has a distinct value for each logical link.";
        reference
          "IEEE Std 802.3, 30.3.7.1.4";
      }
      leaf ompe-onu-frames-with-good-llid-good-crc8 {
        when "../../ompe-mode = 'onu'";
        type yang:counter64;
        units "frames";
        config false;
        description
          "A count of frames received that 1) contain a valid SLD
           field in an ONU, 2) meet the rule for frame acceptance,
           and 3) pass the CRC-8 check.

           The SLD is defined in IEEE Std 802.3, 65.1.3.3.1 or
           76.2.6.1.3.1, as appropriate.
```

```
           The rules for LLID acceptance are defined in
           IEEE Std 802.3, 65.1.3.3.2 or 76.2.6.1.3.2,
           as appropriate.

           The CRC-8 check is defined in IEEE Std 802.3,
           65.1.3.3.3 or 76.2.6.1.3.3, as appropriate.

           This object is applicable for an ONU and has a distinct
                   value for each logical link.";
        reference
          "IEEE Std 802.3, 30.3.7.1.6";
      }
      leaf ompe-olt-frames-with-good-llid-good-crc8 {
        when "../../ompe-mode = 'olt'";
        type yang:counter64;
        units "frames";
        config false;
        description
          "A count of frames received that 1) contain a valid SLD
           field in an OLT, and 2) pass the CRC-8 check.

           The SLD is defined in IEEE Std 802.3, 65.1.3.3.1 or
           76.2.6.1.3.1, as appropriate.

           The frame acceptance are defined in IEEE Std 802.3,
           65.1.3.3.2 or 76.2.6.1.3.2, as appropriate.

           The CRC-8 check is defined in IEEE Std 802.3,
           65.1.3.3.3 or 76.2.6.1.3.3, as appropriate.

           This object is applicable for an OLT and has a distinct
           value for each logical link.";
        reference
          "IEEE Std 802.3, 30.3.7.1.6";
      }
      leaf in-ompe-frames-with-bad-llid {
        when "../../ompe-mode = 'olt'";
        type yang:counter64;
        units "frames";
        config false;
        description
          "A count of frames received that contain a valid SLD
           field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
           76.2.6.1.3.1, as appropriate, and pass the CRC-8 check
           as defined in IEEE Std 802.3, 65.1.3.3.3 or
           76.2.6.1.3.3, as appropriate, but are discarded due to
           the LLID check.

           This object is applicable for an OLT and has a distinct
                   value for each logical link.";
        reference
          "IEEE Std 802.3, 30.3.7.1.8";
      }
      leaf in-ompe-frames-with-good-llid {
        type yang:counter64;
        units "frames";
        config false;
        description
```

```
          "A count of frames received that contain a valid SLD
           field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
           76.2.6.1.3.1 as appropriate, but do not pass the CRC-8
           check as defined in IEEE Std 802.3, 65.1.3.3.3 or
           76.2.6.1.3.3 as appropriate.

           This object is applicable for an OLT and an ONU. It has
           a distinct value for each logical link.";
        reference
          "IEEE Std 802.3, 30.3.7.1.4";
      }
      leaf in-ompe-frames {
        type yang:counter64;
        units "frames";
        config false;
        description
          "A count of frames received that contain a valid SLD
           field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
           76.2.6.1.3.1, as appropriate, and pass the CRC-8
           check as defined in IEEE Std 802.3, 65.1.3.3.3 or
           76.2.6.1.3.3, as appropriate.

           This object is applicable for an OLT and an ONU. It has
           a distinct value for each logical link.";
        reference
          "IEEE Std 802.3, 30.3.7.1.6 (ONU) and 30.3.7.1.7 (OLT)";
      }
      leaf in-ompe-frames-not-match-onu-llid-broadcast {
        when "../../ompe-mode = 'onu'";
        type yang:counter64;
        units "frames";
        config false;
        description
          "A count of frames received that contain a valid SLD
           field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
           76.2.6.1.3.1, as appropriate, pass the CRC-8 check, as
           defined in IEEE Std 802.3, 65.1.3.3.3 or 76.2.6.1.3.3,
           as appropriate, and contain the broadcast bit in the
           LLID and not the ONU's LLID (frame accepted) as defined
           in IEEE Std 802.3,Clause 65 and Clause 76,
           as appropriate.

           This object is applicable for an ONU only.";
        reference
          "IEEE Std 802.3.1,
                   dot3OmpEmulationBroadcastBitNotOnuLlid";
      }
      leaf in-ompe-frames-match-onu-llid-not-broadcast {
        when "../../ompe-mode = 'onu'";
        type yang:counter64;
        units "frames";
        config false;
        description
          "A count of frames received that contain a valid SLD
           field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
           76.2.6.1.3.1, as appropriate, pass the CRC-8 check, as
           defined in IEEE Std 802.3, 65.1.3.3.3 or 76.2.6.1.3.3,
           as appropriate, and contain the ONU's LLID
           (frame accepted) as defined in IEEE Std 802.3, Clause 65
```

```
           and Clause 76, as appropriate.

           This object is applicable for an ONU only.";
        reference
          "IEEE Std 802.3.1, dot3OmpEmulationOnuLLIDNotBroadcast";
      }
      leaf in-ompe-frames-match-onu-llid-broadcast {
        when "../../ompe-mode = 'onu'";
        type yang:counter64;
        units "frames";
        config false;
        description
          "A count of frames received that contain a valid SLD
           field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
           76.2.6.1.3.1, as appropriate, pass the CRC-8 check, as
           defined in IEEE Std 802.3, 65.1.3.3.3 or 76.2.6.1.3.3,
           as appropriate, and contain the broadcast bit in the
           LLID and the ONU's LLID (frame accepted) as defined in
           IEEE Std 802.3, Clause 65 and Clause 76, as appropriate.

           This object is applicable for an ONU only.";
        reference
          "IEEE Std 802.3.1,
                   dot3OmpEmulationBroadcastBitPlusOnuLlid";
      }
      leaf in-ompe-frames-not-match-onu-llid-not-broadcast {
        when "../../ompe-mode = 'onu'";
        type yang:counter64;
        units "frames";
        config false;
        description
          "A count of frames received that contain a valid SLD
           field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
           76.2.6.1.3.1, as appropriate, pass the CRC-8 check,
           as defined in IEEE Std 802.3, 65.1.3.3.3 or
           76.2.6.1.3.3, as appropriate, do not contain the
           broadcast bit in the LLID and do not contain the ONU's
           LLID (frame is NOT accepted) as defined in
           IEEE Std 802.3, Clause 65 and Clause 76, as appropriate.

           This object is applicable for an ONU only.";
        reference
          "IEEE Std 802.3.1,
                   dot3OmpEmulationNotBroadcastBitNotOnuLlid";
      }
    }
    container thresholds-trx {
      if-feature "trx-power-level-reporting-supported";
      description
        "This container defines a set of optical transceiver
         thresholds of an EPON interface as defined in
         IEEE Std 802.3, Clause 60 and Clause 75.";
      reference
        "IEEE Std 802.3.1, dot3ExtPkgOptIfEntry";
      leaf in-trx-power-low-threshold {
        if-feature "trx-power-level-reporting-supported";
        type power-level;
        description
          "This object reflects the current setting of low alarm
```

```
          threshold for the input power into the optical receiver.
          If the value reported in 'in-trx-power' object drops
          below the value set in 'in-trx-power-low-threshold', a
          'in-trx-power-low-threshold-crossing' event is
          generated.

          This object is applicable for an OLT and an ONU. It has
          a distinct value for each logical link.";
      reference
        "IEEE Std 802.3.1,
          dot3ExtPkgOptIfLowerInputPowerThreshold";
    }
    leaf in-trx-power-high-threshold {
      if-feature "trx-power-level-reporting-supported";
      type power-level;
      description
        "This object reflects the current setting of high alarm
          threshold for the input power into the optical receiver.
          If the value reported in 'in-trx-power' object exceeds
          the value set in 'in-trx-power-high-threshold', a
          'in-trx-power-high-threshold-crossing' event is
          generated.

          This object is applicable for an OLT and an ONU. It has
          a distinct value for each logical link.";
      reference
        "IEEE Std 802.3.1,
          dot3ExtPkgOptIfUpperInputPowerThreshold";
    }
    leaf out-trx-power-low-threshold {
      if-feature "trx-power-level-reporting-supported";
      type power-level;
      description
        "This object reflects the current setting of low alarm
          threshold for the output power out of the optical
          transmitter. If the value reported in 'out-trx-power'
          object drops below the value set in
          'out-trx-power-low-threshold', a
          'out-trx-power-low-threshold-crossing' event is
          generated.

          This object is applicable for an OLT and an ONU. It has
          a distinct value for each logical link.";
      reference
        "IEEE Std 802.3.1,
          dot3ExtPkgOptIfLowerOutputPowerThreshold";
    }
    leaf out-trx-power-high-threshold {
      if-feature "trx-power-level-reporting-supported";
      type power-level;
      description
        "This object reflects the current setting of high alarm
          threshold for the output power out of the optical
          transmitter. If the value reported in 'out-trx-power'
          object exceeds the value set in
          'out-trx-power-high-threshold', a
          'out-trx-power-high-threshold-crossing' event is
          generated.
```

```
           This object is applicable for an OLT and an ONU.
                   It has a distinct value for each logical link.";
        reference
          "IEEE Std 802.3.1,
                   dot3ExtPkgOptIfUpperOutputPowerThreshold";
      }
    }
    container statistics-trx {
      if-feature "trx-power-level-reporting-supported";
      status deprecated;
      description
        "This container defines a set of optical transceiver
         statistics counters of an EPON interface as defined in
         IEEE Std 802.3, Clause 60 and Clause 75.";
      reference
        "IEEE Std 802.3.1, dot3ExtPkgOptIfEntry";
      leaf in-trx-power-signal-detect {
        type boolean;
        config false;
        description
          "This object indicates whether a valid optical signal was
           detected (when read as 'true') or not
           (when read as 'false') at the input to the
           optical transceiver.

           This object is applicable for an OLT and an ONU. It has
           a distinct value for each logical link.";
        reference
          "IEEE Std 802.3.1, dot3ExtPkgOptIfSignalDetect";
      }
      leaf in-trx-power {
        type power-level;
        config false;
        description
          "This object reflects the value of the input power, as
           measured at the optical transceiver, expressed in units
           of 0.1 dBm.

           At the ONU, the measurement is performed in a continuous
           manner.

           At the OLT, the measurement is performed in a burst-mode
           manner, for each incoming data burst.

           This object is applicable for an OLT and an ONU. It has
           a distinct value for each logical link.";
        reference
          "IEEE Std 802.3.1, dot3ExtPkgOptIfInputPower";
      }
      leaf in-trx-power-low-15-minutes-bin {
        type power-level;
        config false;
        description
          "This object reflects the lowest value of the input power
           during the period of the last 15 minutes, as measured at
                   the optical transceiver, and expressed in units of
           0.1 dBm.

           At the ONU, the measurement is performed in a continuous
```

```
            manner and stored in a rolling 15-minutes' long
            observation bin.

            At the OLT, the measurement is the average power for
            each incoming data burst, and stored in a rolling
            15-minutes' long observation bin.

            This object is applicable for an OLT and an ONU. It has
            a distinct value for each logical link.";
        reference
          "IEEE Std 802.3.1, dot3ExtPkgOptIfLowInputPower";
      }
      leaf in-trx-power-high-15-minutes-bin {
        type power-level;
        config false;
        description
          "This object reflects the highest value of the input
            power during the period of the last 15 minutes, as
            measured at the optical transceiver, and expressed in
            units of 0.1 dBm.

            At the ONU, the measurement is performed in a continuous
            manner and stored in a rolling 15-minutes' long
            observation bin.

            At the OLT, the measurement is the average power for
            each incoming data burst, and stored in a rolling
            15-minutes' long observation bin.

            This object is applicable for an OLT and an ONU. It has
            a distinct value for each logical link.";
        reference
          "IEEE Std 802.3.1, dot3ExtPkgOptIfHighInputPower";
      }
      leaf out-trx-power-signal-detect {
        type boolean;
        config false;
        description
          "This object indicates whether a valid optical signal was
            detected (when read as 'true') or not
            (when read as 'false') at the output from the
            optical transceiver.

            This object is applicable for an OLT and an ONU. It has
            a distinct value for each logical link.";
        reference
          "IEEE Std 802.3.1, dot3ExtPkgOptIfTransmitAlarm";
      }
      leaf out-trx-power {
        type power-level;
        config false;
        description
          "This object reflects the value of the output power, as
            measured at the optical transceiver, expressed in units
            of 0.1 dBm.

            At the ONU, the measurement is performed in a burst-mode
            manner for each outgoing data burst.
```

```
            At the OLT, the measurement is performed in a continuous
            manner.

            This object is applicable for an OLT and an ONU. It has
            a distinct value for each logical link.";
         reference
           "IEEE Std 802.3.1, dot3ExtPkgOptIfOutputPower";
       }
       leaf out-trx-power-low-15-minutes-bin {
         type power-level;
         config false;
         description
           "This object reflects the lowest value of the output
            power during the period of the last 15 minutes, as
            measured at the optical transceiver, and expressed in
            units of 0.1 dBm.

            At the ONU, the measurement is performed in a burst-mode
            manner and stored in a rolling 15-minutes' long
            observation bin.

            At the OLT, the measurement is the average power for
            each incoming data burst, and stored in a rolling
            15-minutes' long observation bin.

            This object is applicable for an OLT and an ONU. It has
            a distinct value for each logical link.";
         reference
           "IEEE Std 802.3.1, dot3ExtPkgOptIfLowOutputPower";
       }
       leaf out-trx-power-high-15-minutes-bin {
         type power-level;
         config false;
         description
           "This object reflects the highest value of the output
            power during the period of the last 15 minutes, as
            measured at the optical transceiver, and expressed in
            units of 0.1 dBm.

            At the ONU, the measurement is performed in a burst-mode
            manner and stored in a rolling 15-minutes' long
            observation bin.

            At the OLT, the measurement is the average power for
            each incoming data burst, and stored in a rolling
            15-minutes' long observation bin.

            This object is applicable for an OLT and an ONU. It has
            a distinct value for each logical link.";
         reference
           "IEEE Std 802.3.1, dot3ExtPkgOptIfHighOutputPower";
       }
       leaf trx-data-reliable {
         if-feature "trx-power-level-reporting-supported";
         type boolean;
         config false;
         description
           "This object indicates whether data contained in
            individual counters in 'statistics-trx' container are
```

```
           reliable (when read as 'true') or not
           (when read as 'false').

           This object is applicable for an OLT and an ONU. It has
           a distinct value for each logical link.";
        reference
          "IEEE Std 802.3.1, dot3ExtPkgOptIfSuspectedFlag";
      }
    }
    container monitoring-trx {
      if-feature "trx-power-level-reporting-supported";
      description
        "This container defines a set of optical transceiver
         statistics counters of an EPON interface as defined in
         IEEE Std 802.3, Clause 60 and Clause 75.";
      reference
        "IEEE Std 802.3.1, dot3ExtPkgOptIfEntry";
      leaf in-trx-power-signal-detect {
        type boolean;
        config false;
        description
          "This object indicates whether a valid optical signal was
           detected (when read as 'true') or not
           (when read as 'false')at the input to the
           optical transceiver.

           This object is applicable for an OLT and an ONU. It has
           a distinct value for each logical link.";
        reference
          "IEEE Std 802.3.1, dot3ExtPkgOptIfSignalDetect";
      }
      leaf in-trx-power {
        type power-level;
        config false;
        description
          "This object reflects the value of the input power, as
           measured at the optical transceiver, expressed in units
           of 0.1 dBm.

           At the ONU, the measurement is performed in a continuous
           manner.

           At the OLT, the measurement is performed in a burst-mode
           manner, for each incoming data burst.

           This object is applicable for an OLT and an ONU. It has
           a distinct value for each logical link.";
        reference
          "IEEE Std 802.3.1, dot3ExtPkgOptIfInputPower";
      }
      leaf in-trx-power-low-15-minutes-bin {
        type power-level;
        config false;
        description
          "This object reflects the lowest value of the input power
           during the period of the last 15 minutes, as measured at
           the optical transceiver, and expressed in units of
           0.1 dBm.
```

```
            At the ONU, the measurement is performed in a continuous
            manner and stored in a rolling 15-minutes' long
            observation bin.

            At the OLT, the measurement is the average power for
            each incoming data burst, and stored in a rolling
            15-minutes' long observation bin.

            This object is applicable for an OLT and an ONU.
            It has a distinct value for each logical link.";
        reference
          "IEEE Std 802.3.1, dot3ExtPkgOptIfLowInputPower";
      }
      leaf in-trx-power-high-15-minutes-bin {
        type power-level;
        config false;
        description
          "This object reflects the highest value of the input
            power during the period of the last 15 minutes,
            as measured at the optical transceiver,
            and expressed in units of 0.1 dBm.

            At the ONU, the measurement is performed in a continuous
            manner and stored in a rolling 15-minutes' long
            observation bin.

            At the OLT, the measurement is the average power for
            each incoming data burst, and stored in a rolling
            15-minutes' long observation bin.

            This object is applicable for an OLT and an ONU. It has
            a distinct value for each logical link.";
        reference
          "IEEE Std 802.3.1, dot3ExtPkgOptIfHighInputPower";
      }
      leaf out-trx-power-signal-detect {
        type boolean;
        config false;
        description
          "This object indicates whether a valid optical signal was
            detected (when read as 'true') or not
            (when read as 'false') at the output from the
            optical transceiver.

            This object is applicable for an OLT and an ONU. It has
            a distinct value for each logical link.";
        reference
          "IEEE Std 802.3.1, dot3ExtPkgOptIfTransmitAlarm";
      }
      leaf out-trx-power {
        type power-level;
        config false;
        description
          "This object reflects the value of the output power, as
            measured at the optical transceiver, expressed in units
            of 0.1 dBm.

            At the ONU, the measurement is performed in a burst-mode
            manner for each outgoing data burst.
```

```
          At the OLT, the measurement is performed in a continuous
          manner.

          This object is applicable for an OLT and an ONU. It has
          a distinct value for each logical link.";
      reference
        "IEEE Std 802.3.1, dot3ExtPkgOptIfOutputPower";
    }
    leaf out-trx-power-low-15-minutes-bin {
      type power-level;
      config false;
      description
        "This object reflects the lowest value of the output
          power during the period of the last 15 minutes, as
          measured at the optical transceiver, and expressed in
          units of 0.1 dBm.

          At the ONU, the measurement is performed in a burst-mode
          manner and stored in a rolling 15-minutes' long
          observation bin.

          At the OLT, the measurement is the average power for
          each incoming data burst, and stored in a rolling
          15-minutes' long observation bin.

          This object is applicable for an OLT and an ONU. It has
          a distinct value for each logical link.";
      reference
        "IEEE Std 802.3.1, dot3ExtPkgOptIfLowOutputPower";
    }
    leaf out-trx-power-high-15-minutes-bin {
      type power-level;
      config false;
      description
        "This object reflects the highest value of the output
          power during the period of the last 15 minutes, as
          measured at the optical transceiver, and expressed in
          units of 0.1 dBm.

          At the ONU, the measurement is performed in a burst-mode
          manner and stored in a rolling 15-minutes' long
          observation bin.

          At the OLT, the measurement is the average power for
          each incoming data burst, and stored in a rolling
          15-minutes' long observation bin.

          This object is applicable for an OLT and an ONU. It has
          a distinct value for each logical link.";
      reference
        "IEEE Std 802.3.1, dot3ExtPkgOptIfHighOutputPower";
    }
    leaf trx-data-reliable {
      if-feature "trx-power-level-reporting-supported";
      type boolean;
      config false;
      description
        "This object indicates whether data contained in
```

```
          individual counters in 'statistics-trx' container are
          reliable (when read as 'true') or not
          (when read as 'false').

          This object is applicable for an OLT and an ONU. It has
          a distinct value for each logical link.";
       reference
         "IEEE Std 802.3.1, dot3ExtPkgOptIfSuspectedFlag";
     }
   }
   container statistics-pon-fec {
     when "(../fec-capability = 'supported') and
            (../fec-mode = 'enabled-Tx-Rx')";
     if-feature "fec-supported";
     config false;
     description
       "This container defines a set of FEC-related statistics
        counters of an EPON interface, as defined in
        IEEE Std 802.3, Clause 65 and Clause 76.

        Discontinuities in the value of this counter can occur at
        re-initialization of the management system, and at other
        times as indicated by the value of the
        'discontinuity-time' leaf defined in the ietf-interfaces
        YANG module (IETF RFC 8343).";
     reference
       "IEEE Std 802.3.1, dot3OmpEmulationStatEntry";
     leaf fec-code-group-violations {
       type yang:counter64;
       units "code-group";
       config false;
       description
         "For 1G-EPON this is a count of the number of events that
          cause the PHY to indicate â€™Data reception errorâ€™ or
          â€™Carrier Extend Errorâ€™ on the GMII (see IEEE Std 802.3,
          Table 35-1). The contents of this counter is undefined
          when FEC is operating. For 10G-EPON this object is not
          applicable.

          This object is applicable for an OLT and an ONU. At the
          OLT, it has a distinct value for each logical link.";
       reference
         "IEEE Std 802.3, 30.5.1.1.14";
     }
     leaf fec-buffer-head-coding-violations {
       type yang:counter64;
       units "code-group";
       config false;
       description
         "For 1G-EPON PHY, this object represents the count of the
          number of invalid code-groups received directly from the
          link when FEC is enabled. When FEC is disabled this
          counter stops counting.

          For 10G-EPON PHYs, this object is set to zero.

          This object is applicable for an OLT and an ONU. It has
          a distinct value for each logical link.";
       reference
```

```
          "IEEE Std 802.3.1, dot3EponFecBufferHeadCodingViolation";
      }
      leaf fec-code-word-corrected-errors {
        type yang:counter64;
        units "code-group";
        config false;
        description
          "For 1G-EPON or 10G-EPON PHYs, this object represents a
           count of corrected FEC blocks.

           This counter increments by one for each received
           FEC block that contained detected errors and was
           corrected by the FEC function in the PHY.

           This object is applicable for an OLT and an ONU. It has
           a distinct value for each logical link.";
        reference
          "IEEE Std 802.3, 30.5.1.1.17";
      }
      leaf fec-code-word-uncorrected-errors {
        type yang:counter64;
        units "code-group";
        config false;
        description
          "For 1G-EPON or 10G-EPON PHYs, this object represents a
           count of uncorrectable FEC blocks.

           This counter increments by one for each received FEC
           block that contained detected errors and was not
           corrected by the FEC function in the PHY.

           This object is applicable for an OLT and an ONU. It has
           a distinct value for each logical link.";
        reference
          "IEEE Std 802.3, 30.5.1.1.18";
      }
    }
    container mpcp-logical-link-admin-actions {
      description
        "Container of actions.";
      action state-change-action-type {
        description
          "Request a state change on the interface.";
        input {
          leaf state-change-action-type {
            type identityref {
              base state-change-action-type;
            }
            description
              "Type of interface state change requested.";
          }
        }
      }
      action reset-action-type {
        description
          "Request a reset-action of the interface.";
        input {
          leaf reset-action-type {
            type identityref {
```

```
              base reset-action-type;
            }
            description
              "Type of reset action requested of the interface.";
          }
        }
      }
      action register-type {
        description
          "Request a registration action.";
        input {
          leaf register-type {
            type identityref {
              base register-type;
            }
            description
              "Type of registration action requested of the
               interface.";
          }
        }
      }
    }
    list mpcp-queues {
      key "mpcp-queue-index";
      description
        "An instance of this object for each value of
         'mpcp-queue-index' is created when a new logical link is
         registered and deleted when the logical link is
         deregistered.

         All instances of this object in the ONU associated with
         the given logical link are then mapped to a REPORT MPCPDU,
         when generated.

         +------------------------------------+
         |          Destination Address       |
         +------------------------------------+
         |          Source Address            |
         +------------------------------------+
         |          Length/Type               |
         +------------------------------------+
         |          OpCode                    |
         +------------------------------------+
         |          TimeStamp                 |
         +------------------------------------+
         |          Number of Queue Sets      |
         +------------------------------------+   -|
         |          Report bitmap             |    |
         +------------------------------------+    |
         |          Queue 0 report            |    |
         +------------------------------------+    | repeated
         |          Queue 1 report            |    | for every
         +------------------------------------+    | Queue Set
         |          Queue 2 report            |    |
         +------------------------------------+    |
         |          Queue 3 report            |    |
         +------------------------------------+    |
         |          Queue 4 report            |    |
         +------------------------------------+    |
```

```
          |          Queue 5 report           |    |
          +-----------------------------------+    |
          |          Queue 6 report           |    |
          +-----------------------------------+    |
          |          Queue 7 report           |    |
          +-----------------------------------+   -|
          |          Pad/reserved             |
          +-----------------------------------+
          |          FCS                      |
          +-----------------------------------+

          The 'Queue N report' field reports the current occupancy
          of each upstream transmission queue associated with the
          given logical link.

          The 'Number of Queue Sets' field defines the number of
          reported 'Queue N report' sets.

          For each Queue Set, the 'Report bitmap' field defines
          which upstream transmission queues are present in the
          REPORT MPCPDU.
          Although the REPORT MPCPDU can report current occupation
          for up to 8 upstream transmission queues in a single
          REPORT MPCPDU, the actual number is flexible.
          The 'mpcp-queue-group' grouping has a variable size that
          is limited by value of
          'mpcp-maximum-queue-count-per-report' object, allowing
          ONUs report the occupancy of fewer upstream transmission
          queues, as needed.

          This object is applicable for an OLT and an ONU.
          At the OLT, this object has a distinct value for each
          logical link and every queue. At the ONU, it has a
          distinct value for every queue.";
       reference
         "IEEE Std 802.3.1, dot3ExtPkgQueueEntry";
       leaf mpcp-queue-index {
         type uint8 {
           range "0 .. 7" {
             description
               "This object indicates the identity (index) of a
                queue in the ONU. It can have a value between
                0 and 7, limited by the value stored in the
                'mpcp-maximum-queue-count-per-report' object.";
             reference
               "See 'mpcp-maximum-queue-count-per-report' object";
           }
         }
         description
           "An object represents the index of an upstream
            transmission queue storing subscriber packets.
            The size (occupancy) of the upstream transmission queue
            identified by this object is then reported within
            REPORT MPCPDU, defined in IEEE Std 802.3, Clause 64
            and Clause 77.

            This object indicates the identity (index) of a queue in
            the ONU. It can have a value between 0 and 7, limited by
            the value stored in the
```

```
            'mpcp-maximum-queue-count-per-report' object.

            This object is applicable for an OLT and an ONU.
            It has a distinct value for each logical link and each
            queue.
            At the ONU, it has a distinct value for each queue.";
        reference
          "IEEE Std 802.3.1, dot3QueueIndex";
      }
      leaf mpcp-queue-threshold-count {
        type uint8 {
          range "0 .. 7" {
            description
              "This object indicates the identity (index) of a
               queue in the ONU. It can have a value between
               0 and 7, limited by the value stored in the
               'mpcp-maximum-queue-count-per-report' object.";
            reference
              "See 'mpcp-queue-threshold-count-max' object";
          }
        }
        description
          "This object reflects the number of reporting thresholds
           for the specific upstream transmission queue, reflected
           in the REPORT MPCPDU, as defined in IEEE Std 802.3,
           Clause 64 and Clause 77.

           Each 'Queue set' provides information for the specific
           upstream transmission queue occupancy of frames below
           the matching reporting threshold.

           A read of this object reflects the number of reporting
           thresholds for the specific upstream transmission queue.

           This object is applicable for an OLT and an ONU. It has
           a distinct value for each logical link and each queue.
           At the ONU, it has a distinct value for each queue.";
        reference
          "IEEE Std 802.3.1, dot3ExtPkgObjectReportNumThreshold";
      }
      leaf mpcp-queue-threshold-count-max {
        type uint8 {
          range "0 .. 7" {
            description
              "This object can have a value between 0 and 7.";
          }
        }
        config false;
        description
          "This object reflects the maximum number of reporting
           thresholds for the specific upstream transmission queue,
           reflected in the REPORT MPCPDU, as defined in
           IEEE Std 802.3, Clause 64 and Clause 77.

           A read of this object reflects the maximum number of
           reporting thresholds for the specific upstream
           transmission queue.

           This object is applicable for an OLT and an ONU. It has
```

```
            a distinct value for each logical link and each queue.
            At the ONU, it has a distinct value for each queue.";
         reference
           "IEEE Std 802.3.1,
            dot3ExtPkgObjectReportMaximumNumThreshold";
       }
       list mpcp-queue-thresholds {
         when '../mpcp-queue-threshold-count > 0';
         key "mpcp-queue-set-index";
         max-elements 7;
         description
           "An instance of this object for each value of
            'mpcp-queue-index' is created when a new logical link is
            registered and deleted when the logical link is
            deregistered.

            All instances of this object in the ONU associated with
            the given logical link are then mapped to a
            REPORT MPCPDU, when generated.

            +------------------------------------+
            |          Destination Address       |
            +------------------------------------+
            |          Source Address            |
            +------------------------------------+
            |          Length/Type               |
            +------------------------------------+
            |          OpCode                    |
            +------------------------------------+
            |          TimeStamp                 |
            +------------------------------------+
            |          Number of Queue Sets      |
            +------------------------------------+   -|
            |          Report bitmap             |    |
            +------------------------------------+    |
            |          Queue 0 report            |    |
            +------------------------------------+    | repeated for
            |          Queue 1 report            |    | every
            +------------------------------------+    | Queue Set
            |          Queue 2 report            |    |
            +------------------------------------+    |
            |          Queue 3 report            |    |
            +------------------------------------+    |
            |          Queue 4 report            |    |
            +------------------------------------+    |
            |          Queue 5 report            |    |
            +------------------------------------+    |
            |          Queue 6 report            |    |
            +------------------------------------+    |
            |          Queue 7 report            |    |
            +------------------------------------+   -|
            |          Pad/reserved              |
            +------------------------------------+
            |          FCS                       |
            +------------------------------------+

            The 'Queue N report' field reports the current occupancy
            of each upstream transmission queue associated with the
            given logical link.
```

```
           The 'Number of Queue Sets' field defines the number of
           reported 'Queue N report' sets.

           For each Queue Set, the 'Report bitmap' field defines
           which upstream transmission queues are present in the
           REPORT MPCPDU. Although the REPORT MPCPDU can report
           current occupation for up to 8 upstream transmission
           queues in a single REPORT MPCPDU, the actual number is
           flexible.

           The 'mpcp-queue-group' grouping has a variable size that
           is limited by value of
           'mpcp-maximum-queue-count-per-report' object, allowing
           ONUs to report the occupancy of fewer upstream
           transmission queues, as needed.

           This object is applicable for an OLT and an ONU. It has
           a distinct value for each logical link and every queue.
           At the ONU, it has a distinct value for every queue.";
        reference
          "IEEE Std 802.3.1, dot3ExtPkgQueueSetsEntry";
        leaf mpcp-queue-set-index {
          type uint8 {
            range "0 .. 7" {
              description
                "This object indicates the identity (index) of a
                 queue in the ONU. It can have a value between
                 0 and 7, limited by the value stored in the
                 'mpcp-maximum-queue-count-per-report' object.";
              reference
                "See 'mpcp-maximum-queue-count-per-report' object";
            }
          }
          description
            "This object represents the index of the Queue Set for
             the'mpcp-queue-set-group' grouping. The size
             (occupancy) of the upstream transmission queues
             belonging to the given Queue Set is then reported
             within REPORT MPCPDU, definedin IEEE Std 802.3,
             Clause 64 and Clause 77.

             This object can have a value between 0 and 7, limited
             by the value stored in the
             'mpcp-queue-threshold-count-max' object.";
          reference
            "IEEE Std 802.3.1, dot3QueueSetIndex";
        }
        leaf mpcp-queue-set-threshold {
          type uint64;
          units "TQ";
          default "0";
          description
            "This object defines the value of a reporting threshold
             for each Queue Set stored in REPORT MPCPDU defined in
             IEEE Std 802.3, Clause 64 and Clause 77.

             The number of Queue Sets for each upstream
             transmission queue is defined in the
```

```
              'mpcp-queue-threshold-count' object.

              Within REPORT MPCPDU, each Queue Set provides
              information on the current upstream transmission queue
              occupancy for frames below the matching threshold.

              The value stored in this object is expressed in the
              units of Time quanta (TQ), where 1 TQ = 16 ns.

              A read of this object provides the current threshold
              value for the specific upstream transmission queue.

              This object is applicable for an OLT and an ONU. At
              the OLT, it has a distinct value for each logical link,
              each queue, and each Queue Set.

              At the ONU, it has a distinct value for each queue and
              each Queue Set.";
           reference
             "IEEE Std 802.3.1, dot3ExtPkgObjectReportThreshold";
         }
       }
       leaf in-mpcp-queue-frames {
         type yang:counter64;
         config false;
         description
           "A count of the number of times a frame reception event
            results in a frame being queued in (for ONUs) or
            received from (for OLTs) the corresponding queue. This
            object is incremented by one for each frame written to
            (in the case of the ONU) or received for (in case of the
            OLT) the associated queue.

            The queue index matches the queue number in
            REPORT MPCPDU, as defined in IEEE Std 802.3, Clause 64
            and Clause 77.

            This object is applicable for an OLT and an ONU.
            At the OLT,it has a distinct value for each logical link
            and each queue. At the ONU, it has a distinct value for
            each queue.";
         reference
           "IEEE Std 802.3.1, dot3ExtPkgStatRxFramesQueue";
       }
       leaf out-mpcp-queue-frames {
         when "../../mpcp-mode = 'onu'";
         type yang:counter64;
         config false;
         description
           "This object reflects the number of frame transmission
            events from the corresponding upstream transmission
            queue. This object is incremented by one for each frame
            transmitted, when it is output from the associated
            queue.

            The queue index matches the queue number in
            REPORT MPCPDU, as defined in IEEE Std 802.3, Clause 64
            and Clause 77.
```

```
            This object is applicable for an ONU only. At the ONU,
            it has a distinct value for each queue.";
         reference
           "IEEE Std 802.3.1, dot3ExtPkgStatTxFramesQueue";
       }
       leaf mpcp-queue-frames-drop {
         when "../../mpcp-mode = 'onu'";
         type yang:counter64;
         config false;
         description
           "This object reflects the number of frame drop events
            from the corresponding upstream transmission queue.
            This object is incremented by one for each frame dropped
            in the associated queue.

            The queue index matches the queue number in
            REPORT MPCPDU, as defined in IEEE Std 802.3, Clause 64
            and Clause 77.

            This object is applicable for an ONU only. At the ONU,
            it has a distinct value for each queue.";
         reference
           "IEEE Std 802.3.1, dot3ExtPkgStatDroppedFramesQueue";
       }
     }
     list multicast-IDs {
       key "multicast-ID";
       description
         "Multicast-IDs list of multicast IDs
          to be recognized by the device.";
       leaf multicast-ID {
         type uint32;
         description
           "Multicast-IDs to be recognized by the device.";
         reference
           "IEEE Std 802.3, 30.3.5.1.25";
       }
     }
     leaf fec-capability {
       type fec-capability;
       config false;
       description
         "This object is used to identify whether the given
          interface is capable of supporting FEC or not.";
     }
     leaf mpcp-mode {
       type mpcp-mode;
       config false;
       description
         "This object is used to identify the operational state of
          the MultiPoint MAC Control sublayer as defined in
          IEEE Std 802.3, Clause 64 and Clause 77.

          Reading 'olt' for an OLT (controller) mode and 'onu' for
          an ONU (client) mode.

          This object is used to identify the operational mode for
          the MPCP objects.
```

```
           This object is applicable for an OLT, with the same value
           for all logical links, and for an ONU.";
        reference
          "IEEE Std 802.3, 30.3.5.1.3";
      }
      leaf mpcp-sync-time {
        type uint64;
        units "TQ (16ns)";
        config false;
        description
          "This object reports the 'sync lock time' of the OLT
           receiver in units of Time Quanta (TQ; 1 TQ = 16 ns; see
           IEEE Std 802.3, Clause 64 and Clause 77).

           The value returned is equal to [sync lock time ns]/16,
           rounded up to the nearest TQ. If this value exceeds
           4,294,967,295 TQ, the value 4,294,967,295 TQ is returned.

           This object is applicable for an OLT, with distinct values
           for all logical links, and for an ONU.";
        reference
          "IEEE Std 802.3.1, dot3MpcpSyncTime";
      }
      leaf mpcp-logical-link-id {
        type mpcp-supported;
        config false;
        description
          "This object is used to identify the operational state of
           the MultiPoint MAC Control sublayer as defined in
           IEEE Std 802.3, Clause 64 and Clause 77.

           Reading 'olt' for an OLT (controller) mode and 'onu' for
           an ONU (client) mode.

           This object is used to identify the operational mode for
           the MPCP objects.

           This object is applicable for an OLT, with the same value
           for all logical links, and for an ONU.";
        reference
          "IEEE Std 802.3, 30.3.5.1.3";
      }
      leaf mpcp-remote-mac-address {
        type ieee:mac-address;
        config false;
        description
          "This object identifies the source_address parameter of the
           last MPCPDUs passed to the MAC Control. This value is
           updated on reception of a valid frame with:

           1) a destination Field equal to the multicast address
           assigned for MAC Control as specified in
           IEEE Std 802.3, Annex 31A;

           2) the lengthOrType field value equal to the Type assigned
           for MAC Control as specified in IEEE Std 802.3, Annex 31A;

           3) an MPCP Control opcode value equal to the subtype
           assigned for MPCP as specified in IEEE Std 802.3,
```

```
          Annex 31A.

          This object is applicable for an OLT and an ONU. It has a
          distinct value for each logical link.

          The value reflects the MAC address of the remote entity
          and therefore the OLT holds a value for each LLID, which
          is the MAC address of the ONU.

          The ONU has a single value that is the OLT MAC address.";
      reference
        "IEEE Std 802.3, 30.3.5.1.5";
    }
    leaf mpcp-logical-link-state {
      type mpcp-logical-link-state;
      config false;
      description
        "This object identifies the registration state of the
          MultiPoint MAC Control sublayer as defined in
          IEEE Std 802.3, Clause 64 and Clause 77.

          When this object has the enumeration 'unregistered', the
          interface is unregistered and may be used for registering
          a link partner.

          When this object has the enumeration 'registering',
          the interface is in the process of registering a
                  link-partner.

          When this object has the enumeration 'registered', the
          interface has an established link-partner.

          This object is applicable for an OLT and an ONU. It has a
          distinct value for each logical link.";
      reference
        "IEEE Std 802.3, 30.3.5.1.6";
    }
    leaf mpcp-elapsed-time-out {
      type uint64;
      units "TQ (16ns)";
      config false;
      description
        "This object reports the interval from the last MPCP frame
          transmission in increments of Time Quanta
          (TQ; 1 TQ = 16 ns;
          see IEEE Std 802.3, Clause 64 and Clause 77).

          The value returned is equal to [interval from last MPCP
          frame transmission on this EPON interface, expressed
          in ns]/16. If this value exceeds 4,294,967,295 TQ, the
          value 4,294,967,295 TQ is returned.

          This object is applicable for an OLT and an ONU. It has a
          distinct value for each logical link.";
      reference
        "IEEE Std 802.3, 30.3.5.1.19";
    }
    leaf mpcp-elapsed-time-in {
      type uint64;
```

```
        units "TQ (16ns)";
        config false;
        description
          "This object reports the interval from the last MPCP frame
           reception in increments of Time Quanta (TQ; 1 TQ = 16 ns;
           see IEEE Std 802.3, Clause 64 and Clause 77).

           The value returned is equal to [interval from last MPCP
           frame reception on this EPON interface, expressed in
           ns]/16. If this value exceeds 4,294,967,295 TQ, the value
           4,294,967,295 TQ is returned.

           This object is applicable for an OLT and an ONU. It has a
           distinct value for each logical link.";
        reference
          "IEEE Std 802.3, 30.3.5.1.20";
      }
      leaf mpcp-round-trip-time {
        when "../ompe-mode = 'olt'";
        type uint16;
        units "TQ (16ns)";
        config false;
        description
          "This object reports the MPCP round trip time in increments
           of Time Quanta (TQ; 1 TQ = 16 ns; see IEEE Std 802.3,
           Clause 64 and Clause 77).

           The value returned is equal to [round trip time in ns]/16.
           If this value exceeds 65,535 TQ, the value 65,535 TQ is
           returned.

           This object is applicable for an OLT. It has a distinct
           value for each logical link.";
        reference
          "IEEE Std 802.3, 30.3.5.1.21";
      }
      leaf mpcp-maximum-grant-count {
        when "../ompe-mode = 'onu'";
        type uint8;
        config false;
        description
          "This object reports the maximum number of grants that an
           ONU can store for handling. The maximum number of grants
           that an ONU can store for handling has a range of
           0 to 255.

           This object is applicable for an ONU and has a distinct
           value for each logical link.";
        reference
          "IEEE Std 802.3, 30.3.5.1.24";
      }
      leaf mpcp-logical-link-count {
        type mpcp-llid-count;
        units "LLID";
        config false;
        description
          "This object reflects the number of logical links
           registered on the OLT / ONU. The LLID field, as defined in
           the IEEE Std 802.3, Clause 65 and Clause 76, is a 2-byte
```

```
          register (15-bit field and a broadcast bit) limiting the
          number of logical links to 32,768.

          This object is initialized to the value of 0 when the
          OLT / ONU is powered up.

          This object is applicable for an OLT and an ONU. It has
                  the same value for all logical links.";
        reference
          "IEEE Std 802.3.1, dot3ExtPkgObjectNumberOfLLIDs";
      }
      leaf mpcp-maximum-queue-count-per-report {
        when "../ompe-mode = 'olt'";
        type mpcp-maximum-queue-count-per-report;
        config false;
        description
          "This object reflects the maximum number of queues (0-7)
           that can be accepted by the OLT in a single REPORT MPCPDU,
           as defined in IEEE Std 802.3, Clause 64 and Clause 77.

           This object is applicable for an OLT and has a distinct
           value for each logical link.";
        reference
          "IEEE Std 802.3.1, dot3ExtPkgObjectReportMaximumNumQueues";
      }
      leaf ompe-mode {
        type ompe-mode;
        config false;
        description
          "This object indicates the mode of operation of the
           Reconciliation Sublayer for Point-to-Point Emulation (see
           IEEE Std 802.3, 65.1 or 76.2 as appropriate).

           The value of 'unknown' is assigned in initialization; true
           state or type is not yet known.

           The value of 'olt' is assigned when the sublayer is
           operating in OLT mode.

           The value of 'onu' is assigned when the sublayer is
           operating in ONU mode.

           This object is applicable for an OLT and an ONU. It has
           the same value for each logical link.";
        reference
          "IEEE Std 802.3, 30.3.7.1.2";
      }
    }
  }
```

8. YANG module for Ethernet Link OAM

8.1 Introduction

IEEE Std 802.3, Clause 57 includes management capabilities for Ethernet-like interfaces to provide some basic operations, administration and maintenance (OAM) functions. The defined functionality includes discovery, error signaling, loopback, and link monitoring. This clause defines a portion of the YANG module for use with NETCONF or RESTCONF to manage these Ethernet-like interface capabilities.

8.2 Overview

Ethernet OAM is composed of a core set of functions and a set of optional functional groups as described in Clause 57 of IEEE Std 802.3. The core functions include discovery operations (determining if the other end of the link is OAM capable and what OAM functions it supports), state machine implementation, and some critical event flows.

Ethernet OAM provides single-hop functionality in that it works only between two directly connected Ethernet stations. Ethernet OAM has three functional objectives, which are detailed in 8.2.1 through 8.2.3. The definition of a basic Ethernet OAM protocol data unit is given in 8.2.4.

8.2.1 Remote fault indication

Remote fault indication provides a mechanism for one end of an Ethernet link to signal the other end that the receive path is non-operational. Some Ethernet Physical Layers offer mechanisms to signal this condition at the Physical Layer. Ethernet OAM added a mechanism so that some Ethernet Physical Layers can operate in unidirectional mode, allowing frames to be transmitted in one direction even when the other direction is non-operational. Traditionally, Ethernet PHYs do not allow frame transmission in one direction if the other direction is not operational. Using this mode, Ethernet OAM allows frame-based signaling of remote fault conditions while still not allowing higher layer applications to be aware of the unidirectional capability. This clause includes mechanisms for capturing that fault information and reflecting such information in data nodes and notifications within the NETCONF management framework.

8.2.2 Link monitoring

Ethernet OAM includes event signaling capability so that one end of an Ethernet link can indicate the occurrence of certain important events to the other end of the link. This happens via layer 2 protocols. This clause defines methods for incorporating the occurrence of these events, at both the local end and the far end of the link, into the YANG-based management framework.

Ethernet OAM also includes mechanisms for one Ethernet station to query another directly connected Ethernet station about the status of its Ethernet interface variables and status. This clause does not include mechanisms for controlling how one Ethernet endpoint may use this functionality to query the status or statistics of a peer Ethernet entity.

8.2.3 Remote loopback

Remote loopback is a link state where the peer Ethernet entity echoes every received packet (without modifications) back onto the link. Remote loopback is intrusive in that the other end of the link is not forwarding traffic from higher layers out over the link. This clause defines data nodes controlling loopback operation and reading the status of the loopback state.

8.2.4 Ethernet OAM protocol data units

An Ethernet OAM protocol data unit (OAMPDU) is a valid Ethernet frame with a destination Media Access Control (MAC) address equal to the MAC address assigned for Slow Protocols (see IEEE Std 802.3, Annex 57A), a lengthOrType field equal to the Type assigned for Slow Protocols, and a Slow Protocols subtype equal to that of the subtype assigned for Ethernet OAM.

OAMPDU is used throughout this clause as an abbreviation for Ethernet OAM protocol data unit. OAMPDUs are the mechanism by which two directly connected Ethernet interfaces exchange OAM information.

8.3 Security considerations for Ethernet operations, administration, and maintenance (OAM) module

The readable data nodes in this module can provide information about network traffic, and therefore, they may be considered sensitive. In particular, OAM provides mechanisms for reading the Clause 30 IEEE 802.3 management attributes from a link partner via a layer 3 protocol. IEEE Std 802.3 OAM does not include encryption or authentication mechanisms. It should not be used in environments where this interface information is considered sensitive, and where the facility terminations are unprotected. By default, OAM is disabled on Ethernet-like interfaces and is therefore not a risk.

IEEE Std 802.3 OAM is designed to support deployment in access and enterprise networks. In access networks, one end of a link is the CO-side, and the other is the CPE-side, and the facilities are often protected in wiring cages or closets. In such deployments, it is often the case that the CO-side is protected from access from the CPE-side. Within IEEE Std 802.3 OAM, this protection from remote access is accomplished by configuring the CPE-side in passive mode using the mode leaf. This prevents the CPE from accessing functions and information at the CO-side of the connection. In enterprise networks, read-only interface information is often considered non-sensitive.

The frequency of OAM PDUs on an Ethernet interface does not adversely affect data traffic, as OAM is a slow protocol with very limited bandwidth potential, and it is not required for normal link operation. Although there are a number of objects in this module with read-write or read-create MAX-ACCESS, they have limited effects on user data.

The loopback capability of OAM can have potentially disruptive effects; when remote loopback is enabled, the remote station automatically transmits all received traffic back to the local station except for OAM traffic. This completely disrupts all higher layer protocols such as bridging, IP, and NETCONF/RESTCONF.

The administrative state and mode are also configuration nodes. Disabling OAM can interrupt management activities between peer devices, potentially causing serious problems. Setting the mode node to an undesired value can allow access to Ethernet monitoring, events, and functions that may not be acceptable in a particular deployment scenario. In addition to loopback functionality, Ethernet interface statistics and events can be accessed via the OAM protocol, which may not be desired in some circumstances.

OAM event configuration also contains configuration nodes. These nodes control whether events are sent, and at what thresholds. Note that the frequency of event communication is limited by the frequency limits of Slow Protocols on Ethernet interfaces. Also, the information available via OAM events is also available via OAM Variable Requests. Access to this information via either OAM events or Variable Requests is controlled by the admin and mode nodes. As mentioned previously, inadequate protection of these variables can result in access to link information and functions.

8.4 Mapping of IEEE 802.3 managed objects

This subclause contains the mapping between the YANG data nodes defined in this clause and the attributes defined in IEEE Std 802.3, Clause 30. Table 8–1 provides the mapping between the *ieee802-ethernet-link-oam* module data nodes and the OAM attributes of IEEE Std 802.3, Clause 30.

Table 8–1—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-link-oam* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-link-oam* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
oOAM	aOAMAdminState	30.3.6.1.2	interfaces/interface/ethernet/link-oam	admin	R/W
	dot3OamOperStatus aOAMDiscoveryState aOAMLocalFlagsField aOAMRemoteFlagsField	IEEE Std 802.3.1 30.3.6.1.4 30.3.6.1.10 30.3.6.1.11	interfaces/interface/ethernet/link-oam/discovery-info/local	operational-status	R
	aOAMLocalState	30.3.6.1.14		loopback-mode	R
	aOAMMode	30.3.6.1.3		mode	RW
	aOAMLocalRevision	30.3.6.1.12		revision	R
	aOAMLocalPDUConfiguration	30.3.6.1.8		oammtu	R
	aOAMRemoteConfiguration	30.3.6.1.7	interfaces/interface/ethernet/link-oam/discovery-info/remote/functions-supported	uni-directional-link-fault	R/W
				loopback	R/W
				mib-retrieval	R/W
	aOAMLocalConfiguration	30.3.6.1.6	interfaces/interface/ethernet/link-oam/discovery-info/local/functions-supported/link-monitor	link-monitoring	R/W
	aOAMLocalErrSymPeriodConfig aOAMLocalErrFrameConfig aOAMLocalErrFramePeriodConfig aOAMLocalErrFrameSecsSummaryConfig aOAMLocalErrSymPeriodConfig	30.3.6.1.34 30.3.6.1.36 30.3.6.1.38 30.3.6.1.40 30.3.6.1.42	interfaces/interface/ethernet/link-oam/link-monitor/event-type	threshold-type window threshold	R/W R/W R/W
	aOAMRemoteMACAddress	30.3.6.1.5	interfaces/interface/ethernet/link-oam/discovery-info/remote	mac-address	R
	aOAMRemoteVendorOUI	30.3.6.1.16		vendor-oui	R

Table 8–1—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-link-oam* YANG data nodes *(continued)*

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-link-oam* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
	aOAMRemoteVendorSpecificInfo	30.3.6.1.17		vendor-info	R
	aOAMRemoteState	30.3.6.1.15		loopback-mode	R
	aOAMMode	30.3.6.1.3		mode	R
	aOAMRemoteRevision	30.3.6.1.13		revision	R
	aOAMRemotePDUConfiguration	30.3.6.1.9		oammtu	R

Table 8–1—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-link-oam* YANG data nodes *(continued)*

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-link-oam* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
OAM	aOAMLocalConfiguration	30.3.6.1.6	interfaces/interface/ethernet/link-oam/discovery-info/local/functions-supported	uni-directional-link-fault	R
				loopback	R
				link-monitoring	R
				mib-retrieval	R
	dot3OamEventLogEntry	IEEE Std 802.3.1	interfaces-state/ interface/ethernet/link-oam/event-log/event-log-entry	index	R
				oui	R
				timestamp	R
				location	R
				event-type	R
				running-total	R
				event-total	R
	aOAMLocalErrSymPeriodEvent aOAMLocalErrFrameEvent aOAMLocalErrFramePeriodConfig aOAMLocalErrFrameSecsSummaryEvent aOAMRemoteErrSymPeriodEvent aOAMRemoteErrFrameEvent aOAMRemoteErrFramePeriodEvent aOAMRemoteErrFrameSecsSummaryEvent	30.3.6.1.35 30.3.6.1.37 30.3.6.1.38 30.3.6.1.41 30.3.6.1.42 30.3.6.1.43 30.3.6.1.44 30.3.6.1.45	interfaces/interface/ethernet/link-oam/event-log/event-log-entry/threshold	threshold-event-type window threshold value	R R R R
	Dot3OamStatsEntry	RFC-4878	interfaces/interface/ethernet/link-oam/statistics	out-information	R
	aOAMInformationTx	30.3.6.1.20			

Table 8–1—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-link-oam* YANG data nodes *(continued)*

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-link-oam* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
	aOAMInformationRx	30.3.6.1.21		in-information	R
	aOAMUniqueEventNotificationTx	30.3.6.1.22		out-unique-event-notification	R
	aOAMUniqueEventNotificationRx	30.3.6.1.24		in-unique-event-notification	R
	aOAMDuplicateEventNotificationTx	30.3.6.1.23		out-duplicate-event-notification	R
	aOAMDuplicateEventNotificationRx	30.3.6.1.25		in-duplicate-event-notification	R
	aOAMLoopbackControlTx	30.3.6.1.26		out-loopback-control	R
	aOAMLoopbackControlRx	30.3.6.1.27		in-loopback-control	R
	aOAMVariableRequestTx	30.3.6.1.28		out-variable-request	R

Table 8–1—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-link-oam* YANG data nodes *(continued)*

IEEE Std 802.3, Clause 30		Reference	Corresponding *ieee802-ethernet-link-oam* YANG data nodes		
Managed object(s)	**Attribute(s)**		**Container(s)**	**Data node(s)**	**R/W**
OAM	aOAMVariableRequestRx	30.3.6.1.29	interfaces/interface/ethernet/link-oam/statistics	variable-requeste-rx	R
	aOAMVariableResponseTx	30.3.6.1.30		out-variable-response	R
	aOAMVariableResponseRx	30.3.6.1.31		in-variable-response	R
	aOAMOrganizationSpecificTx	30.3.6.1.32		out-org-specific	R
	aOAMOrganizationSpecificRx	30.3.6.1.33		in-org-specific	R
	aOAMUnsupportedCodesTx	30.3.6.1.18		unsupported-condes-tx	R
	aOAMUnsupportedCodesRx	30.3.6.1.19		in-unsupported-codes	R
	aFramesLostDueToOAMError	30.3.6.1.46		frames-lost-due-to-oam	R
	aOAMLocalErrSymPeriodEvent, Errored Symbols	30.3.6.1.35		local-error-symbol-period-log-entries	R
	aOAMLocalErrFrameEvent, Errored Frames	30.3.6.1.37		local-error-frame-log-entries	R
	aOAMLocalErrFramePeriodEvent, Errored Frames	30.3.6.1.39		local-error-frame-period-log-entries	R
	aOAMLocalErrFrameSecsSummar-yEvent, Errored Frame Seconds Summary	30.3.6.1.41		local-error-frame-second-log-entries	R
	aOAMRemoteErrSymPeriodEvent, Errored Symbols	30.3.6.1.42		remote-error-symbol-period-log-entries	R
	aOAMRemoteErrFrameEven, Errored Frames	30.3.6.1.43		remote-error-frame-log-entries	R
	aOAMRemoteErrFramePeriodEvent, Errored Frames	30.3.6.1.44		remote-error-frame-period-log-entries	R
	aOAMRemoteErrFrameSecsSummar-yEvent, Errored Frame Seconds Summary	30.3.6.1.45		remote-error-frame-second-log-entries	R

8.5 YANG module definition[m]

The YANG module tree hierarchy uses terms defined in IETF RFC 8407.

8.5.1 Tree hierarchy

8.5.1.1 ieee802-ethernet-link-oam

```
module: ieee802-ethernet-link-oam

  augment /if:interfaces/if:interface:
    +--rw link-oam!
       +---x remote-loopback {remote-loopback-initiate}?
       |  +---w input
       |  |  +---w enable    boolean
       |  +--ro output
       |     +--ro success          boolean
       |     +--ro error-message?   string
       +---x reset-stats
       |  +--ro output
       |     +--ro success          boolean
       |     +--ro error-message?   string
       +---n non-threshold-event
       |  +-- oui              vendor-oui
       |  +-- timestamp        uint64
       |  +-- location         event-location
       |  +-- event-type       identityref
       |  +-- running-total    yang:counter64
       |  +-- event-total      yang:counter64
       +---n threshold-event {link-monitoring-local or
link-monitoring-remote}?
       |  +-- oui              vendor-oui
       |  +-- timestamp        uint64
       |  +-- location         event-location
       |  +-- event-type       identityref
       |  +-- running-total    yang:counter64
```

[m]Copyright release for YANG modules: Users of this standard may freely reproduce the YANG module contained in this subclause so that it can be used for its intended purpose.

```
       |  +-- event-total      yang:counter64
       |  +-- threshold {link-monitoring-local or
link-monitoring-remote}?
       |     +-- threshold-event-type    threshold-event-enum
       |     +-- window                  uint64
       |     +-- threshold               uint64
       |     +-- value                   uint64
       +--rw admin?                 admin-state
       +--rw discovery-info
       |  +--rw local
       |  |  +--ro operational-status     operational-state
       |  |  +--ro loopback-mode          loopback-status {remote-loopback-initiate or
remote-loopback-respond}?
       |  |  +--rw mode?                  mode
       |  |  +--rw functions-supported
       |  |  |  +--rw uni-directional-link-fault?   boolean {uni-directional-link-fault}?
       |  |  |  +--rw loopback?                    boolean {remote-loopback-initiate}?
       |  |  |  +--rw link-monitor {link-monitoring-remote or
link-monitoring-local}?
       |  |  |  |  +--rw link-monitoring?   boolean
       |  |  |  |  +--rw event-type* [threshold-type] {link-monitoring-local}?
       |  |  |  |     +--rw threshold-type    threshold-event-enum
       |  |  |  |     +--rw window?           uint64
       |  |  |  |     +--rw threshold?        uint64
       |  |  |  x--rw mib-retrieval?                boolean {remote-mib-retrieval-initiate or
remote-mib-retrieval-respond}?
       |  |  |  +--rw data-retrieval?               boolean {remote-data-retrieval-initiate or
remote-mib-retrieval-respond}?
       |  |  +--ro revision?              uint64
       |  |  +--ro oammtu?                uint16
       |  +--ro remote
       |     +--ro mac-address?           ieee:mac-address
       |     +--ro vendor-oui?            vendor-oui
       |     +--ro vendor-info?           uint64
       |     +--ro loopback-mode          loopback-status
       |     +--ro mode?                  mode
       |     +--ro functions-supported
       |     |  +--ro uni-directional-link-fault?   boolean
       |     |  +--ro loopback?                    boolean
```

```
      |     |  +--ro link-monitoring?             boolean
      |     |  +--ro mib-retrieval?               boolean
      |     +--ro revision?             uint64
      |     +--ro oammtu?               uint16
      +--ro event-log
      |  +--ro event-log-entry* [index]
      |     +--ro index           uint64
      |     +--ro oui             vendor-oui
      |     +--ro timestamp       uint64
      |     +--ro location        event-location
      |     +--ro event-type      identityref
      |     +--ro running-total   yang:counter64
      |     +--ro event-total     yang:counter64
      |     +--ro threshold {link-monitoring-local or
link-monitoring-remote}?
      |        +--ro threshold-event-type    threshold-event-enum
      |        +--ro window                  uint64
      |        +--ro threshold               uint64
      |        +--ro value                   uint64
      +--ro statistics
         +--ro out-information                          yang:counter64
         +--ro in-information                           yang:counter64
         +--ro out-unique-event-notification            yang:counter64 {link-monitoring-local}?
         +--ro in-unique-event-notification             yang:counter64 {link-monitoring-remote}?
         +--ro out-duplicate-event-notification         yang:counter64 {link-monitoring-local}?
         +--ro in-duplicate-event-notification          yang:counter64 {link-monitoring-remote}?
         +--ro out-loopback-control                     yang:counter64 {remote-loopback-initiate}?
         +--ro in-loopback-control                      yang:counter64 {remote-loopback-respond}?
         +--ro out-variable-request                     yang:counter64 {remote-data-retrieval-initiate}?
         +--ro in-variable-request                      yang:counter64 {remote-mib-retrieval-respond}?
         +--ro out-variable-response                    yang:counter64 {remote-mib-retrieval-respond}?
         +--ro in-variable-response                     yang:counter64 {remote-data-retrieval-initiate}?
         +--ro out-org-specific                         yang:counter64
         +--ro in-org-specific                          yang:counter64
         +--ro out-unsupported-codes                    yang:counter64
         +--ro in-unsupported-codes                     yang:counter64
         +--ro frames-lost-due-to-oam                   yang:counter64
         +--ro local-error-symbol-period-log-entries    yang:counter64
         +--ro local-error-frame-log-entries            yang:counter64
```

```
+--ro local-error-frame-period-log-entries      yang:counter64
+--ro local-error-frame-second-log-entries      yang:counter64
+--ro remote-error-symbol-period-log-entries    yang:counter64 {link-monitoring-remote}?
+--ro remote-error-frame-log-entries            yang:counter64 {link-monitoring-remote}?
+--ro remote-error-frame-period-log-entries     yang:counter64 {link-monitoring-remote}?
+--ro remote-error-frame-second-log-entries     yang:counter64 {link-monitoring-remote}?
```

8.5.2 YANG module

In the YANG module definition, should any discrepancy between the text of the description for individual YANG nodes in the YANG modules of this clause and the corresponding definition in 8.2 through 8.5 of this clause occur, the definitions in the YANG modules attached to this standard shall take precedence.

An ASCII text version of the YANG module can be found at the following URL:[n]

> ***Editor's Note (to be removed prior to publication):***
>
> *Yang files contained in https://github.com/YangModels/yang/tree/main/standard/ieee/published/802.3 are IEEE 802.3.2-2019 version and will be updated at the publication time.*

https://github.com/YangModels/yang/tree/master/standard/ieee/published/802.3

> ***Editor's Note (to be removed prior to publication):***
>
> *Pretty printing of ieee802-ethernet-link-oam.yang file may change the appearance by adding whitespace and reformatting lines*

> ***Editor's Note (to be removed prior to publication):***
>
> IEEE Std 802.3.1 and IEEE Std 802.3.2 to be updated at the publication time

```
module ieee802-ethernet-link-oam {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-link-oam";
  prefix ieee802-link-oam;

  import ieee802-types {
    prefix ieee;
    reference
      "IEEE 802 types";
  }
  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
  }
  import iana-if-type {
    prefix ianaift;
    reference
      "http://www.iana.org/assignments/yang-parameters/
       iana-if-type@2023-01-26.yang";
  }
  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }

  organization
    "IEEE 802.3 Ethernet Working Group
     Web URL: http://www.ieee802.org/3/";
```

[n]Copyright release for YANG modules: Users of this standard may freely reproduce the YANG module contained in this subclause so that it can be used for its intended purpose.

```
contact
  "Web URL: http://www.ieee802.org/3/";
description
  "This module contains a collection of YANG definitions
   for managing the Ethernet Link OAM feature defined by IEEE
   802.3. It provides functionality roughly equivalent to that of
   the DOT3-OAM-MIB defined in IETF RFC 4878.";

revision 2025-04-17 {
  description
    "Updates under IEEE Std 802.3.2-202x, Draft D3.3";
  reference
    "IEEE Std 802.3-2022, unless dated explicitly";
}

feature uni-directional-link-fault {
  description
    "This feature means the device supports Uni Directional Link
     Fault detection.";
  reference
    "IEEE Std 802.3, 57.1.2:a, 30.3.6.1.6 aOAMLocalConfiguration
     and 30.3.6.1.7 aOAMRemoteConfiguration";
}

feature remote-loopback-initiate {
  description
    "This feature means the device supports being the initiator
     of remote loopback.";
  reference
    "IEEE Std 802.3, 57.1.2:b, 30.3.6.1.6
     aOAMLocalConfiguration";
}

feature remote-loopback-respond {
  description
    "This feature means the device supports responding to remote
     loopback control OAMPDUs received from the peer";
  reference
    "IEEE Std 802.3, 57.1.2:b, 30.3.6.1.7
     aOAMRemoteConfiguration";
}

feature link-monitoring-local {
  description
    "This feature means the device monitors the link at the local
     side and can generate Link Event OAMPDUs to the peer
     device.";
  reference
    "IEEE Std 802.3, 57.1.2:c:1, 30.3.6.1.6
     aOAMLocalConfiguration, and 30.3.6.1.7
     aOAMRemoteConfiguration";
}

feature link-monitoring-remote {
  description
    "This feature means the device can process Link Event OAMPDUs
     received from the peer device and report itself about this
     event on its own management interface.";
  reference
```

```
      "IEEE Std 802.3, 57.1.2:c:1,
       30.3.6.1.6 aOAMLocalConfiguration,
       and 30.3.6.1.7 aOAMRemoteConfiguration";
  }

  feature remote-mib-retrieval-initiate {
    status deprecated;
    description
      "remote-mib-retrieval-initiate is deprecated and changed name
       to feature remote-data-retrieval-initiate. This feature
       means the device supports data retrieval from the peer
       device. I.e. the device can send Variable Requests OAMPDUs
       to the peer side and process the received Variable Response
       OAMPDUs.";
    reference
      "IEEE Std 802.3, 57.1.2:c:2,
       30.3.6.1.6 aOAMLocalConfiguration,
       and 30.3.6.1.7 aOAMRemoteConfiguration";
  }

  feature remote-data-retrieval-initiate {
    description
      "This feature means the device supports data retrieval from
       the peer device. I.e. the device can send Variable Requests
       OAMPDUs to the peer side and process the received Variable
       Response OAMPDUs.";
    reference
      "IEEE Std 802.3, 57.1.2:c:2,
       30.3.6.1.6 aOAMLocalConfiguration,
       and 30.3.6.1.7 aOAMRemoteConfiguration";
  }

  feature remote-mib-retrieval-respond {
    description
      "This feature means the device allows the peer device to
       retrieve data from the managed device. I.e. the device can
       process received Variable Requests OAMPDUs and respond with
       Variable Response OAMPDUs.";
    reference
      "IEEE Std 802.3, 57.1.2:c:2,
       30.3.6.1.6 aOAMLocalConfiguration,
       and 30.3.6.1.7 aOAMRemoteConfiguration";
  }

  identity event-type {
    description
      "Base identity for all Link OAM event types.";
  }

  identity threshold-event-type {
    base event-type;
    description
      "Event type for a Link Monitoring threshold event.";
  }

  identity link-fault-event {
    if-feature "uni-directional-link-fault";
    base event-type;
    description
```

```
      "Event type for a uni-directional link fault event.";
    reference
      "IEEE Std 802.3, 57.2.10.1";
  }

  identity dying-gasp-event {
    base event-type;
    description
      "Event type for a dying gasp event.";
    reference
      "IEEE Std 802.3, 57.2.10.1";
  }

  identity critical-event {
    base event-type;
    description
      "Event type for a critical event.";
    reference
      "IEEE Std 802.3, 57.2.10.1";
  }

  typedef threshold-event-enum {
    type enumeration {
      enum symbol-period-event {
        value 1;
        description
          "Errored symbol period event.";
      }
      enum frame-period-event {
        value 2;
        description
          "Errored frame period event.";
      }
      enum frame-event {
        value 3;
        description
          "Errored frame event";
      }
      enum frame-seconds-event {
        value 4;
        description
          "Errored frame seconds event.";
      }
    }
    description
      "Enumeration of the valid threshold event types.";
    reference
      "IEEE Std 802.3, 57.5.3";
  }

  typedef mode {
    type enumeration {
      enum passive {
        value 0;
        description
          "Ethernet Link OAM Passive mode.";
      }
      enum active {
        value 1;
```

```
        description
          "Ethernet Link OAM Active mode.";
      }
    }
    description
      "Enumeration of the valid modes in which Link OAM may run.";
    reference
      "IEEE Std 802.3, 57.2.9 and 30.3.6.1.3.";
  }

  typedef event-location {
    type enumeration {
      enum event-location-local {
        value 1;
        description
          "A local event.";
      }
      enum event-location-remote {
        value 2;
        description
          "A remote event.";
      }
    }
    description
      "The location of the event that caused a log entry.";
  }

  typedef loopback-status {
    type enumeration {
      enum none {
        value 1;
        description
          "Loopback is not being performed.";
      }
      enum initiating {
        value 2;
        description
          "Initiating master loopback.";
      }
      enum master-loopback {
        value 3;
        description
          "In master loopback mode.";
      }
      enum terminating {
        value 4;
        description
          "Terminating master loopback mode.";
      }
      enum local-loopback {
        value 5;
        description
          "In slave loopback mode.";
      }
      enum unknown {
        value 6;
        description
          "Parser and multiplexer combination unexpected.";
      }
```

```
    }
    description
      "The loopback mode of an OAM interface.";
    reference
      "IEEE Std 802.3, 57.2.11";
  }

  typedef operational-state {
    type enumeration {
      enum disabled {
        value 1;
        description
          "IEEE Std 802.3 OAM is disabled.";
      }
      enum link-fault {
        value 2;
        description
          "IEEE Std 802.3 OAM has encountered a link fault.";
      }
      enum passive-wait {
        value 3;
        description
          "Passive OAM entity waiting to see if peer is
           OAM capable.";
      }
      enum active-send-local {
        value 4;
        description
          "Active OAM entity trying to determine if peer
           is OAM capable.";
      }
      enum send-local-and-remote {
        value 5;
        description
          "OAM discovered peer but still to accept or
           reject peer configuration.";
      }
      enum send-local-and-remote-ok {
        value 6;
        description
          "OAM peering is allowed by local device.";
      }
      enum peering-locally-rejected {
        value 7;
        description
          "OAM peering rejected by local device.";
      }
      enum peering-remotely-rejected {
        value 8;
        description
          "OAM peering rejected by remote device.";
      }
      enum operational {
        value 9;
        description
          "IEEE Std 802.3 OAM is operational.";
      }
      enum operational-half-duplex {
        value 10;
```

```
          description
            "IEEE Std 802.3 OAM is operating in half-duplex mode.";
        }
      }
      description
        "Operational state of an interface.";
      reference
        "IEEE Std 802.3, 30.3.6.1.4,
         30.3.6.1.10, and 30.3.6.1.11";
    }

    typedef vendor-oui {
      type string {
        length "6";
      }
      description
        "Organizationally Unique Identifier (OUI) as assigned to
         vendor by the IEEE Registration Authority";
      reference
        "IEEE Std 802, subclause 8.2.2";
    }

    typedef admin-state {
      type enumeration {
        enum enabled {
          value 1;
          description
            "IEEE Std 802.3, Clause 57 OAM is in the
             enabled admin state.";
        }
        enum disabled {
          value 2;
          description
            "IEEE Std 802.3, Clause 57 OAM is in the
             disabled admin state.";
        }
      }
      description
        "Admin state of the OAM function on an interface.";
      reference
        "IEEE Std 802.3, 30.3.6.1.2 and 30.3.6.2";
    }

    grouping event-details {
      description
        "Nodes describing an event, used in the event log and in
         notifications.";
      reference
        "IETF RFC 4878, Dot3OamEventLogEntry";
      leaf oui {
        type vendor-oui;
        mandatory true;
        description
          "Organizationally Unique Identifier for the device that
           generated the event.";
      }
      leaf timestamp {
        type uint64;
        units "milliseconds";
```

```
      mandatory true;
      description
        "Timestamp in milliseconds since Unix epoch for when the
         event occurred.";
    }
    leaf location {
      type event-location;
      mandatory true;
      description
        "Where the event occurred (local or remote).";
    }
    leaf event-type {
      type identityref {
        base event-type;
      }
      mandatory true;
      description
        "Type of event that occurred.";
      reference
        "IEEE Std 802.3, 30.3.6.1.10 and 30.3.6.11";
    }
    leaf running-total {
      type yang:counter64;
      mandatory true;
      description
        "The running total number of errors seen since OAM was
         enabled on the interface. For threshold events, this is
         the total number of times that particular type of error
         (e.g. symbol error) has occurred, which may be greater
         than the number of threshold-crossing event notifications
         of that type generated during that time (which is conveyed
         by the event-total leaf).";
    }
    leaf event-total {
      type yang:counter64;
      mandatory true;
      description
        "Total number of times this event has occurred since OAM
         was enabled on the interface. For threshold events this is
         the number of events generated of this type (as opposed to
         the total number of errors of that type, which may be
         greater, and is conveyed by the running-total leaf.";
    }
  }

  grouping threshold-event-details {
    description
      "Nodes describing a threshold event, used in the event
       log and in notifications";
    reference
      "IETF RFC 4878, Dot3OamEventLogEntry";
    container threshold {
      when
        "../event-type = 'ieee802-link-oam:threshold-event-type'" {
        description
          "These nodes only apply to threshold event types";
      }
      if-feature
        "link-monitoring-local or
```

```
          link-monitoring-remote";
        description
          "Nodes specific to threshold (link monitoring) events";
        leaf threshold-event-type {
          type threshold-event-enum;
          mandatory true;
          description
            "The type of threshold event";
          reference
            "IEEE Std 802.3, 57.5.3";
        }
        leaf window {
          type uint64;
          mandatory true;
          description
            "Size of the window in which the event was generated.
             Units are dependent on the threshold event type.";
        }
        leaf threshold {
          type uint64;
          mandatory true;
          description
            "Size of the threshold that was breached during the
             window. Units are dependent on the threshold
             event type.";
        }
        leaf value {
          type uint64;
          mandatory true;
          description
            "Breaching value. Units are dependent on the threshold
             event type, and match that of the threshold.";
        }
      }
    }

    grouping statistics-common {
      description
        "Collection of Link OAM event/packet counters.";
      reference
        "IETF RFC 4878, Dot3OamStatsEntry";
      leaf out-information {
        type yang:counter64;
        mandatory true;
        description
          "Number of information OAMPDUs transmitted.";
        reference
          "IEEE Std 802.3, 30.3.6.1.20";
      }
      leaf in-information {
        type yang:counter64;
        mandatory true;
        description
          "Number of information OAMPDUs received.";
        reference
          "IEEE Std 802.3, 30.3.6.1.21";
      }
      leaf out-unique-event-notification {
        if-feature "link-monitoring-local";
```

```
        type yang:counter64;
        mandatory true;
        description
          "Number of unique event notification OAMPDUs transmitted.";
        reference
          "IEEE Std 802.3, 30.3.6.1.22";
      }
      leaf in-unique-event-notification {
        if-feature "link-monitoring-remote";
        type yang:counter64;
        mandatory true;
        description
          "Number of unique event notification OAMPDUs received.";
        reference
          "IEEE Std 802.3, 30.3.6.1.24";
      }
      leaf out-duplicate-event-notification {
        if-feature "link-monitoring-local";
        type yang:counter64;
        mandatory true;
        description
          "Number of duplicate event notification OAMPDUs
                   transmitted.";
        reference
          "IEEE Std 802.3, 30.3.6.1.23";
      }
      leaf in-duplicate-event-notification {
        if-feature "link-monitoring-remote";
        type yang:counter64;
        mandatory true;
        description
          "Number of duplicate event notification OAMPDUs
           received.";
        reference
          "IEEE Std 802.3, 30.3.6.1.25";
      }
      leaf out-loopback-control {
        if-feature "remote-loopback-initiate";
        type yang:counter64;
        mandatory true;
        description
          "Number of loopback control OAMPDUs transmitted.";
        reference
          "IEEE Std 802.3, 30.3.6.1.26";
      }
      leaf in-loopback-control {
        if-feature "remote-loopback-respond";
        type yang:counter64;
        mandatory true;
        description
          "Number of loopback control OAMPDUs received.";
        reference
          "IEEE Std 802.3, 30.3.6.1.27";
      }
      leaf out-variable-request {
        if-feature "remote-data-retrieval-initiate";
        type yang:counter64;
        mandatory true;
        description
```

```
          "Number of variable request OAMPDUs transmitted.";
        reference
          "IEEE Std 802.3, 30.3.6.1.28";
      }
      leaf in-variable-request {
        if-feature "remote-mib-retrieval-respond";
        type yang:counter64;
        mandatory true;
        description
          "Number of variable request OAMPDUs received.";
        reference
          "IEEE Std 802.3, 30.3.6.1.29";
      }
      leaf out-variable-response {
        if-feature "remote-mib-retrieval-respond";
        type yang:counter64;
        mandatory true;
        description
          "Number of variable response OAMPDUs transmitted.";
        reference
          "IEEE Std 802.3, 30.3.6.1.30";
      }
      leaf in-variable-response {
        if-feature "remote-data-retrieval-initiate";
        type yang:counter64;
        mandatory true;
        description
          "Number of variable response OAMPDUs received.";
        reference
          "IEEE Std 802.3, 30.3.6.1.31";
      }
      leaf out-org-specific {
        type yang:counter64;
        mandatory true;
        description
          "Number of organization specific OAMPDUs transmitted.";
        reference
          "IEEE Std 802.3, 30.3.6.1.32";
      }
      leaf in-org-specific {
        type yang:counter64;
        mandatory true;
        description
          "Number of organization specific OAMPDUs received.";
        reference
          "IEEE Std 802.3, 30.3.6.1.33";
      }
      leaf out-unsupported-codes {
        type yang:counter64;
        mandatory true;
        description
          "Number of OAMPDUs with unsupported codes transmitted.";
        reference
          "IEEE Std 802.3, 30.3.6.1.18";
      }
      leaf in-unsupported-codes {
        type yang:counter64;
        mandatory true;
        description
```

```
        "Number of OAMPDUs with unsupported codes received.";
      reference
        "IEEE Std 802.3, 30.3.6.1.19";
    }
    leaf frames-lost-due-to-oam {
      type yang:counter64;
      mandatory true;
      description
        "A count of the number of frames that were dropped by the
         OAM multiplexer. Since the OAM multiplexer has multiple
         inputs and a single output, there may be cases where
         frames are dropped due to transmit resource contention.
         This counter is incremented whenever a frame is dropped by
         the OAM layer.";
      reference
        "IEEE Std 802.3, 30.3.6.1.46";
    }
  }

  grouping discovery-remote {
    description
      "Nodes describing the discovery process remote end of a
           link.";
    leaf mode {
      type mode;
      description
        "Mode (passive/active).";
      reference
        "IEEE Std 802.3, 30.3.6.1.3";
    }
    container functions-supported {
      description
        "The Link OAM functions supported by this interface.";
      reference
        "IEEE Std 802.3, 30.3.6.1.7";
      leaf uni-directional-link-fault {
        type boolean;
        description
          "Unidirectional link fault support.";
      }
      leaf loopback {
        type boolean;
        description
          "Remote Loopback support.";
      }
      leaf link-monitoring {
        type boolean;
        description
          "Link monitoring support.";
      }
      leaf mib-retrieval {
        type boolean;
        description
          "MIB variable retrieval support.";
      }
    }
    leaf revision {
      type uint64;
      config false;
```

```
        description
          "Configuration revision.";
        reference
          "IEEE Std 802.3, 30.3.6.1.12 and 30.3.6.1.13";
      }
      leaf oammtu {
        type uint16;
        units "octets";
        config false;
        description
          "The maximum OAMPDU size for the remote node.
           The peer OAM entities exchange the maximum size they can
           support and negotiate to use the smaller of the two maximum
           OAMPDU sizes.";
        reference
          "IEEE Std 802.3, 30.3.6.1.8 and 30.3.6.1.9";
      }
    }

    grouping discovery-local {
      description
        "Nodes describing the local end discovery process of a
             link.";
      leaf mode {
        type mode;
        description
          "This object configures the mode of OAM operation as active
           or passive. Active mode provides capabilities to initiate
           monitoring activities with the remote OAM peer eneity,
           while passive mode waits for the peer to initiate actions
           with it. Changing this value results in incrementing the
           revision field of locally generated OAMPDUs
           (see IEEE Std 802.3, 30.3.6.1.12) and triggers the
           OAM discovery process if the operational state was
           already 'operational'.  The default value is
           implementation-dependent.";
        reference
          "IEEE Std 802.3, 30.3.6.1.3";
      }
      container functions-supported {
        description
          "The Link OAM functions supported by this interface.";
        reference
          "IEEE Std 802.3, 30.3.6.1.7";
        leaf uni-directional-link-fault {
          if-feature "uni-directional-link-fault";
          type boolean;
          description
            "Unidirectional link fault support.
             This affects the setting of the 'Unidirectional Support'
             bit in the OAM configuration field put in the
             Information OAMPDU.
             This bit indicates to the peer device that it can send
             OAM PDUs on links that are operating in unidirectional
             mode (traffic flowing in one direction only).";
        }
        leaf loopback {
          if-feature "remote-loopback-initiate";
          type boolean;
```

```
        default "true";
        description
          "Remote Loopback support.";
      }
      container link-monitor {
        if-feature
          "link-monitoring-remote or
           link-monitoring-local";
        description
          "Configure link monitor parameters.";
        reference
          "IEEE Std 802.3, 57.1.2:c";
        leaf link-monitoring {
          type boolean;
          default "true";
          description
            "Enable or disable monitoring.
             This affects the setting of the 'Link Events' bit in
             the OAM configuration field put in the Information
             OAMPDU. This bit indicates to the peer device that the
             OAM entity can send and receive Event Notification
             OAMPDUs.";
        }
        list event-type {
          if-feature "link-monitoring-local";
          key "threshold-type";
          description
            "A list containing at most one entry for each of the
             threshold event types. If there is no entry for a
             particular event type, the default values are used for
             both window size and threshold.";
          leaf threshold-type {
            type threshold-event-enum;
            description
              "The type of threshold event for which this list
               entry is specifying the configuration.";
            reference
              "IEEE Std 802.3, 57.5.3";
          }
          leaf window {
            type uint64;
            description
              "The size of the window to use when monitoring for
               this threshold event. The units, default and upper
               and lower bounds depend on the threshold type as
               follows:

               Symbol Period:
                 Units:   number of symbols
                 Default: number of symbols in one second for the
                          underlying physical layer
                 Min:     number of symbols in one second for the
                          underlying physical layer
                 Max:     number of symbols in one minute for the
                          underlying physical layer

               Frame:
                 Units:   deciseconds
                 Default: 1 second
```

```
                Min:     1 second
                Max:     1 minute

              Frame Period:
                Units:   number of frames
                Default: number of minFrameSize frames in one
                         second for the underlying physical layer
                Min:     number of minFrameSize frames in one
                         second for the underlying physical layer
                Max:     number of minFrameSize frames in one
                         minute for the underlying physical layer

              Frame Seconds:
                Units:   deciseconds
                Default: 60 seconds
                Min:     10 seconds
                Max:     900 seconds";
           reference
             "IEEE Std 802.3, 30.3.6.1.34, 30.3.6.1.36, 30.3.6.1.38,
              and 30.3.6.1.40";
         }
         leaf threshold {
           type uint64 {
             range "1..max";
           }
           default "1";
           description
             "The threshold value to use when determining whether
              to generate an event given the number of errors that
              occurred in a given window. The units depend on the
              threshold type as follows:

              Symbol Period: number of errored symbols
              Frame:         number of errored frames
              Frame Period:  number of errored frames
              Frame Seconds: number of seconds containing at least
              1 frame error";
           reference
             "IEEE Std 802.3, 30.3.6.1.34, 30.3.6.1.36, 30.3.6.1.38,
              and 30.3.6.1.40";
         }
       }
     }
     leaf mib-retrieval {
       if-feature
         "remote-mib-retrieval-initiate or
          remote-mib-retrieval-respond";
       type boolean;
       status deprecated;
       description
         "leaf mib-retrieval is deprecated and changed name to
          data-retrieval.
          MIB variable retrieval support.
          This affects the setting of the 'Variable Retrieval' bit
          in the OAM configuration field put in the Information
          OAMPDU. This bit indicates to the peer device that the
          OAM entity can send and receive Variable Request and
          Response OAMPDUs.";
     }
```

```
      leaf data-retrieval {
        if-feature
          "remote-data-retrieval-initiate or
           remote-mib-retrieval-respond";
        type boolean;
        description
          "Variable retrieval support.
           This affects the setting of the 'Variable Retrieval' bit
           in the OAM configuration field put in the Information
           OAMPDU. This bit indicates to the peer device that the
           OAM entity can send and receive Variable Request and
           Response OAMPDUs.";
      }
    }
    leaf revision {
      type uint64;
      config false;
      description
        "Configuration revision.";
      reference
        "IEEE Std 802.3, 30.3.6.1.12 and 30.3.6.1.13";
    }
    leaf oammtu {
      type uint16;
      units "octets";
      config false;
      description
        "The maximum OAMPDU size for the local node. The peer OAM
         entities exchange the maximum size they can support and
         negotiate to use the smaller of the two maximum OAMPDU
         sizes.";
      reference
        "IEEE Std 802.3, 30.3.6.1.8 and 30.3.6.1.9";
    }
  }

  grouping discovery-info {
    description
      "Information relating to the discovery process.";
    container local {
      description
        "Properties of the local device.";
      leaf operational-status {
        type operational-state;
        config false;
        mandatory true;
        description
          "Operational status.";
        reference
          "IETF RFC 4878, dot3OamOperStatus; IEEE Std 802.3,
           30.3.6.1.4, 30.3.6.1.10, and 30.3.6.1.11";
      }
      leaf loopback-mode {
        if-feature
          "remote-loopback-initiate or
           remote-loopback-respond";
        type loopback-status;
        config false;
        mandatory true;
```

```
          description
            "The loopback mode the interface is in.";
          reference
            "IEEE Std 802.3, 30.3.6.1.14";
        }
        uses discovery-local;
      }
      container remote {
        config false;
        description
          "Properties of the remote (peer) device.";
        leaf mac-address {
          type ieee:mac-address;
          description
            "Remote MAC address.";
          reference
            "IEEE Std 802.3, 30.3.6.1.5";
        }
        leaf vendor-oui {
          type vendor-oui;
          description
            "Remote vendor OUI.";
          reference
            "IEEE Std 802.3, 30.3.6.1.16";
        }
        leaf vendor-info {
          type uint64;
          description
            "Remote vendor info. The semantics of this value are
             proprietary and specific to the vendor.";
          reference
            "IEEE Std 802.3, 30.3.6.1.17";
        }
        leaf loopback-mode {
          type loopback-status;
          mandatory true;
          description
            "The loopback mode the interface is in.";
          reference
            "IEEE Std 802.3, 30.3.6.1.15";
        }
        uses discovery-remote;
      }
    }

    augment "/if:interfaces/if:interface" {
      when "derived-from-or-self(if:type, 'ianaift:ethernetCsmacd') or
            derived-from-or-self(if:type, 'ianaift:ptm') " {
        description
          "Augments the interface model with nodes
           specific to Ethernet Link OAM.";
      }
      description
        "Augments Ethernet interface model with nodes
         specific to Ethernet Link OAM.";
      container link-oam {
        presence "Implies Link OAM is configured on the interface.";
        description
          "Interface operational state for Ethernet Link OAM.";
```

```
      action remote-loopback {
        if-feature "remote-loopback-initiate";
        description
          "Start/stop remote loopback on the specified interface.";
        reference
          "IEEE Std 802.3, 57.1.2:b";
        input {
          leaf enable {
            type boolean;
            mandatory true;
            description
              "Whether to enable or disable remote loopback.";
          }
        }
        output {
          leaf success {
            type boolean;
            mandatory true;
            description
              "True if the operation was successful,
               false otherwise.";
          }
          leaf error-message {
            type string;
            description
              "If the operation failed, optionally used to
               provide extra details.";
          }
        }
      }
      action reset-stats {
        description
          "Reset Ethernet Link OAM statistics on this interface.";
        output {
          leaf success {
            type boolean;
            mandatory true;
            description
              "True if the operation was successful,
               false otherwise.";
          }
          leaf error-message {
            type string;
            description
              "If the operation failed, optionally used to provide
               extra details.";
          }
        }
      }
      notification non-threshold-event {
        description
          "This notification is sent when a local or remote
           non-threshold crossing event is detected.";
        uses event-details {
          refine "event-type" {
            must ". != 'ieee802-link-oam:threshold-event-type'" {
              description
                "This leaf is not set to
                 'threshold-event-type'.";
```

```
            }
          }
        }
      }
      notification threshold-event {
        if-feature
          "link-monitoring-local or
           link-monitoring-remote";
        description
          "This notification is sent when a local or remote
           threshold crossing event is detected.";
        uses event-details {
          refine "event-type" {
            must ". = 'ieee802-link-oam:threshold-event-type'" {
              description
                "This leaf is set to 'threshold-event-type'.";
            }
          }
        }
        uses threshold-event-details;
      }
      leaf admin {
        type admin-state;
        default "disabled";
        description
          "This object is used to provision the default
           administrative OAM mode for this interface. This object
           represents the desired state of OAM for this interface.
           It starts in the disabled state until an explicit
           management action or configuration information retained
           by the system causes a transition to the enabled(1)
           state. When enabled(1), Ethernet OAM will attempt to
           operate over this interface. The default value is
           implementation-dependent.";
      }
      container discovery-info {
        description
          "Information relating to the discovery process.";
        uses discovery-info;
      }
      container event-log {
        config false;
        description
          "List of Ethernet Link OAM event log entries on the
           interface.";
        list event-log-entry {
          key "index";
          description
            "Ethernet Link OAM event log entry.";
          leaf index {
            type uint64;
            description
              "Index of this event in the event log.";
          }
          uses event-details;
          uses threshold-event-details;
        }
      }
      container statistics {
```

```
        config false;
        description
          "Statistics for an 802.3 OAM interface.";
        uses statistics-common;
        leaf local-error-symbol-period-log-entries {
          type yang:counter64;
          mandatory true;
          description
            "Number of local error symbol period log entries.";
        }
        leaf local-error-frame-log-entries {
          type yang:counter64;
          mandatory true;
          description
            "Number of local error frame log entries.";
        }
        leaf local-error-frame-period-log-entries {
          type yang:counter64;
          mandatory true;
          description
            "Number of local error frame period log entries.";
        }
        leaf local-error-frame-second-log-entries {
          type yang:counter64;
          mandatory true;
          description
            "Number of local error frame second log entries.";
        }
        leaf remote-error-symbol-period-log-entries {
          if-feature "link-monitoring-remote";
          type yang:counter64;
          mandatory true;
          description
            "Number of remote error symbol period log entries.";
        }
        leaf remote-error-frame-log-entries {
          if-feature "link-monitoring-remote";
          type yang:counter64;
          mandatory true;
          description
            "Number of remote error frame log entries.";
        }
        leaf remote-error-frame-period-log-entries {
          if-feature "link-monitoring-remote";
          type yang:counter64;
          mandatory true;
          description
            "Number of remote error frame period log entries.";
        }
        leaf remote-error-frame-second-log-entries {
          if-feature "link-monitoring-remote";
          type yang:counter64;
          mandatory true;
          description
            "Number of remote error frame second log entries.";
        }
      }
    }
  }
```

```
}
```

NOTE—IEEE Std 802.3, Annex K defines optional alternative terminology for “master” and “slave”.

Annex A

(informative)

Bibliography

The following referenced documents are informative and not needed in the implementation of this standard, rather left here for the benefit of the reader. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

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[B5] IETF RFC 3635, *Definitions of Managed Objects for the Ethernet-like Interface Types*, J. Flick, September 2003.

[B6] IETF RFC 6991, *Common YANG Data Types*, Schoenwaelder J., July 2013.

[B7] IETF RFC 8342, *Network Management Datastore Architecture (NMDA)*, M. Bjorklund, J. Schoenwaelder, P. Shafer, K. Watsen, and R. Wilton, March 2018.

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[p]IEEE publications are available from the Institute of Electrical and Electronics Engineers (http://standards.ieee.org/).

[q]Internet Requests for Comments (RFCs) are available on the World Wide Web at the following ftp site: venera.isi.edu; logon: anonymous; password: user's e-mail address; directory: in-inotes.